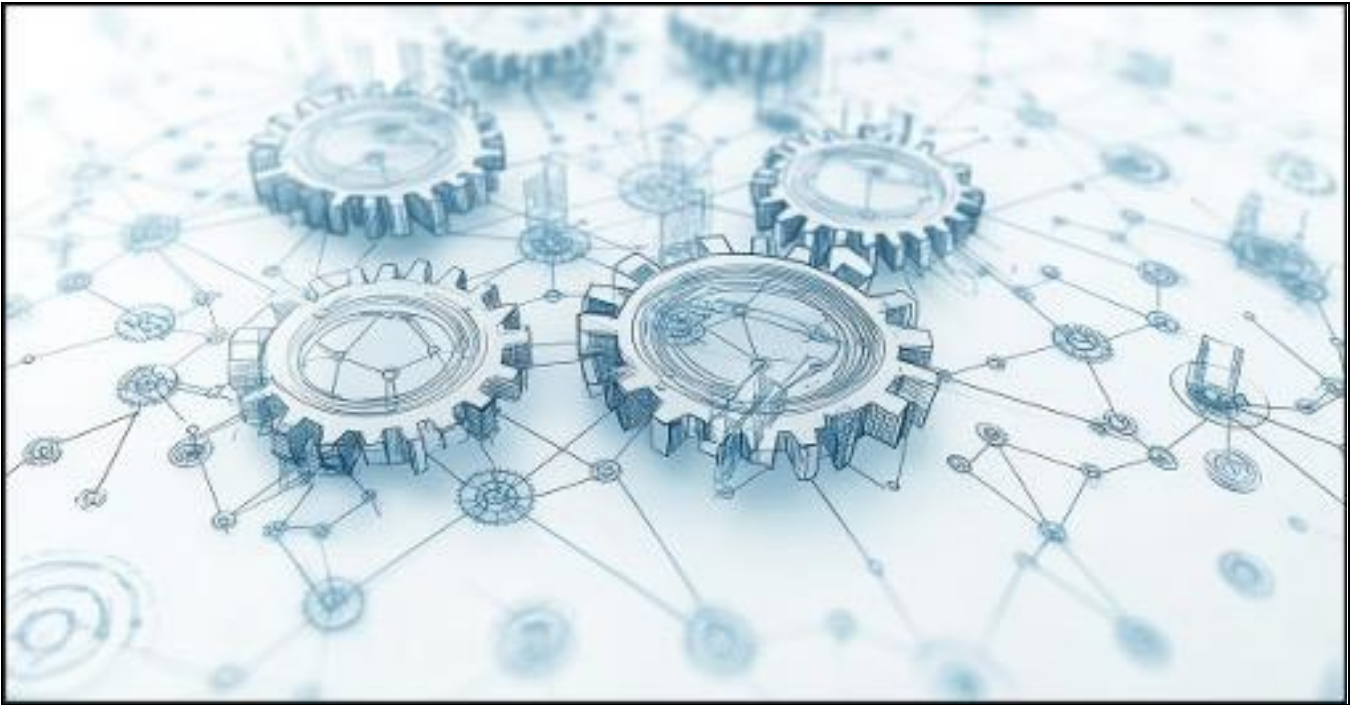


Watsonx Orchestrate

Hands-on lab guide: Creating agents with flow and document processing



Version 3.3

July 2026

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Introduction

Lab overview

Welcome to the watsonx Orchestrate flow and document processing lab guide. This guide is designed to help both beginners and experienced users alike, providing a step-by-step overview of the current capabilities of watsonx Orchestrate's flow and document processing capabilities.

When complete, the agent created will be able to classify documents, distinguish invoices from contracts and extract specific fields from these documents depending on their type.

What you will learn

Once you have completed this lab you will have developed skills enabling you to complete the following tasks:

- How to set up agents that use document classification
- How to identify and extract key data fields from uploaded documents
- Prompt the Model to identify key data in contracts and invoices
- Create flows inside your agent to manage complex task routing
- Test the agent you have created

How to get support

If you encounter an issue with this lab,

- IBM employees can open an IBM Support ticket at <http://ibm.biz/support>.
- Business Partners can request help at the [Partner Plus Support](#) website.

Prerequisites

You should have completed the other labs in this learning plan:

- Completed the preparation guide activities
- Have access to a watsonx Orchestrate SaaS instance
- You have downloaded and extracted the following archive on your local computer: [Sample Documents](#)

This archive contains the following documents

- Contract.pdf
- Utility_bill_training.pdf
- Utility_bill_2.png
- Utility_bill_4.png
- Utility_bill_5.png

Known Issues and Workarounds

As of the time of writing this lab, we have identified several issues that may affect your experience. Please read the following carefully to understand how to work around them:

- **Adding a node to a link:** In certain cases, the '+' icon used to add a node to a link may not appear—especially after adding the first extractor to the workflow. Instead, small moving arrows may be displayed, preventing you from adding any elements to the link.
Workaround: Exit the workflow editor and reopen it.
- **Extracting fields from documents:** After creating a field or updating its definition, the UI may take time to display the newly extracted value—or may not show it at all. The same issue can occur when adding examples and clicking the *Show in document* button.
Workaround: Wait a few seconds for the screen to refresh or click your browser's refresh button.
- **Fields showing as “Not found”:** Occasionally, fields may appear as “Not found in the document” or Null even though they were previously extracted correctly.
Workaround: Wait a few seconds to allow the LLM to return the correct values and for the UI to update.

Use case

This lab focuses on a practical use case involving document classification, followed by key data extraction from both unstructured documents (primarily contracts) and structured ones (such as invoices). At the outset, the guide outlines the sequence of activities required to support the user journey of extracting essential information from uploaded documents.

In this lab scenario, a payment recovery organization processes numerous contracts and invoices weekly, extracting critical information like payment terms, amounts, and due dates. Currently, this is done manually, consuming significant employee time and increasing the risk of errors. With growing document volumes, the firm faces scalability challenges and inefficiencies in their processes.

Challenges:

- **Manual Process:** Employees spend hours reviewing documents and entering data, which slows down operations.
- **Errors:** Human mistakes lead to discrepancies, delays, and potential client issues.
- **Scalability:** As the business expands, the volume of documents grows, requiring more manual labor, which is costly and unsustainable.

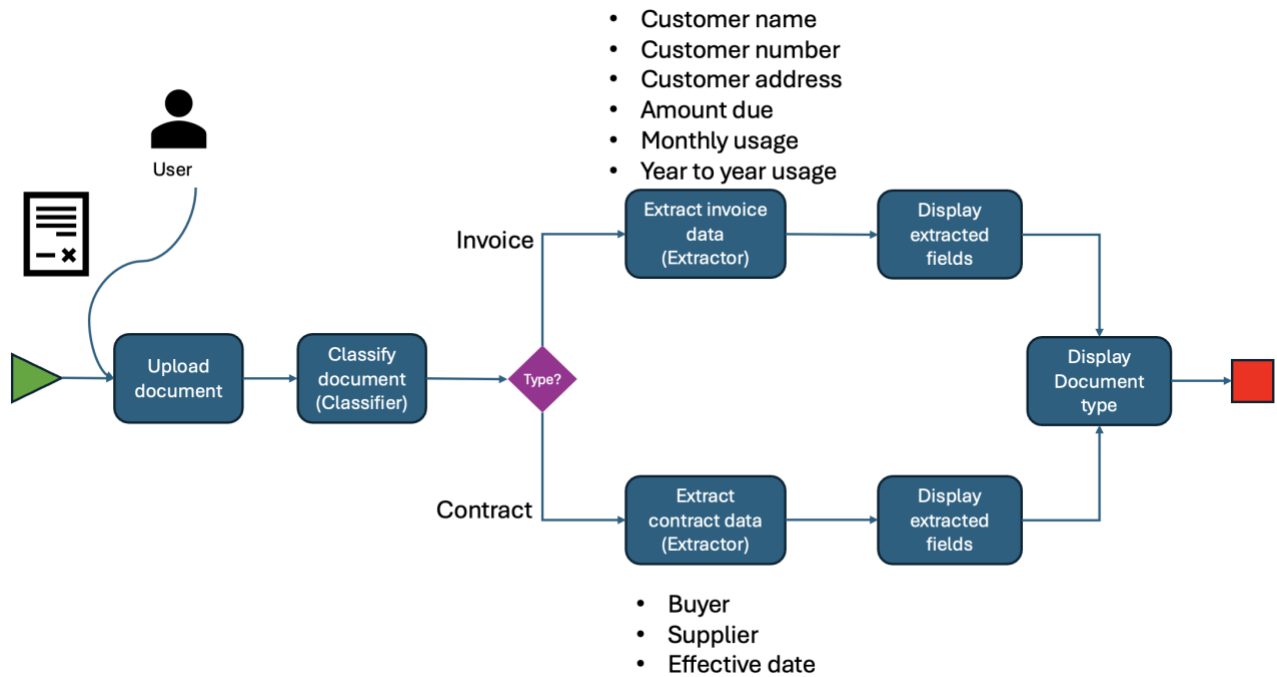
Solution: Watsonx Orchestrate

The firm implements Watsonx Orchestrate to automate the extraction of key data from contracts and invoices, streamlining the process and reducing errors.

- **Automated Extraction:** Watsonx Orchestrate automatically extracts relevant data from documents, eliminating manual entry.
- **Workflow Integration:** The extracted data is seamlessly integrated into the firm's systems (e.g., billing and accounting), speeding up the process.
- **Confidence Review:** If the extraction confidence is low, a review step ensures data accuracy without delays.

The process begins with configuring the document classifier and extractor tools to identify the type of document being ingested. Based on the classification, the system then determines which key-value pairs to extract. Finally, the flow concludes with a summary of the main fields retrieved by the agent during the extraction process, completing the end-to-end user journey.

The diagram below presents the workflow supporting this use case:



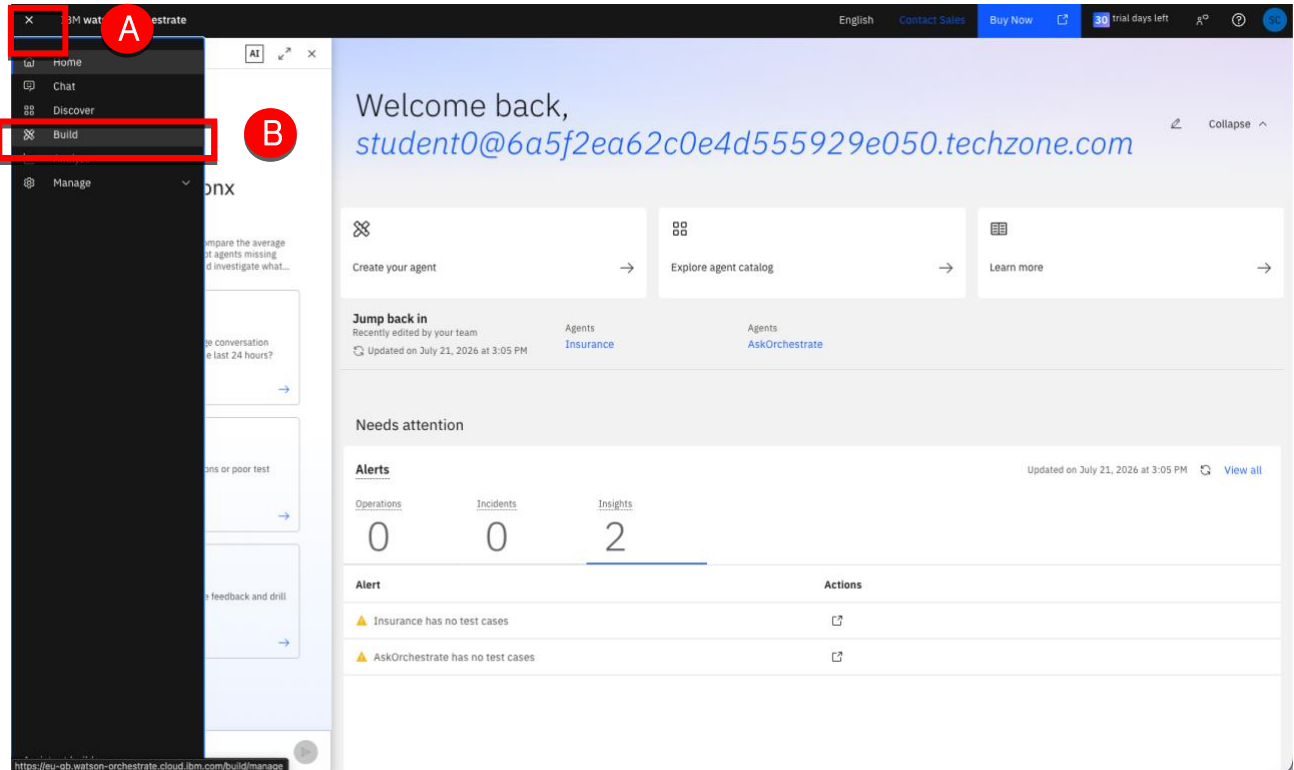
This lab guide will guide you through the different steps to create this flow.

Let's get started.

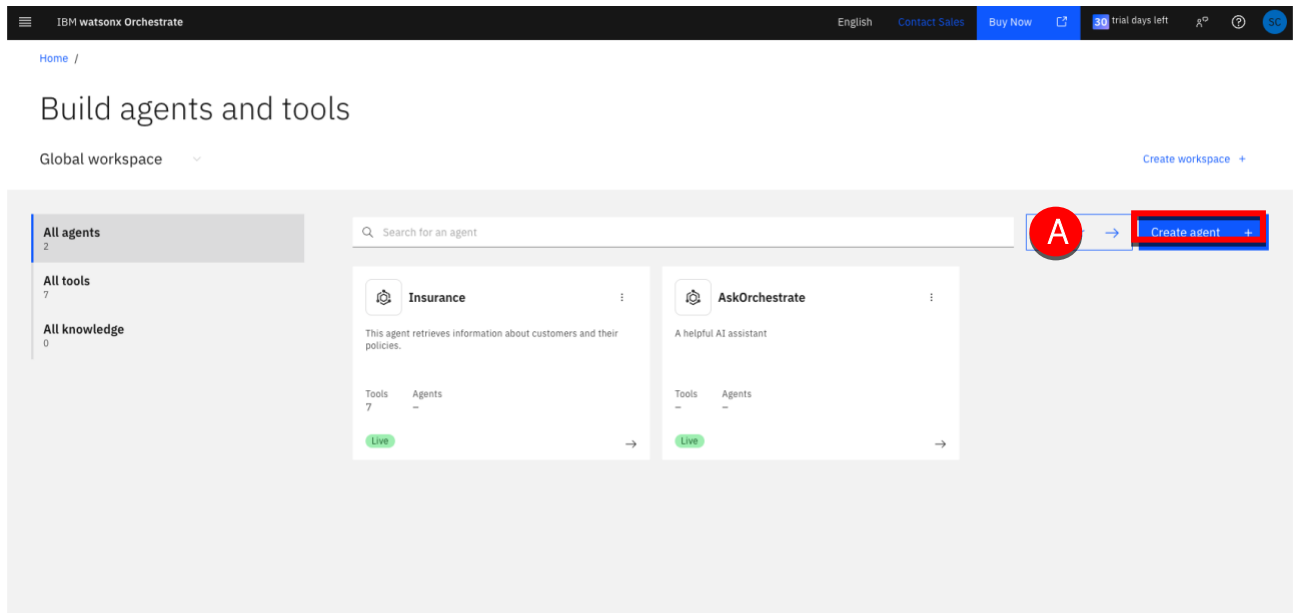
Create an Agent

In this section, you will create a document processing agent. This agent will implement the process previously described using the watsonx Orchestrate flow capabilities, as well as a Document Classifier and Document extractors.

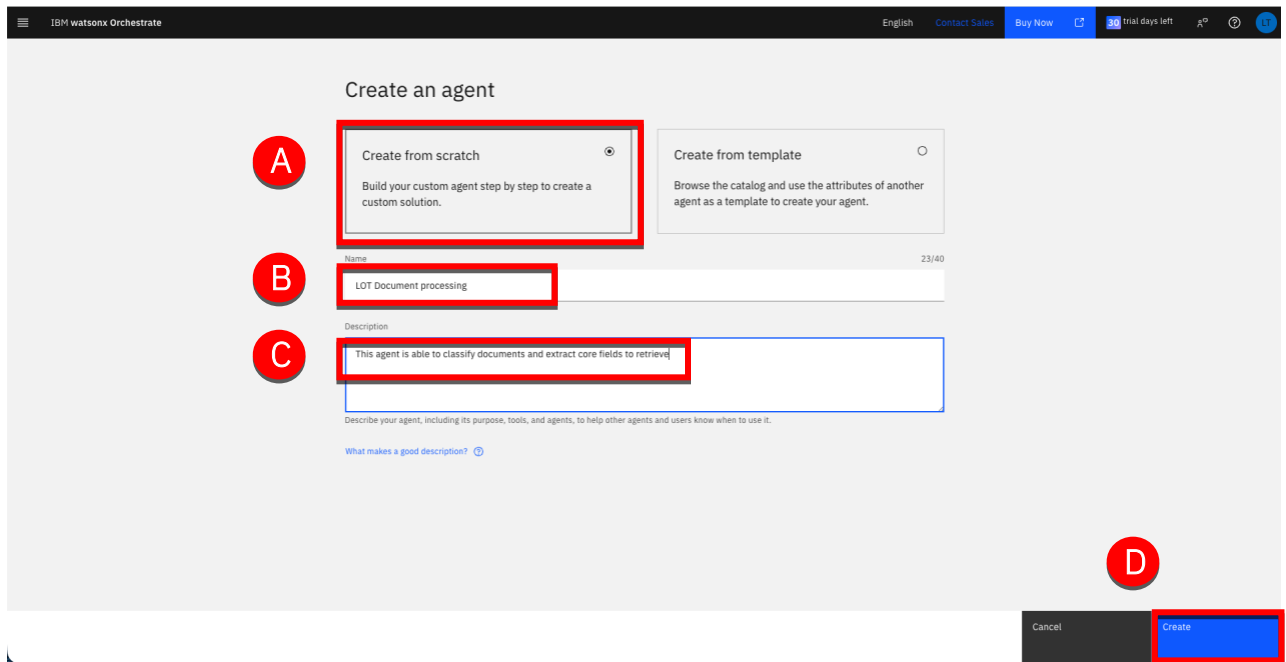
1. Launch your watsonx Orchestrate SaaS instance.
2. Click the Main menu icon (A) then click **Build** (B).



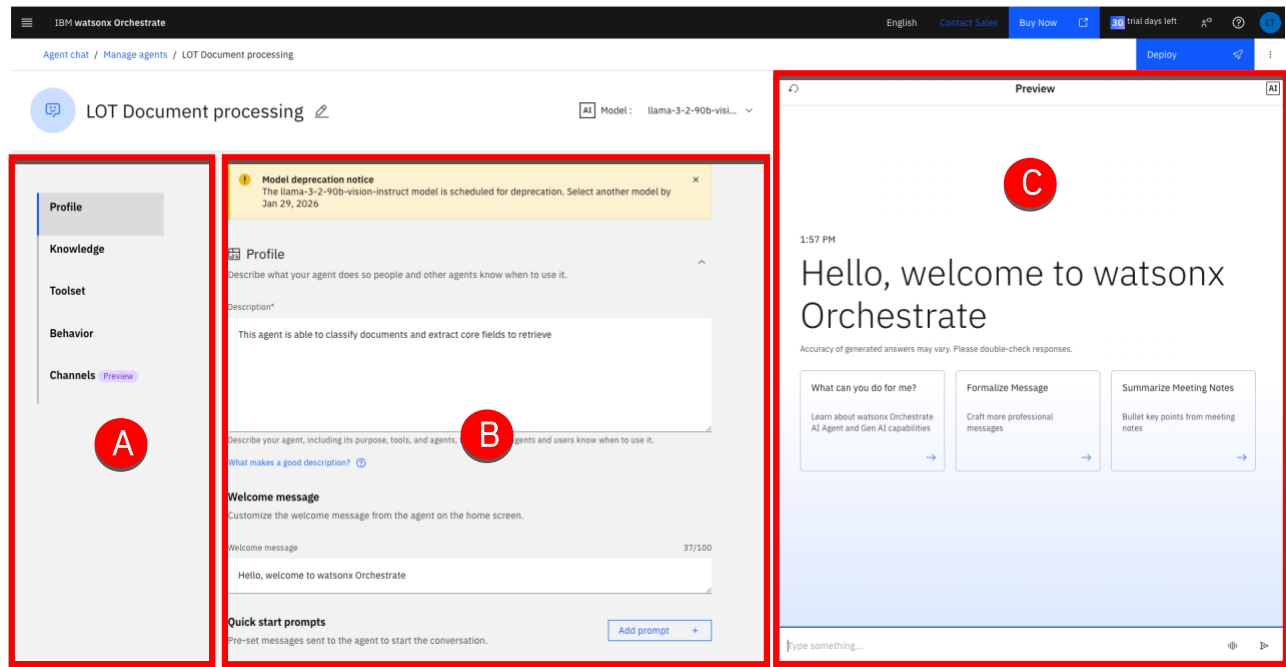
3. Click **Create Agent** (A).



4. Select **Create from scratch** (A). Type "[YOUR INITIALS] Document processing" in the **Name** field (B). Type "This agent is able to classify documents and extract core fields to retrieve" in the **Description** field (C). Click **Create** (D).



5. Note the different sections of the Agent builder tool.



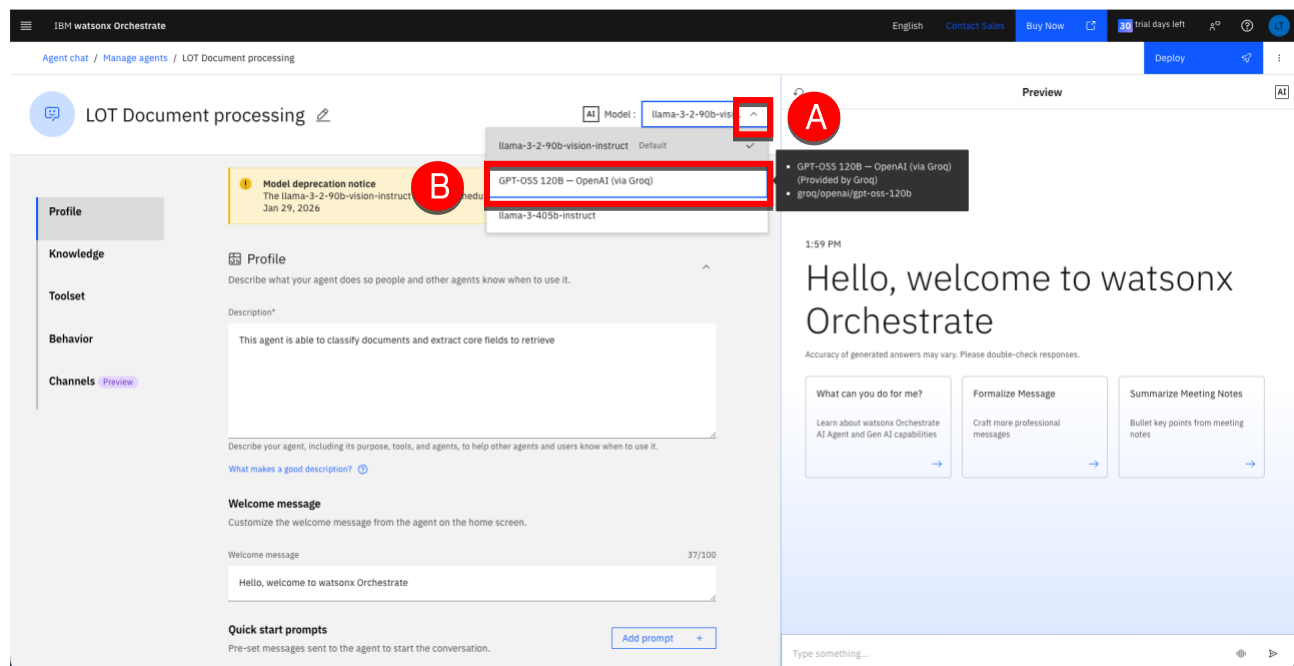
The Agent Builder opens; the screen is divided into three key areas:

- (A) **Navigation Panel** – move between different sections of the Agent Builder.
- (B) **Configuration** – set up and customize the core functionality of your agent.
- (C) **Preview** – preview and test your agent.

Below is a description of each section:

- **Profile:** Define your agent's purpose, usage scenarios, and interaction style. Describe what the agent does, when it should be used (especially in multi-agent configurations), and choose its style—Default or ReAct—to guide how it interprets requests, plans and uses tools.
- **Knowledge:** This is where you equip your agent with knowledge by uploading files or connecting to conversational search platforms such as Milvus, Elasticsearch, or custom-built sources. This ensures the agent can generate accurate, contextual responses by drawing on relevant content.
- **Toolset:** Provide your agent with tools to perform tasks. Tools can be added from the Catalog, imported from OpenAPI specification files and MCP servers, or built with custom flows. Tools extend the agent's capabilities, enabling it to automate actions such as retrieving data or sending emails.
- **Behavior:** Here, you define how the agent interacts with users, formats data and how it handles requests. Add rules and instructions to shape its tone, response style, and overall behavior during interactions.
- **Channels:** This is where you connect your agent to communication platforms like Slack or embed the agent in a website.

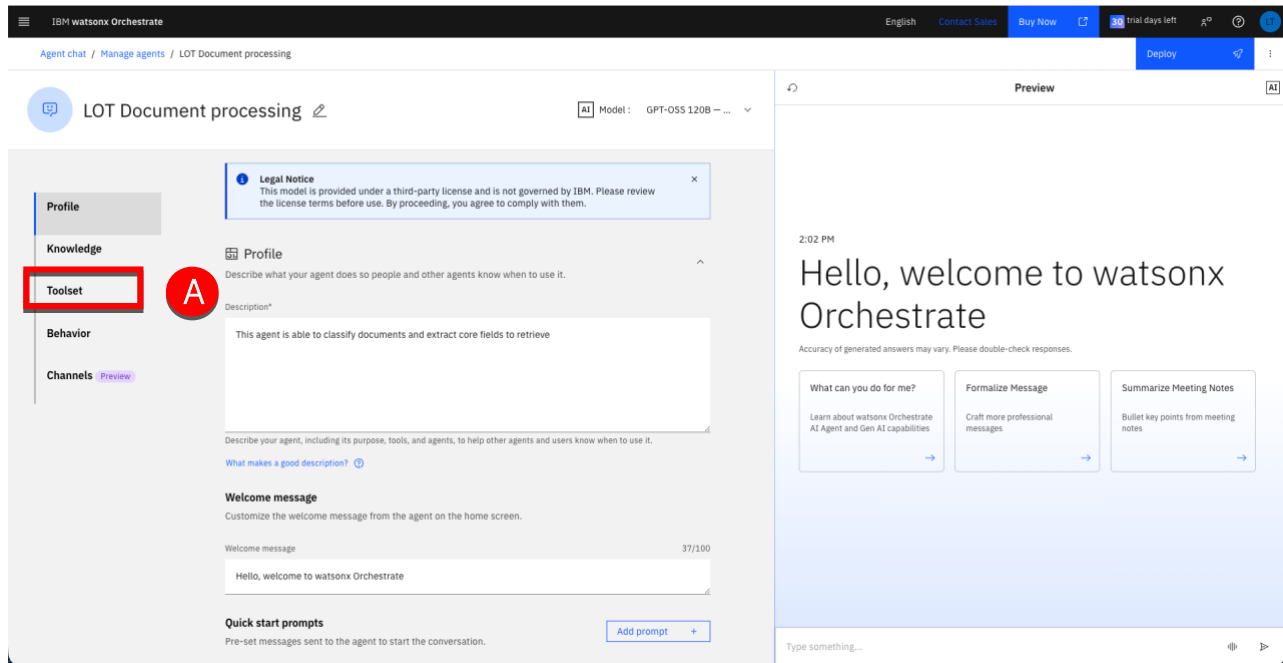
6. Expand the **AI Model** menu (A), Select **GPT-OSS** model (B).



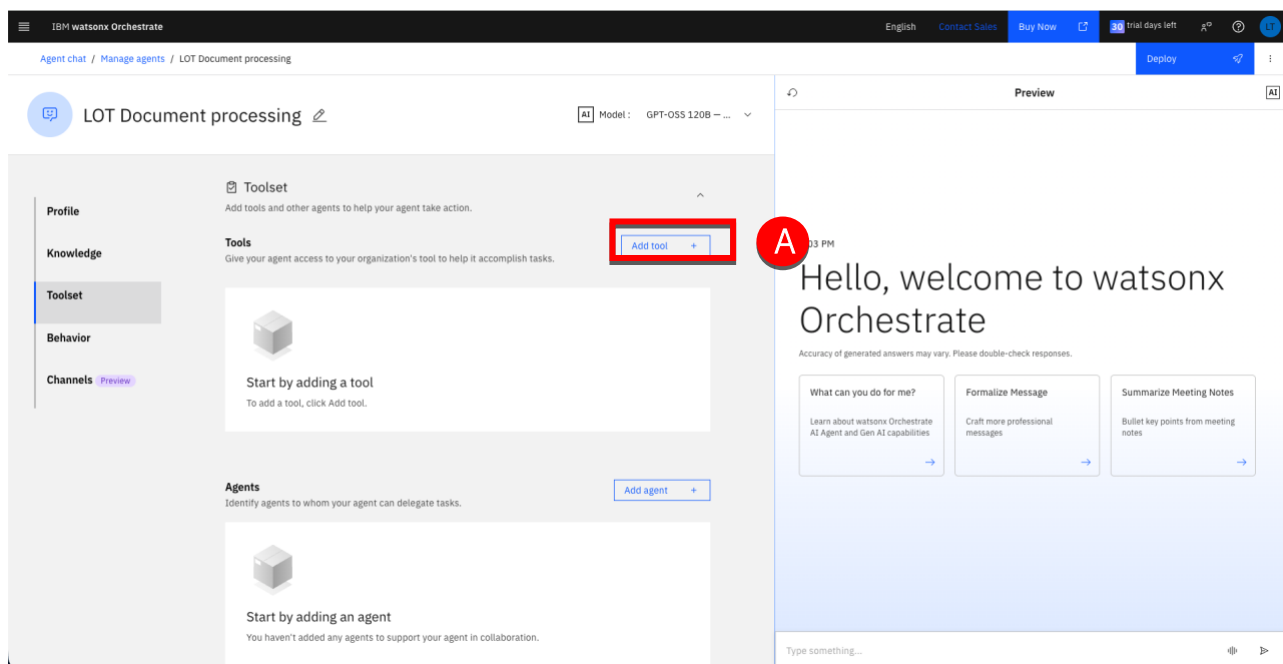
Add tools

In this section you will add a tool to your agent. This tool will contain the process described in the introduction of this lab. You will start by creating the tool and implementing the tool behavior as a flow.

1. Click **Toolset** (A).

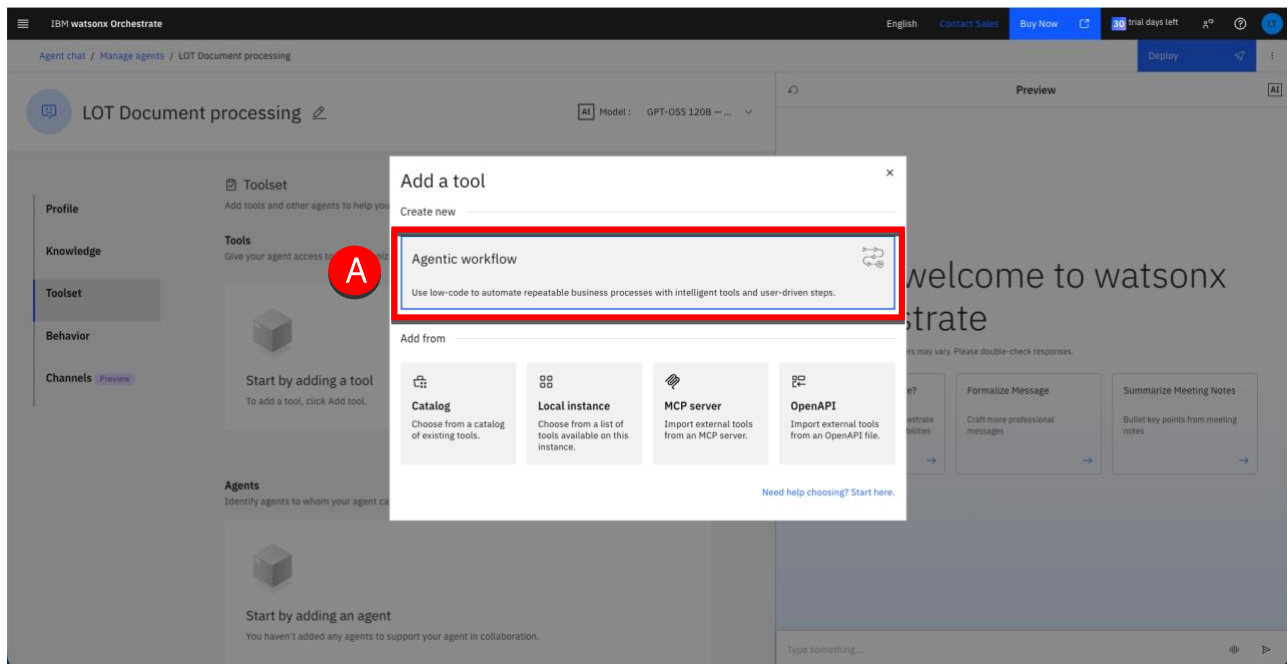


2. Click **Add tool** (A).

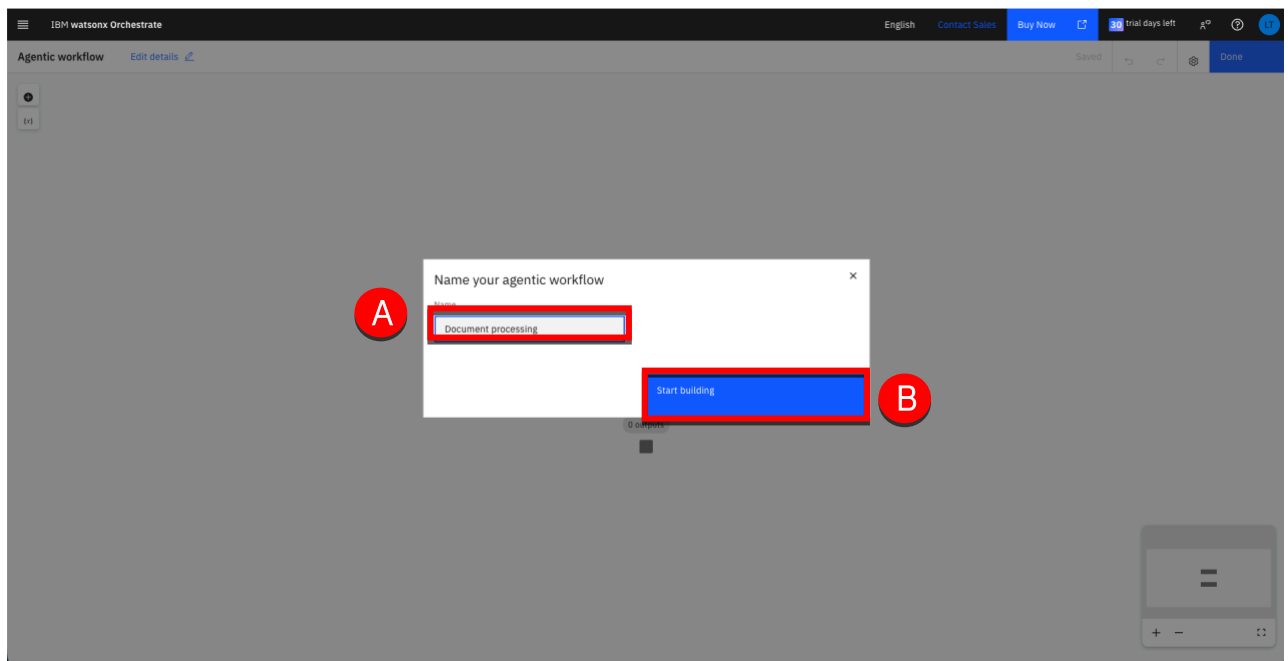


Create the workflow

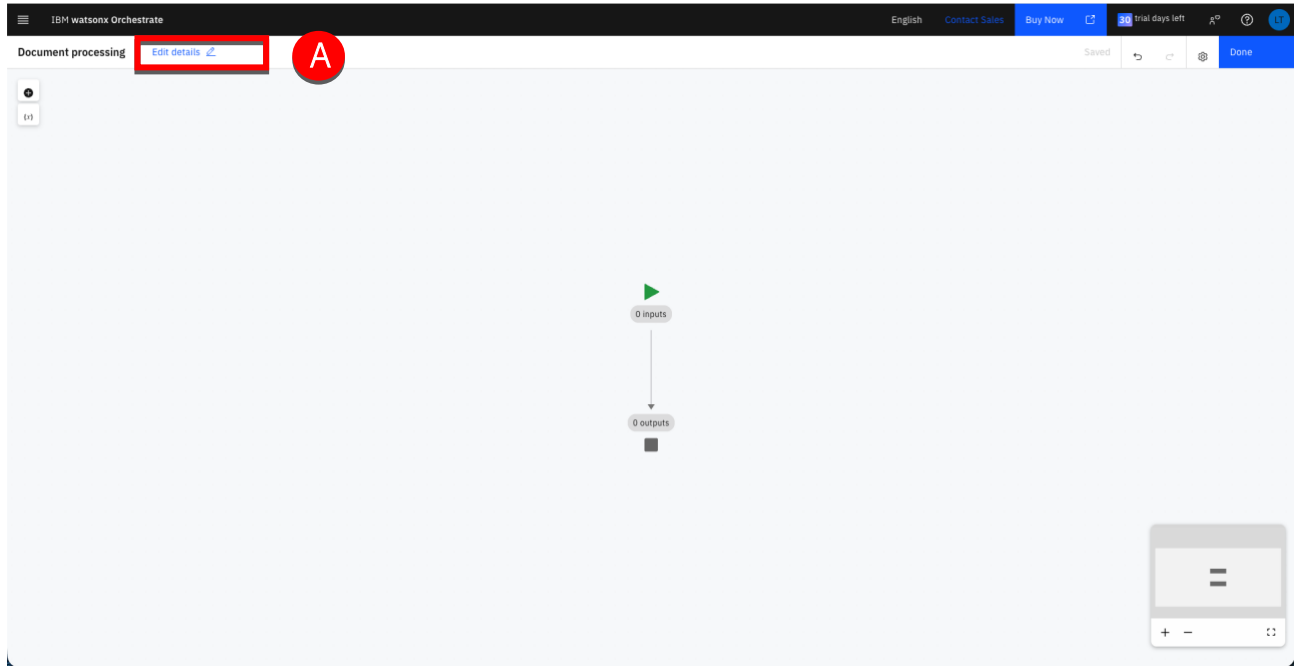
1. Click **Agentic workflow** (A).



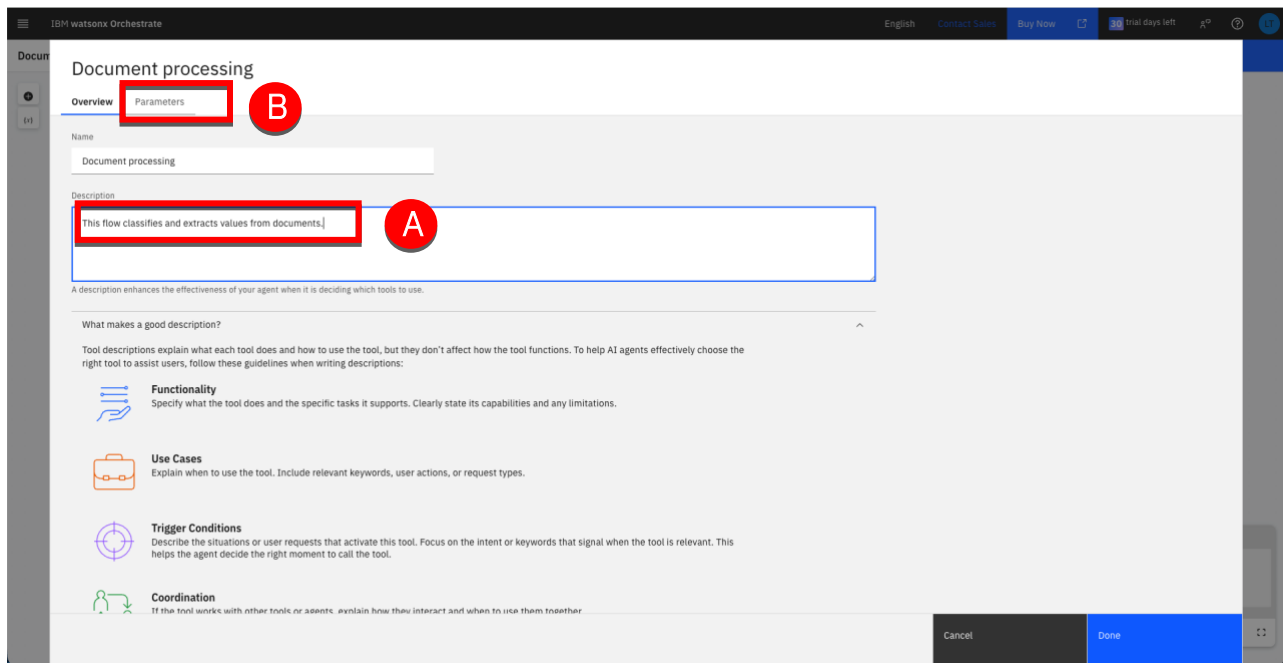
2. Type 'Document processing' in the **Name** (A). Click **Start building** (B).



3. Click **Edit details** (A).



4. Type "This flow classifies and extracts values from documents." In the **Description** (A). Click **Parameters** (B).

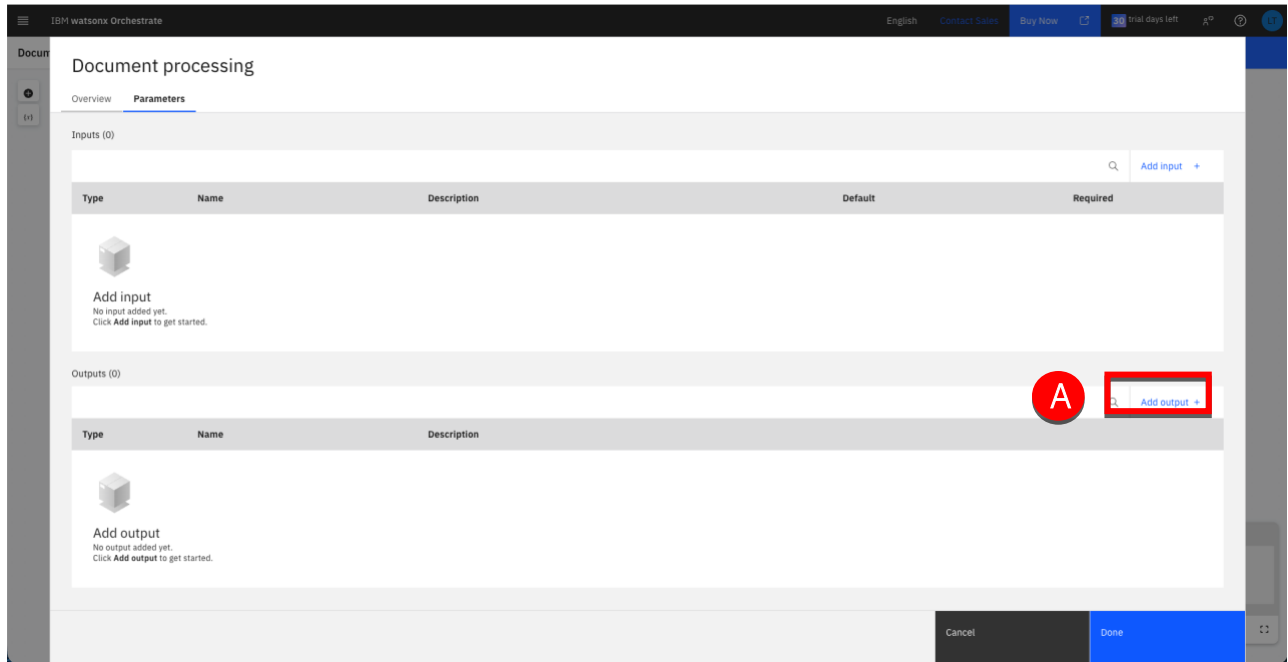


Define the process output

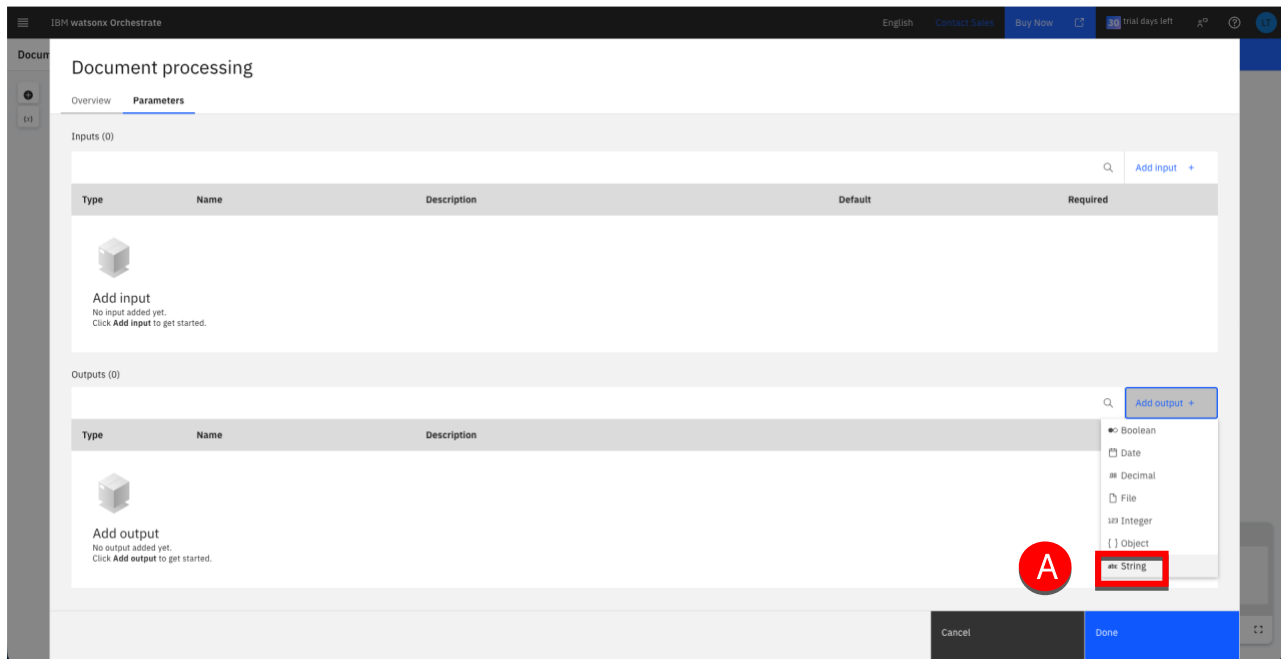
From the Parameters tab, you'll find the Input and Output sections, which apply to the entire flow. In this case, no inputs are required since they are already handled within the flow itself through the user

activity file upload. For the output, instead of using a text output node, the flow can return the classification result, defined as a string.

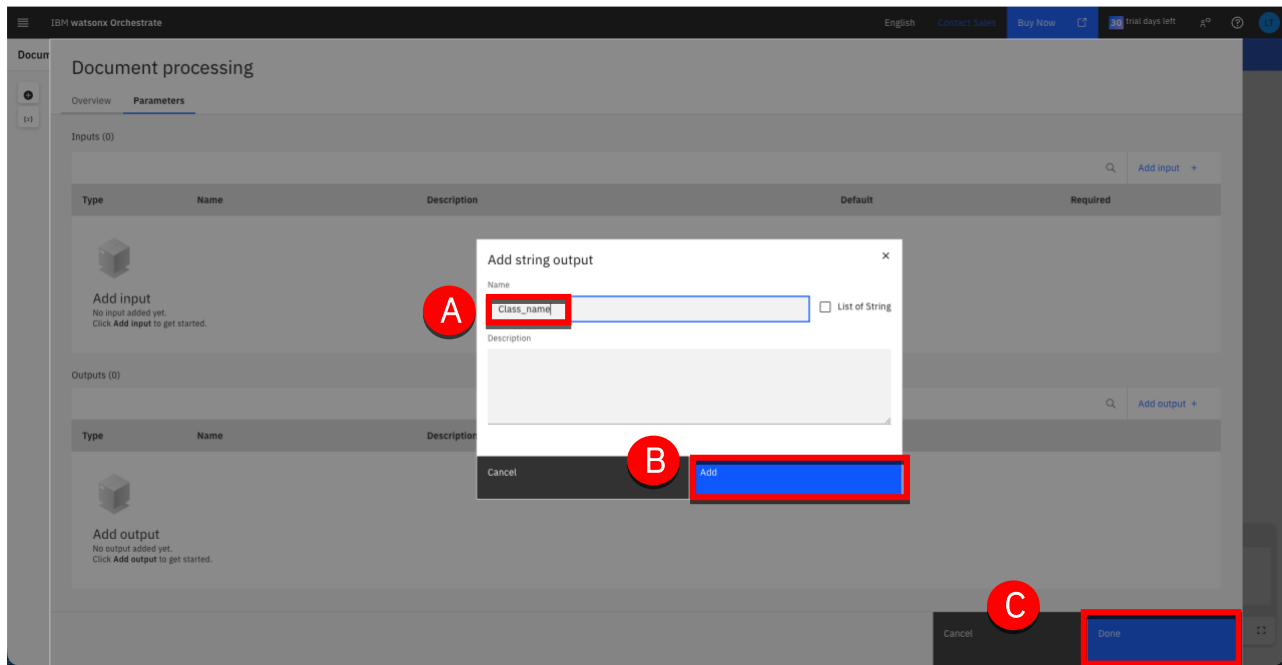
1. Click **Add output** (A).



2. Select **String** (A).



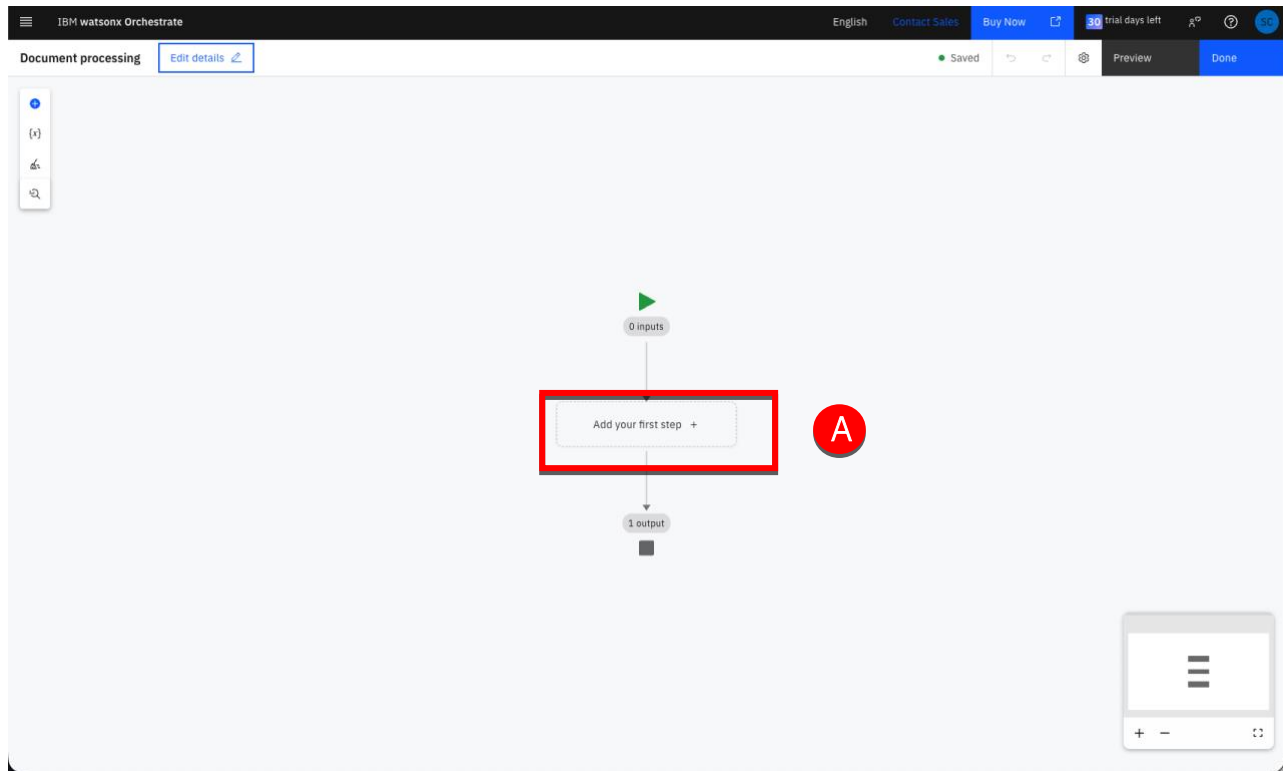
3. Type “Class_name” in the **Name** field (A). Click **Add** (B). Then click **Done** (C).



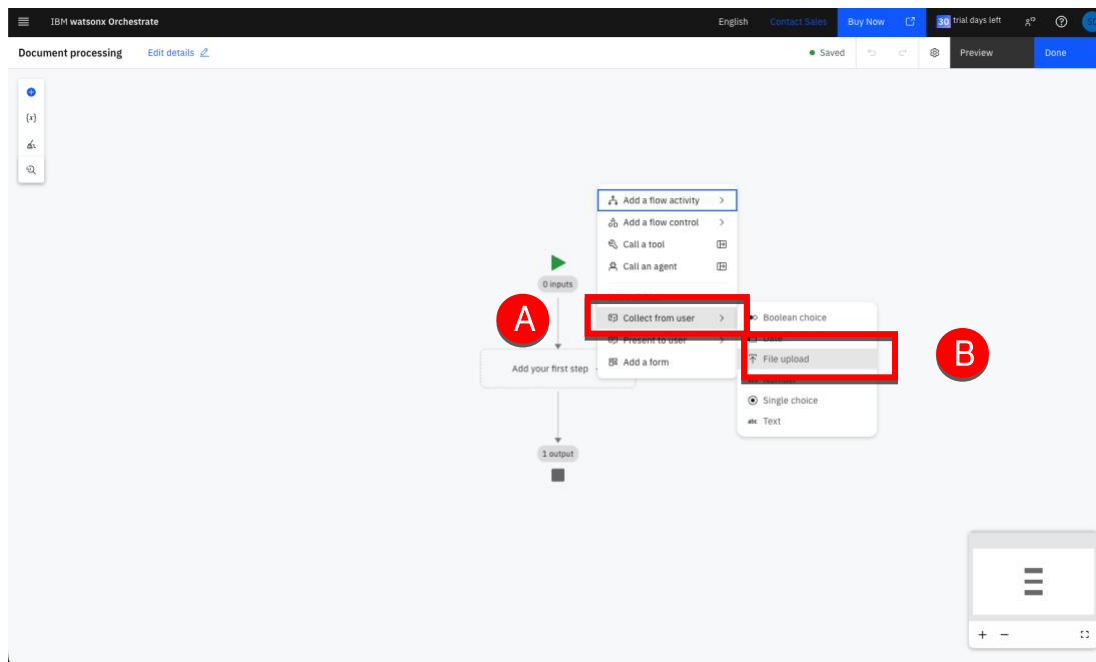
Design the workflow diagram

In this section, you will add the different workflow activities including the document classifier and the two extractors.

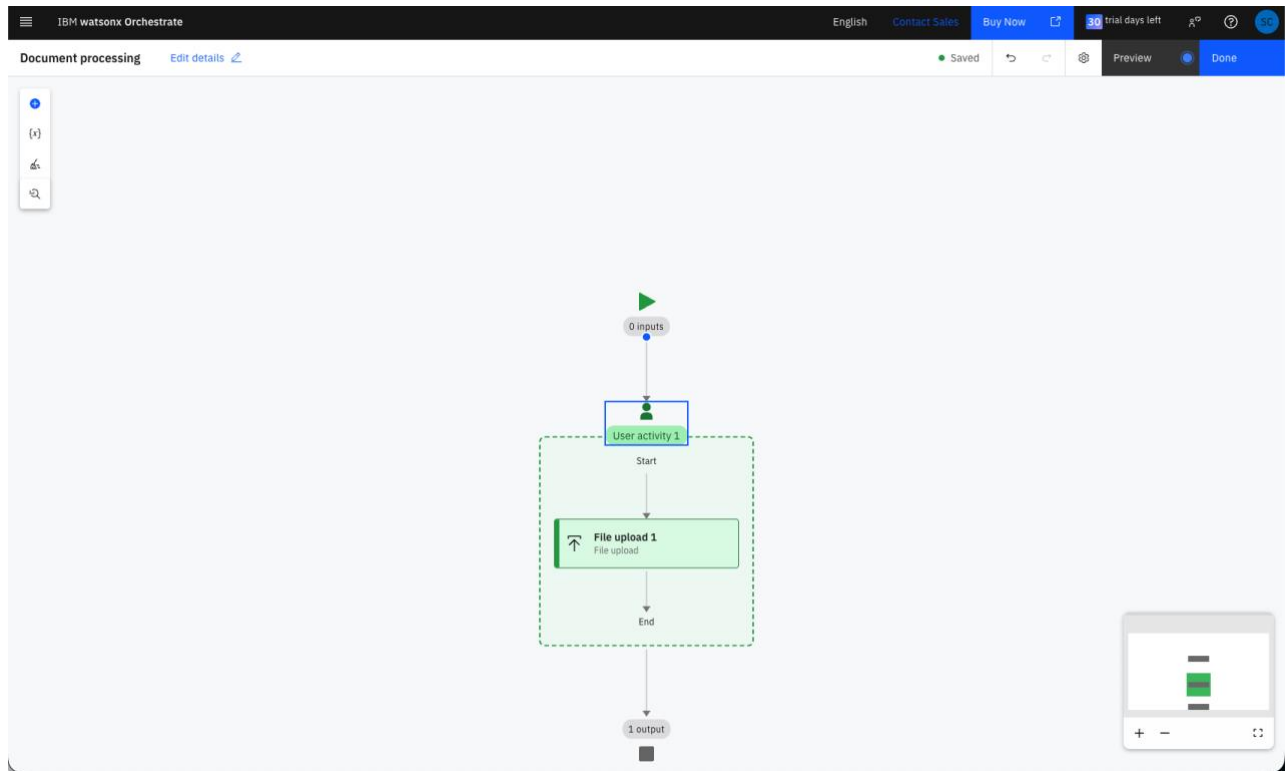
1. Click the **Add your first step +** to add an activity (A). The first activity of the process consists in uploading the document to process. It will be implemented as a user activity.



2. Select **Collect from user** (A) and then select **File upload** (B).



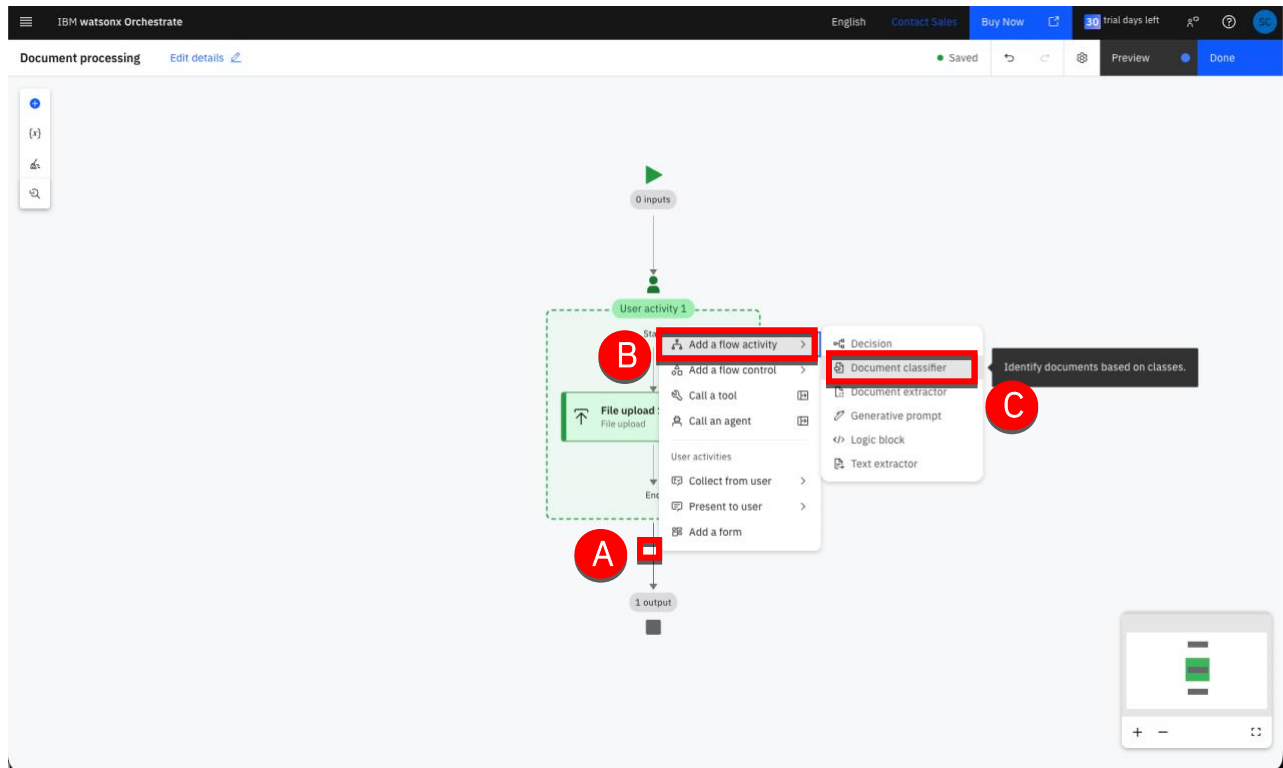
The first activity of the process is now defined. The next step is to classify the document the user will upload. To do this, you will add a watsonx Orchestrate Document Classifier.



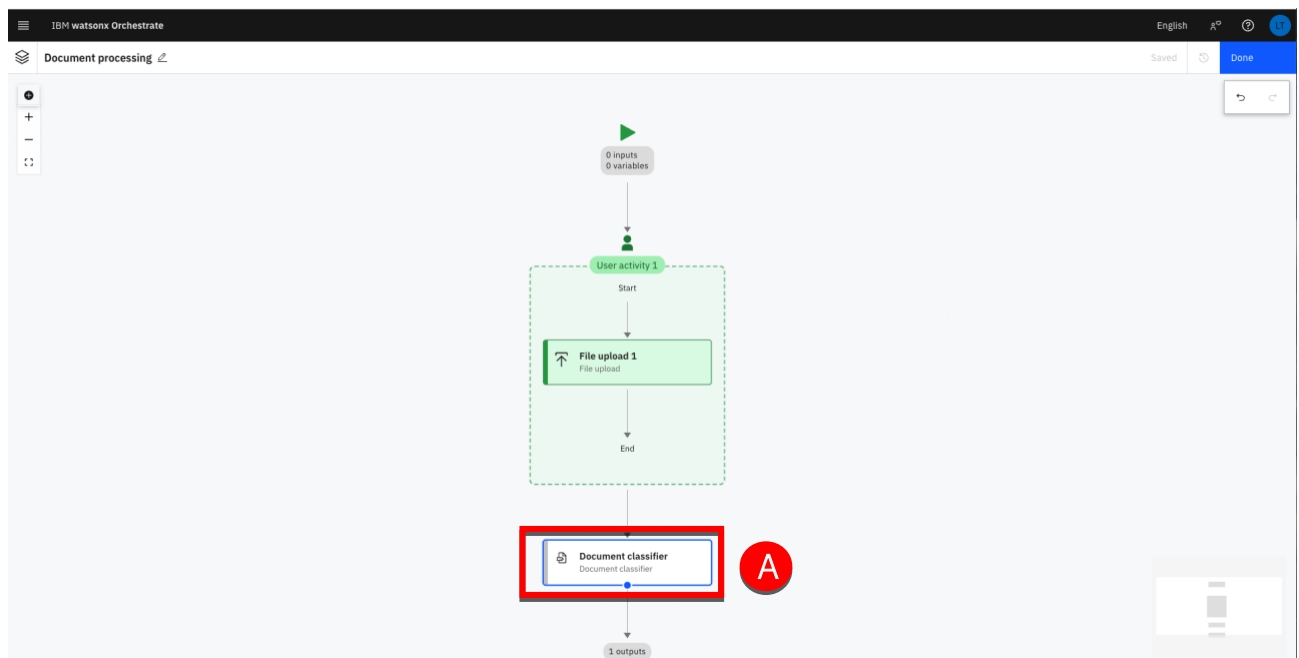
Add a document classifier

In this section, you will add a document classifier to distinguish Invoice documents from the Contract ones.

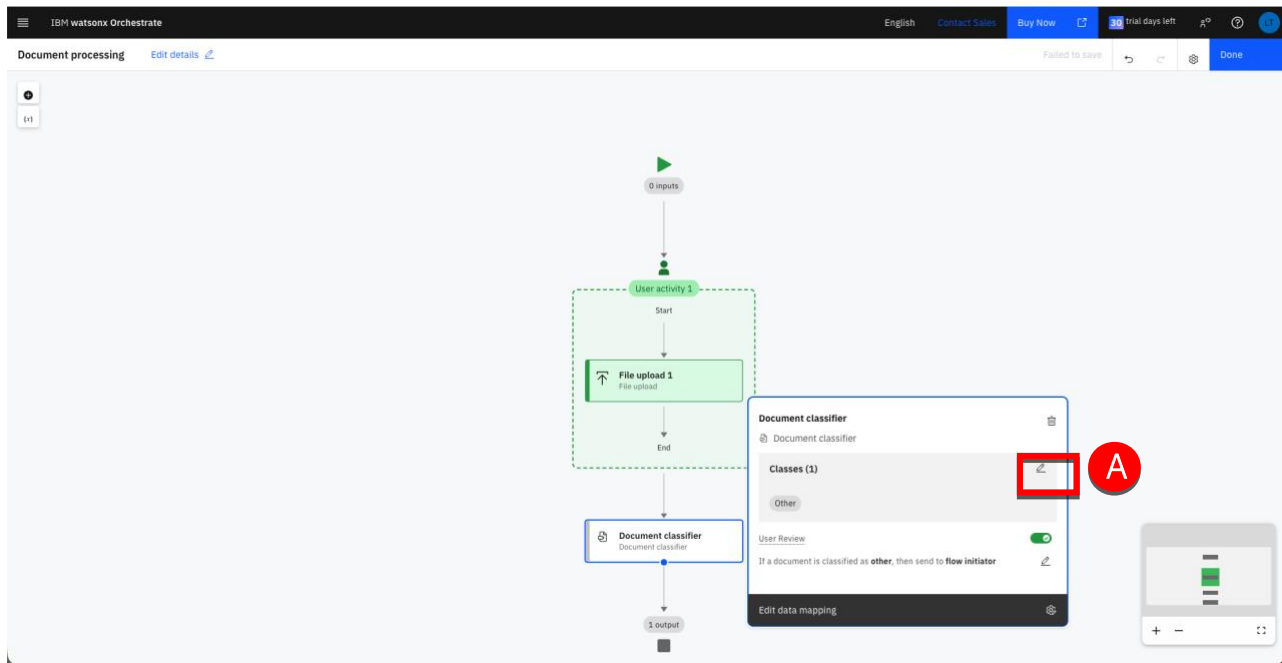
1. Move the cursor over the last link of the process and click + (A). Select **Add a flow activity** (B) then click **Document classifier** (C).



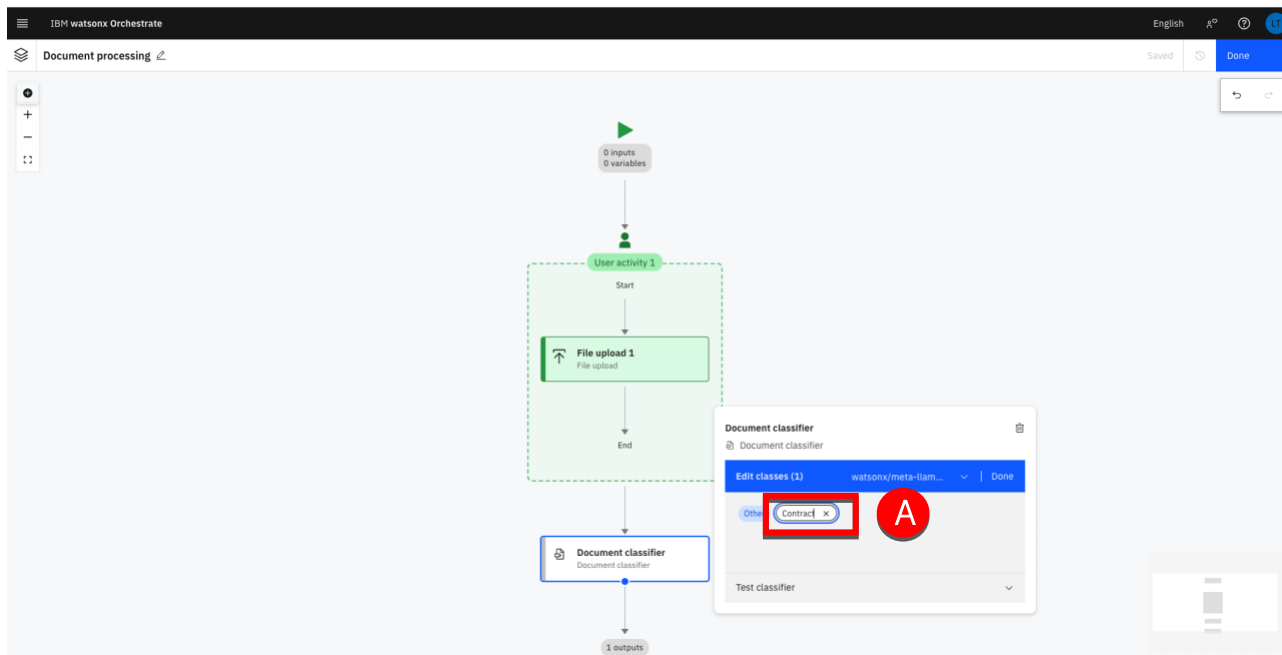
2. Click the **Document classifier** node (A).



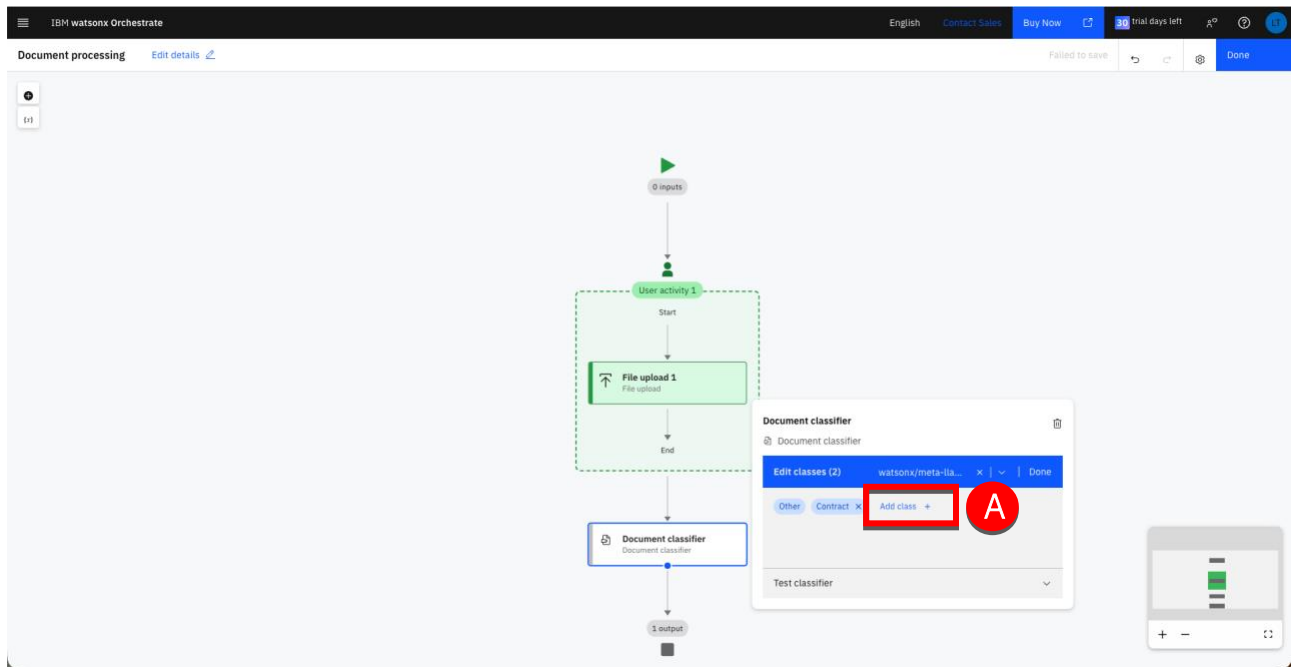
3. Click the edit icon (A).



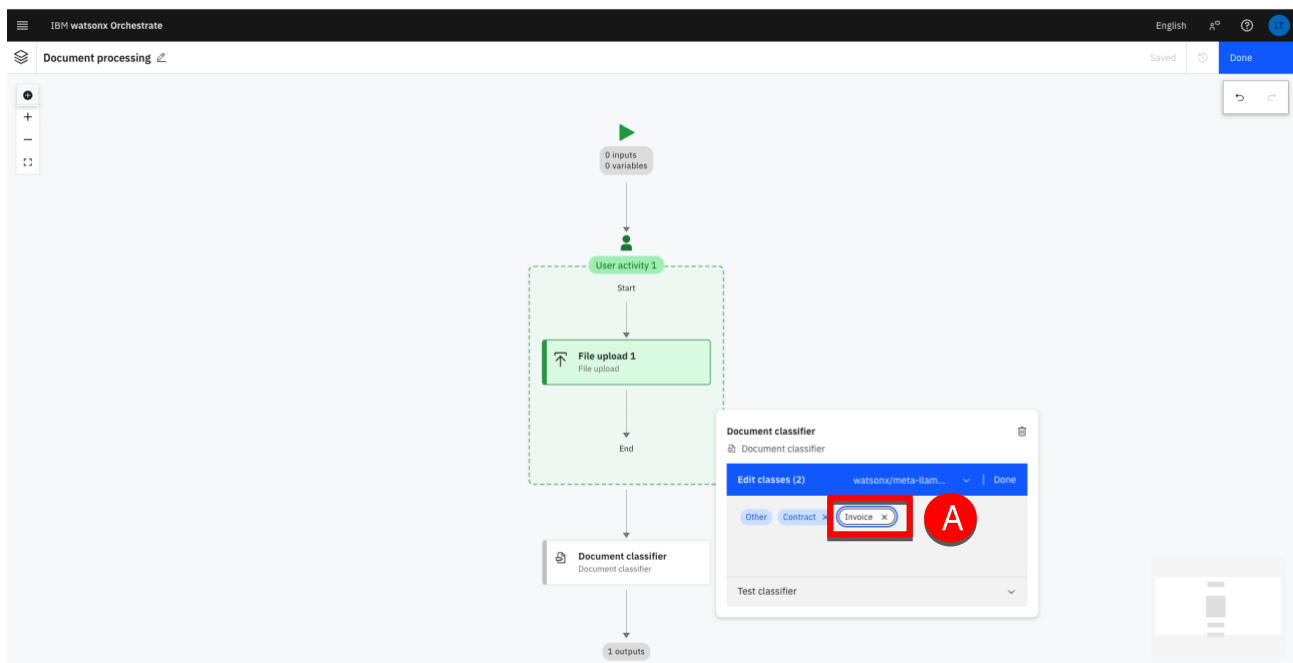
4. Type 'Contract' (A), then Enter.



5. Click **Add class** (A).

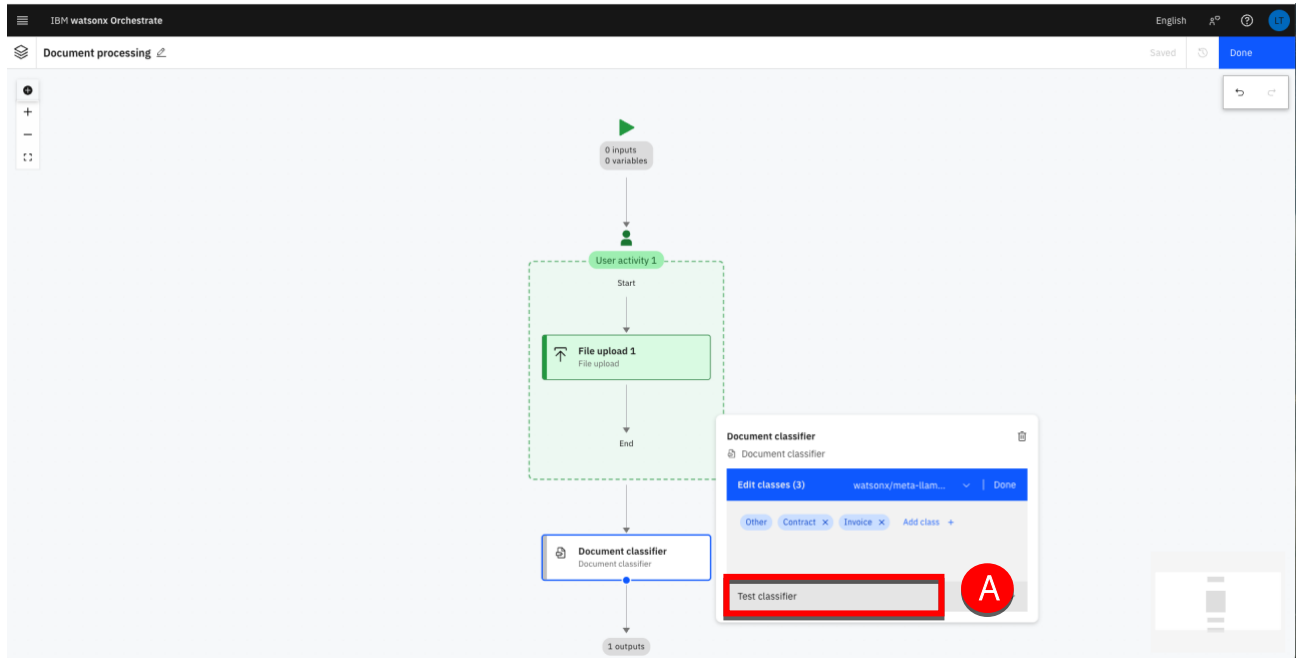


6. Type 'Invoice' into the new field (A), then TAB out of the field.

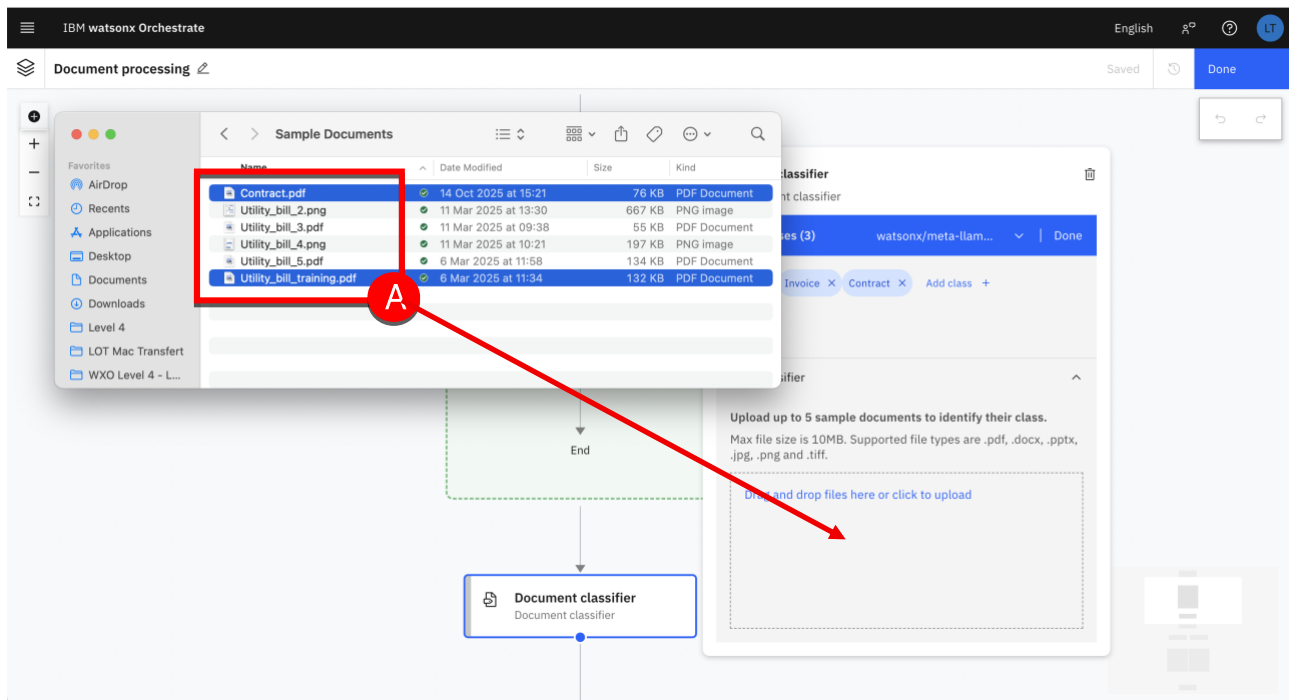


You are now ready to test your document classifier.

7. Click **Test classifier** (A).



8. Drag and drop the two sample files, **Utility_bill_training.pdf** and **Contract.pdf** from your local machine location (A). Note: ensure you select both files before dropping in the upload area, if you upload a single document you will need to click the **Upload** icon.



9. Wait for the documents to be analyzed and classified (A).

The screenshot shows the IBM watsonx Orchestrate interface. On the left, a flow diagram for 'Document processing' is visible. It starts with a 'User activity 1' block, followed by a 'File upload 1' block, and then a 'Document classifier' block. The 'Document classifier' block is highlighted with a red box. On the right, a 'Document classifier' panel is open. It shows 'Edit classes (3)' with 'Contract' and 'Invoice' selected. Below, the 'Test classifier' section displays 'Predicted results' for 'Contract.pdf' (classified as 'Contract') and 'Utility_bill_training.pdf' (classified as 'Invoice'). A red box highlights the 'Contract' and 'Invoice' labels in the predicted results, and a red circle with the letter 'A' is next to it.

Note. The two documents have been successfully classified, Contract.pdf has been classified as a Contract Invoice.pdf has been classified as an Invoice.

10. Click **Done** (A).

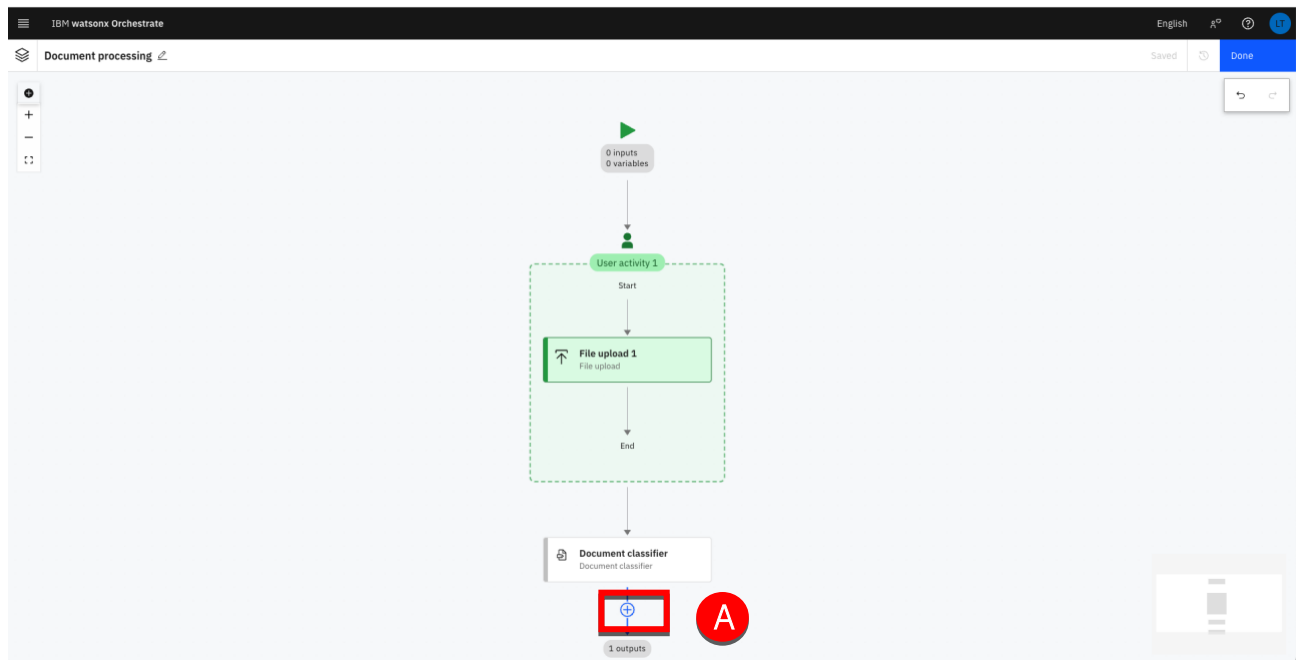
This screenshot is similar to the previous one, showing the same flow diagram and document classifier panel. However, the 'Done' button in the 'Edit classes (3)' section of the 'Document classifier' panel is now highlighted with a red box. A red circle with the letter 'A' is next to it, indicating the action to be taken.

Note: You can click the 'Upload' icon seen above to test out the classifier with more document types and verify its accuracy.

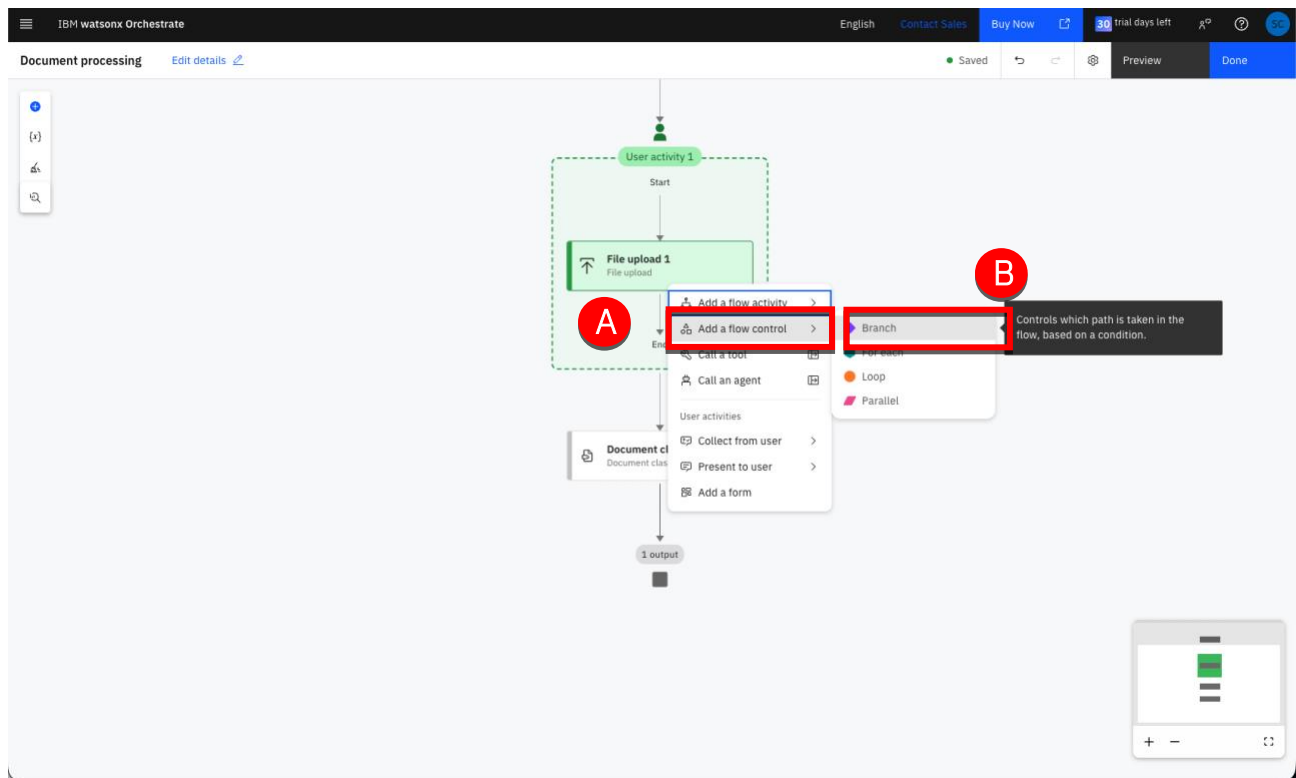
The next step is to route the process on the right extractor path depending on the document type. To do this, you will add a branch node responsible for the triage.

Add a Branch node

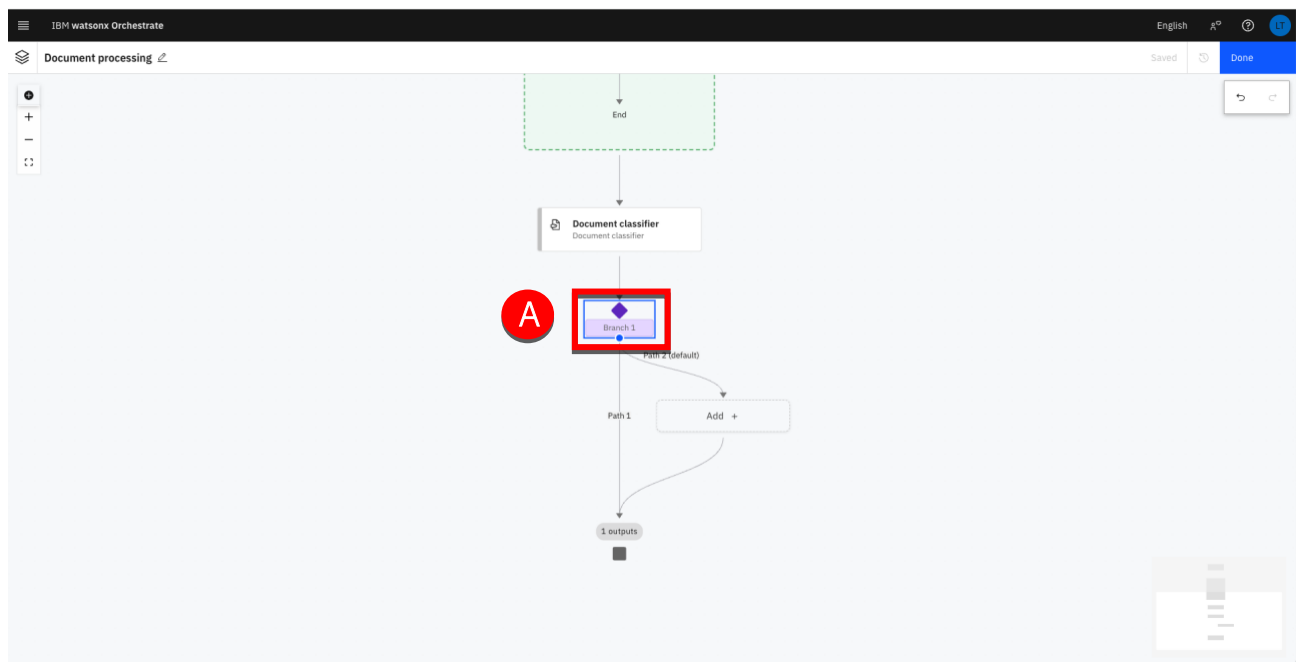
1. Move the cursor over the last link of the diagram then click + (A).



- Click **Add flow control** (A) then click **Branch** (B).

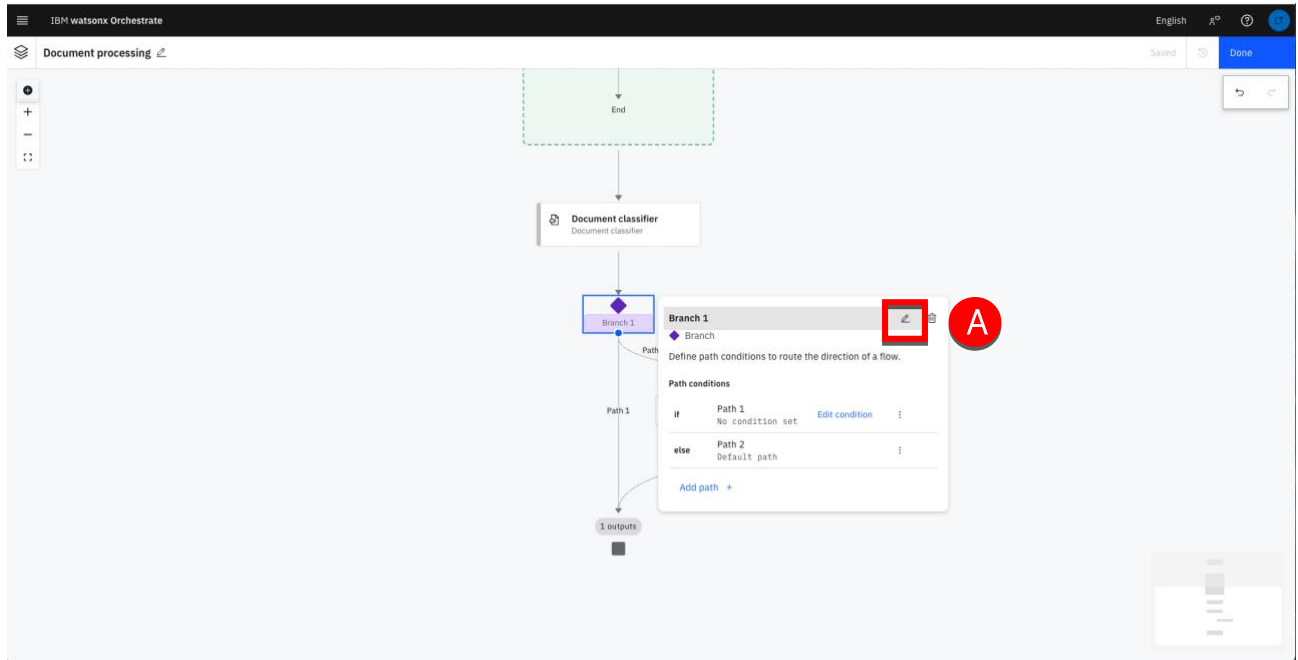


- Click the **Branch 1** node (A).

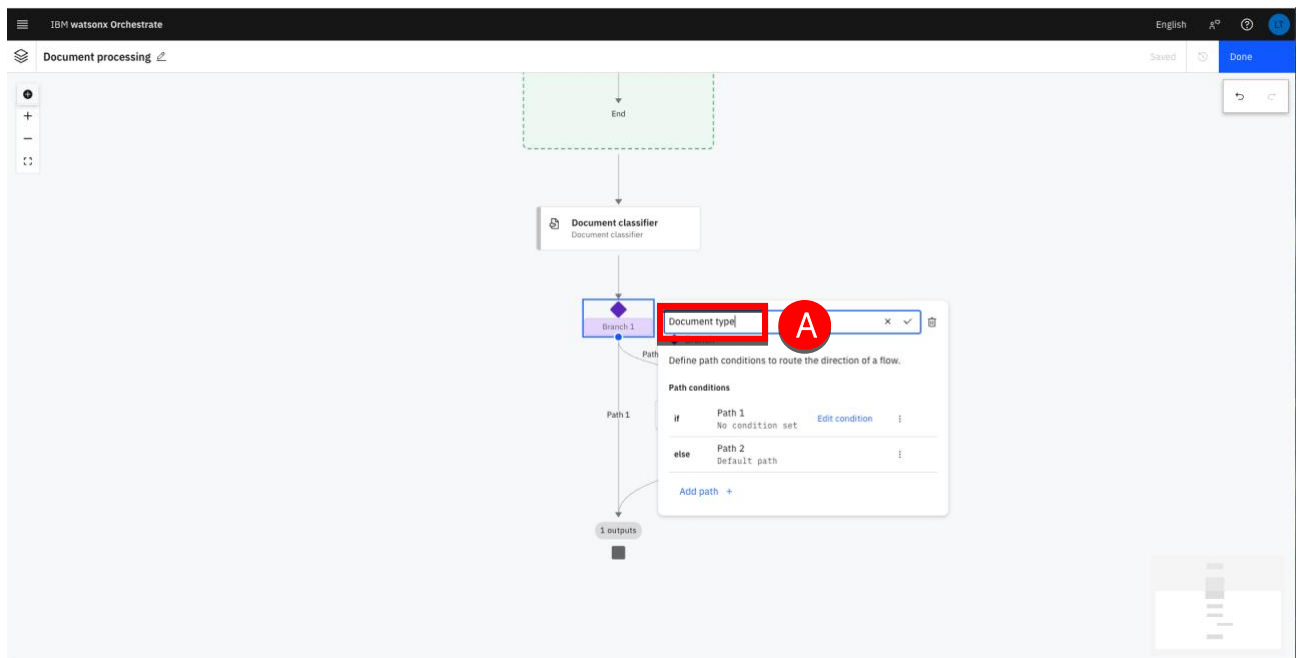


Note: When creating a branch from scratch, two paths are generated by default, but additional paths can be added as needed.

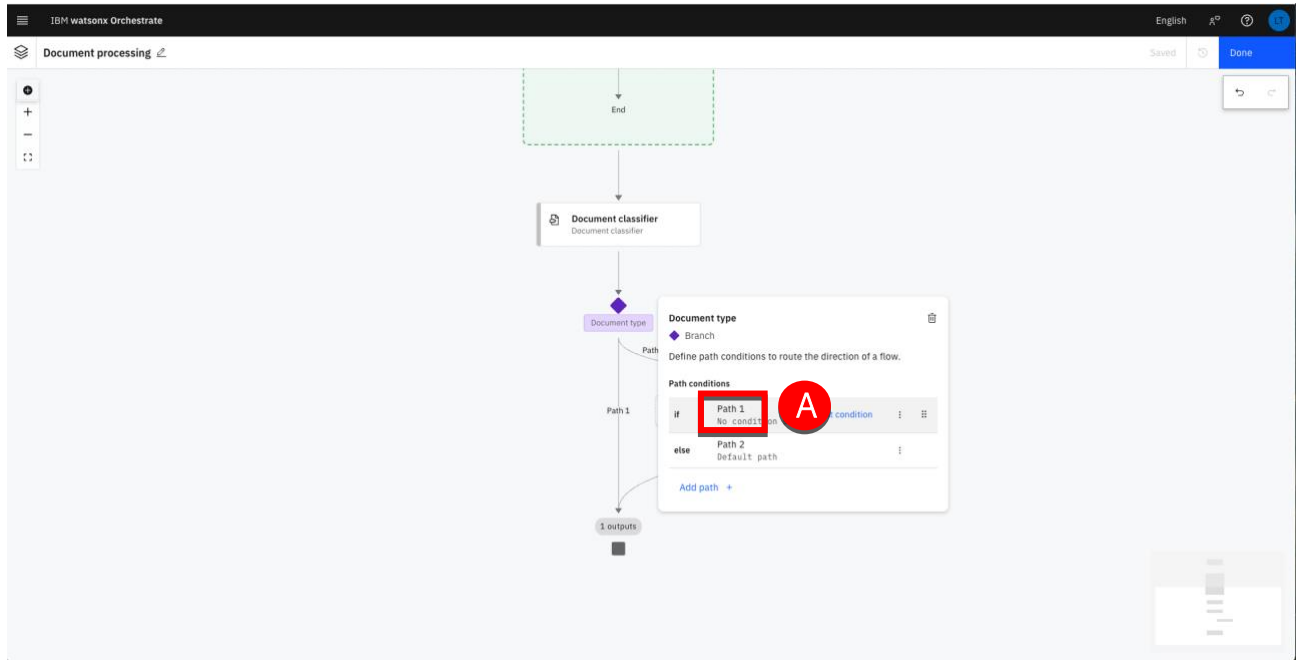
4. Move the cursor over the **Branch 1** name then click the edit icon (A)



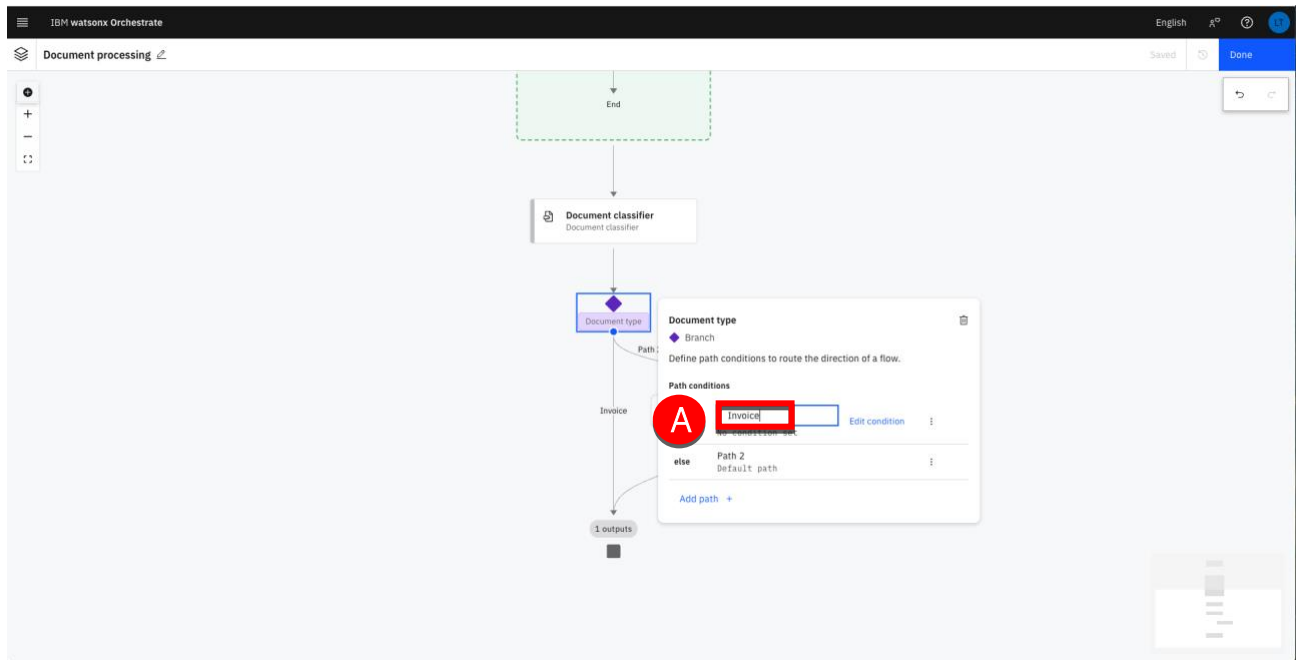
5. Type 'Document type' in the node name field (A) then hit Enter.



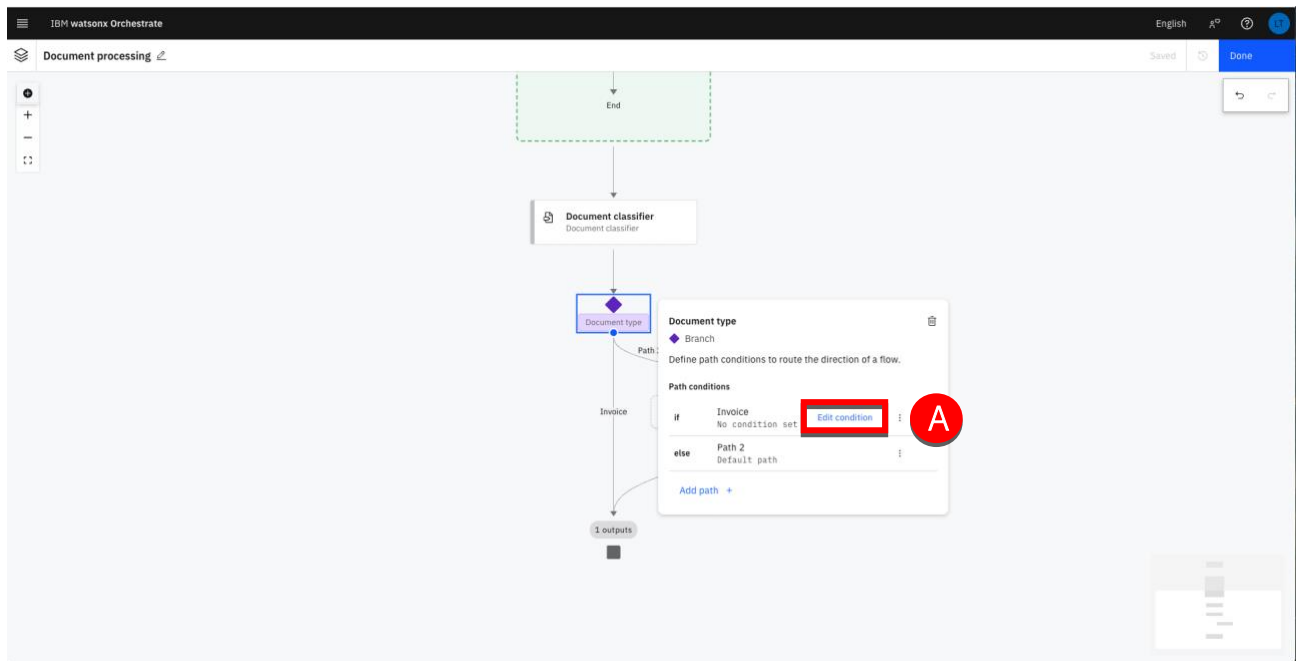
6. Click over **Path 1** (A).



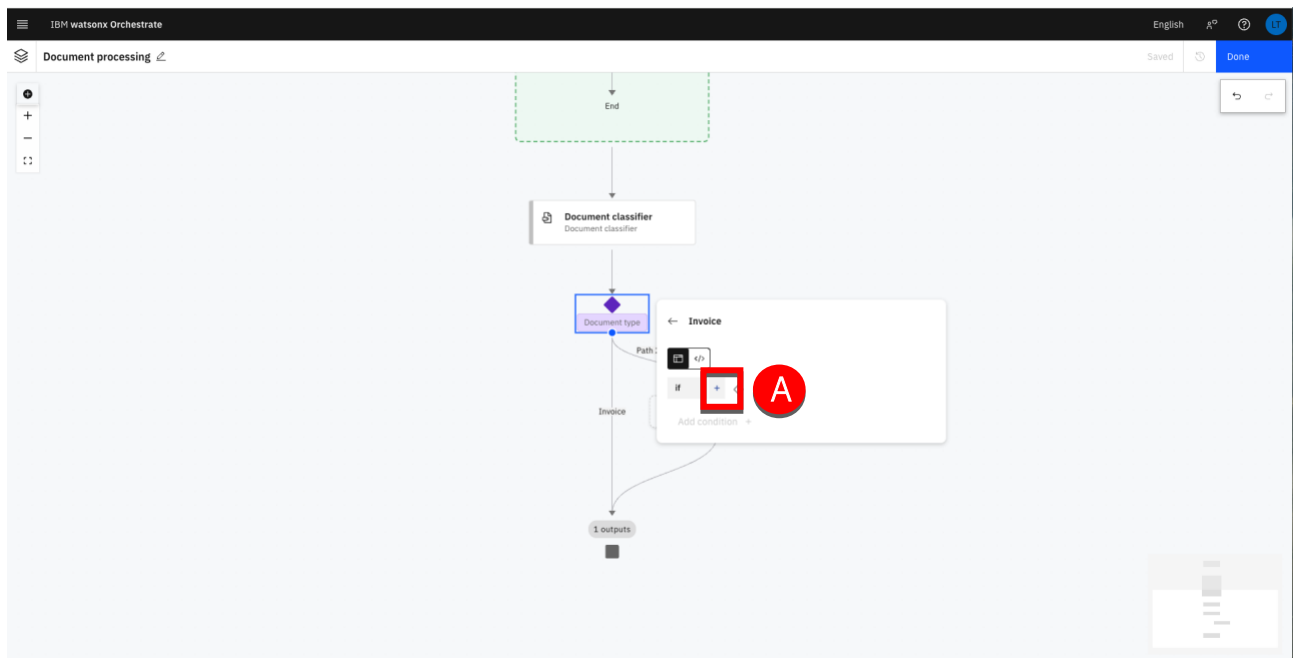
7. Type 'Invoice' (A).



8. Click **Edit condition** (A) to edit the condition and select invoice documents.

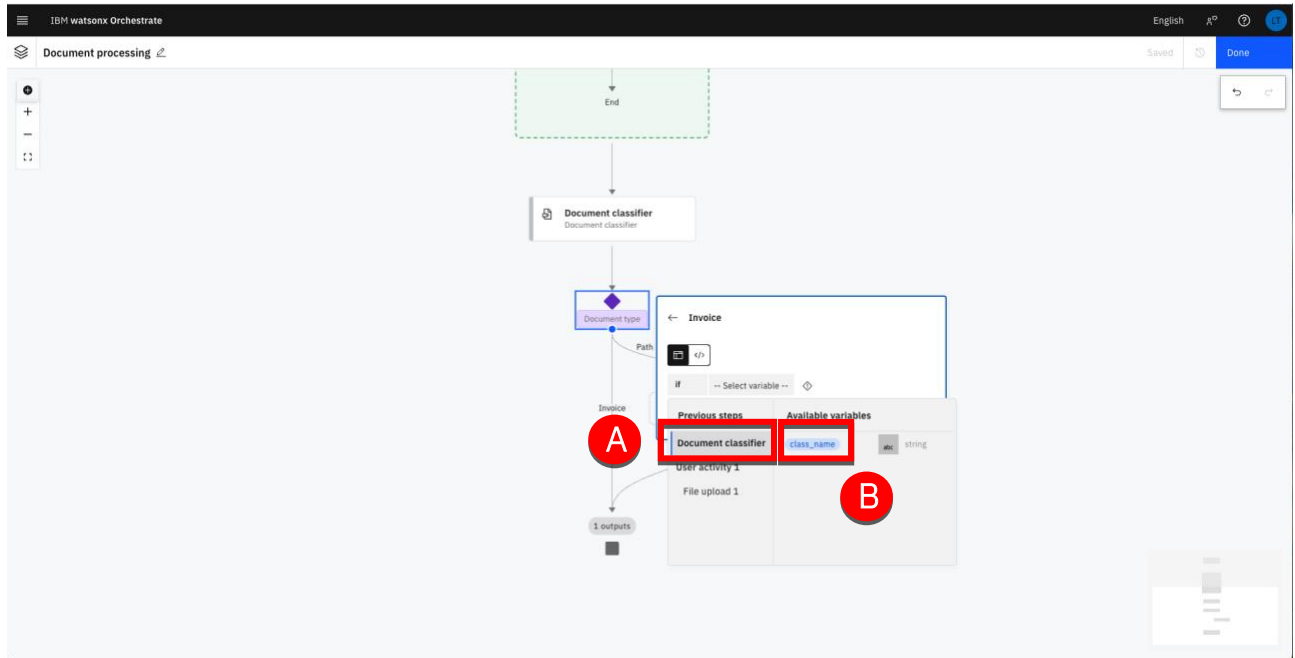


9. Click the + icon (A) to add conditions.

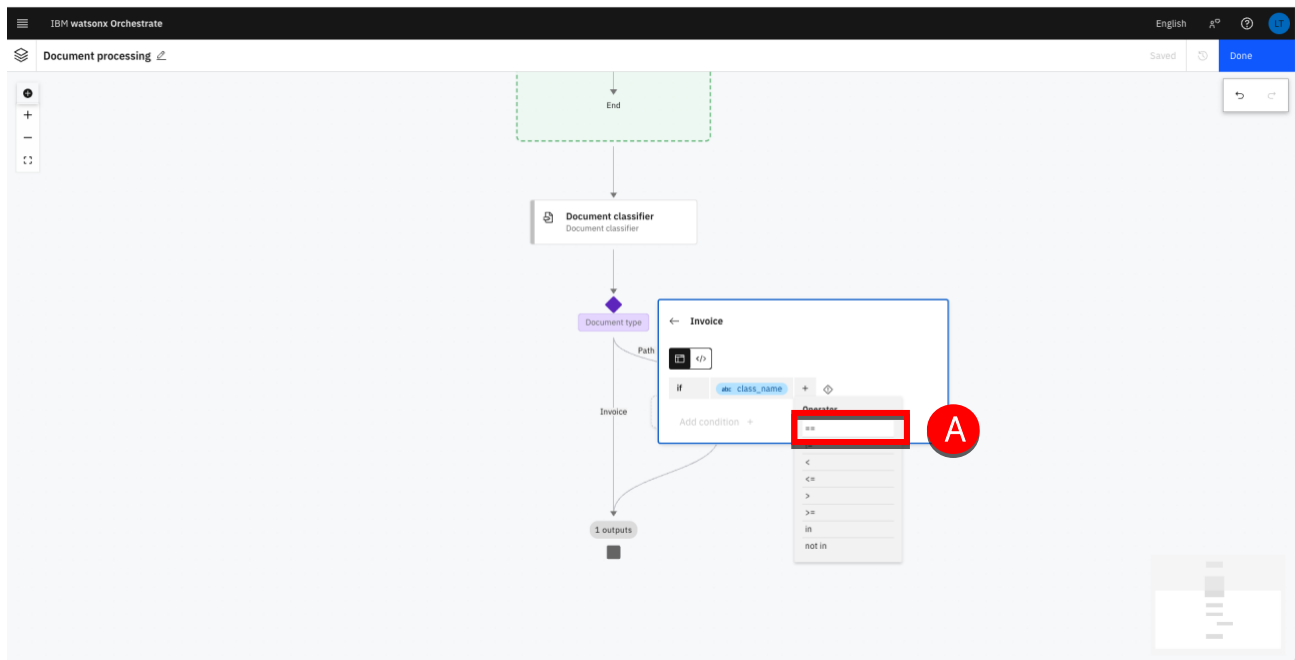


You will use the result of the Document classifier step as the variable to evaluate for the routing.

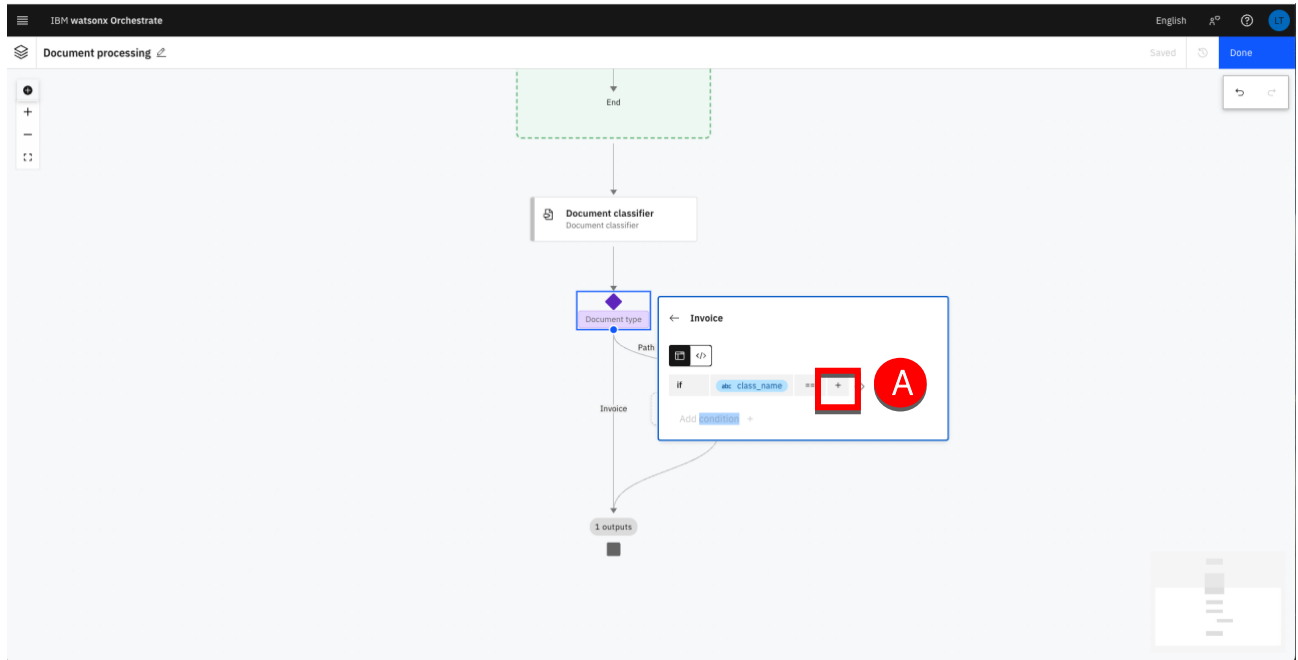
10. Click **Document classifier** (A) and then click **class_name** (B).



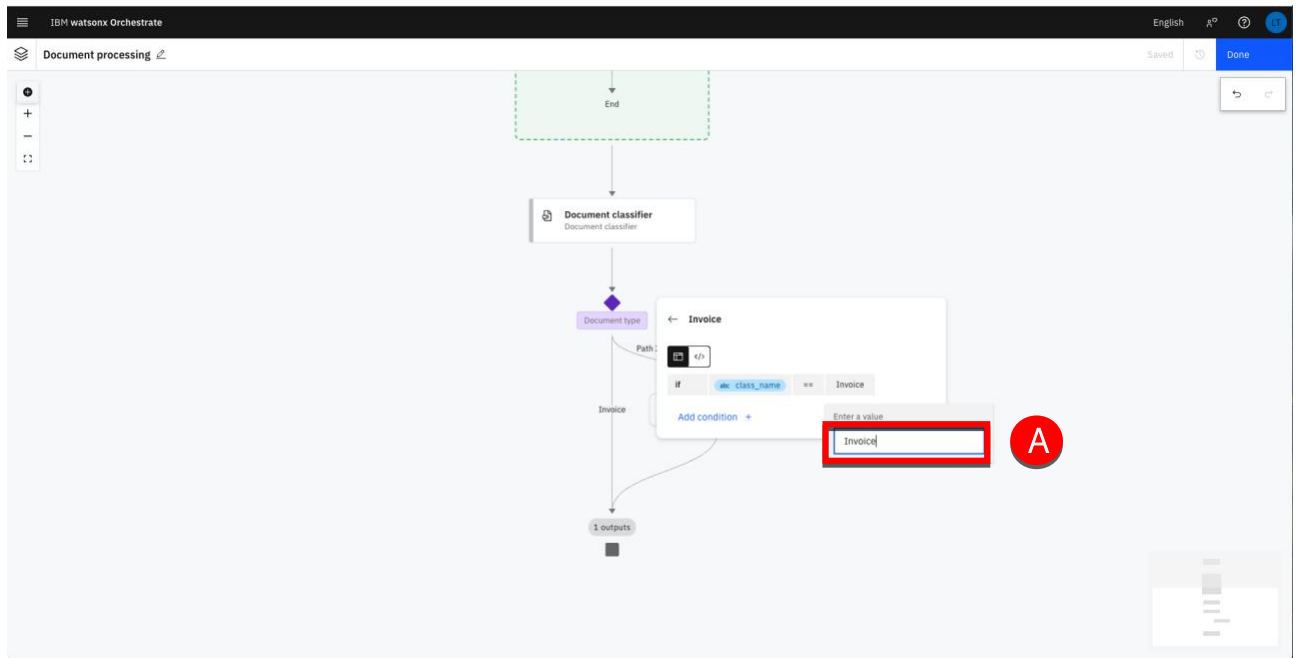
11. Select the '=' operator (A).



12. Click + to specify the value to check (A).



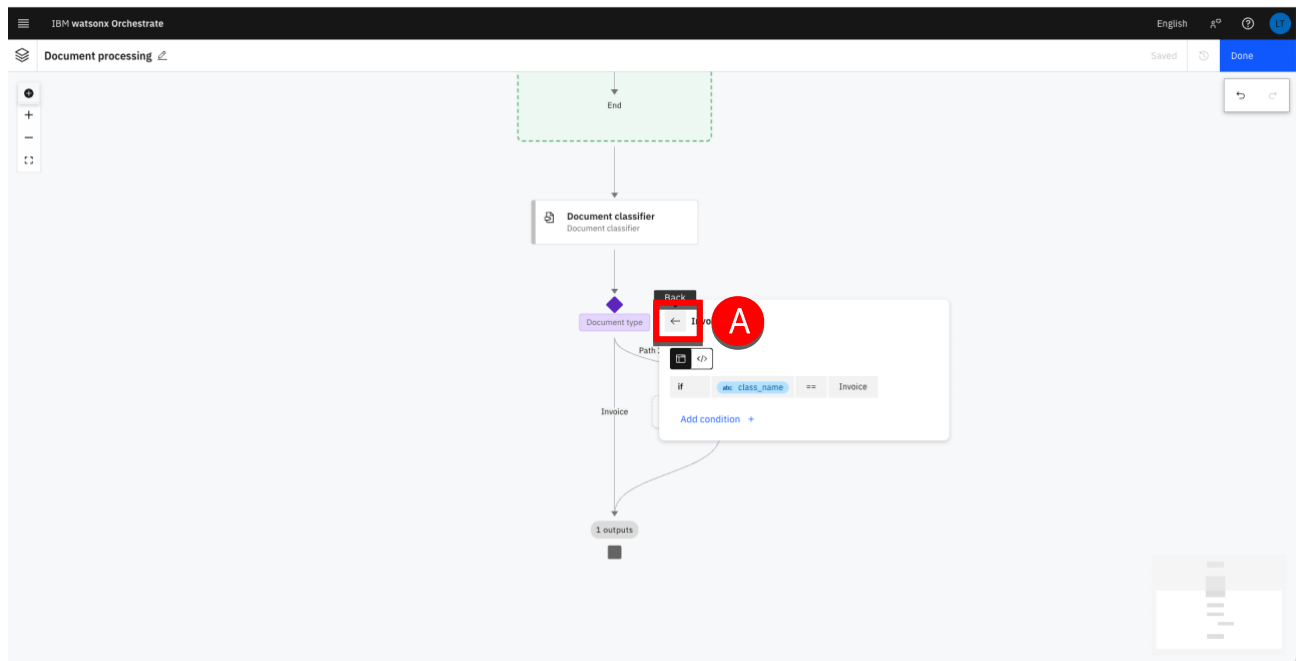
13. Type 'Invoice' (A) then hit Enter.



The process will be routed in this branch if the document type is Invoice.

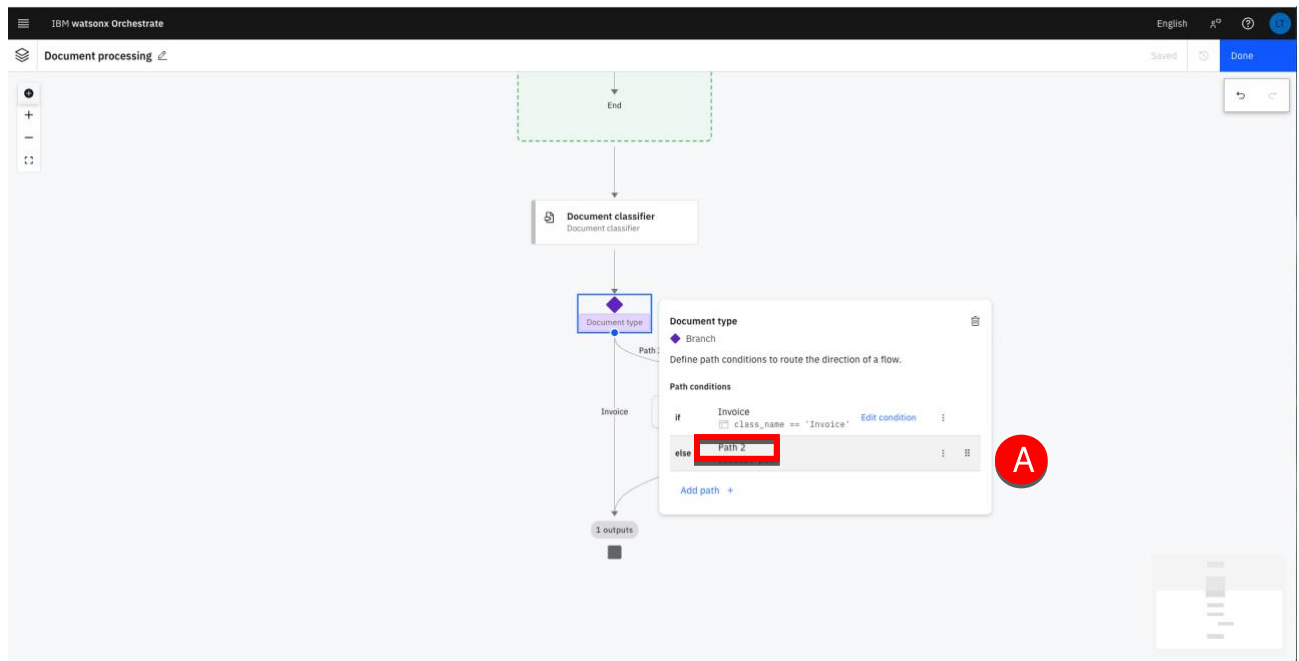
Note: Conditions can also be edited with the expression editor (</> icon). The equivalent expression to enter using the expression editor will then be: `flow["Document classifier"].output.class_name == "Contract"`

14. Click the **Back** icon (A).

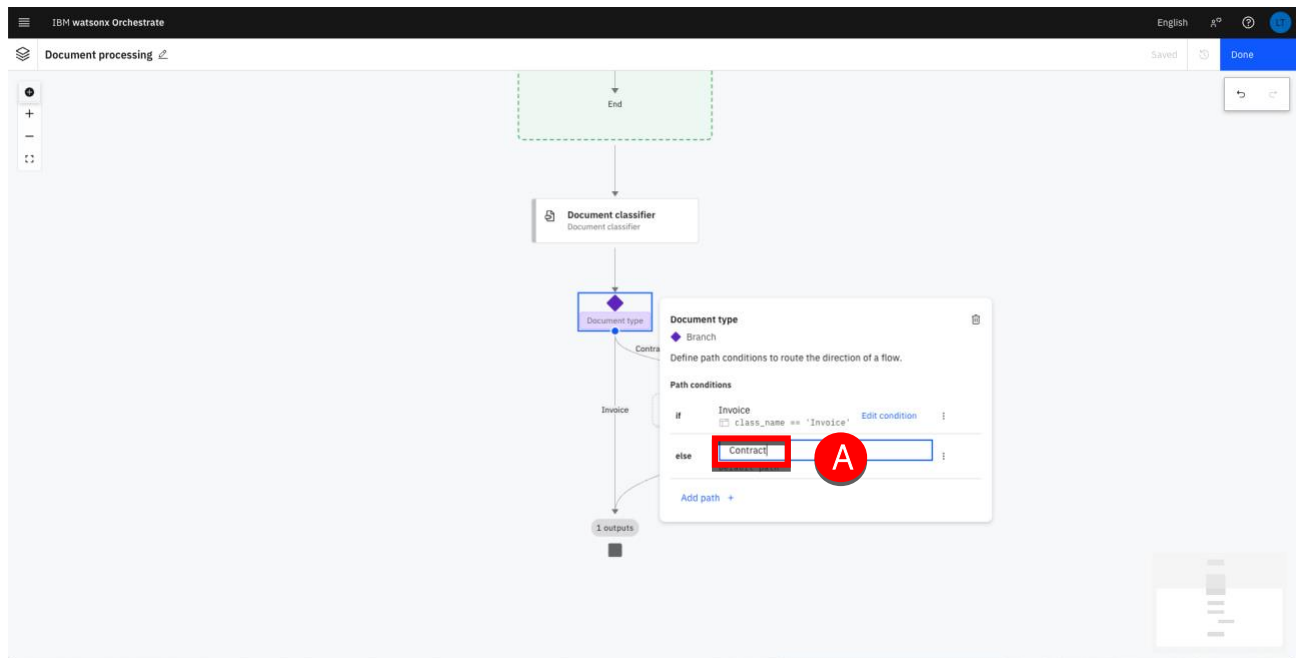


Let's just rename the second path Contract. As you will just have 2 types of documents, any document that is not an invoice (i.e. contracts) will be routed to this branch.

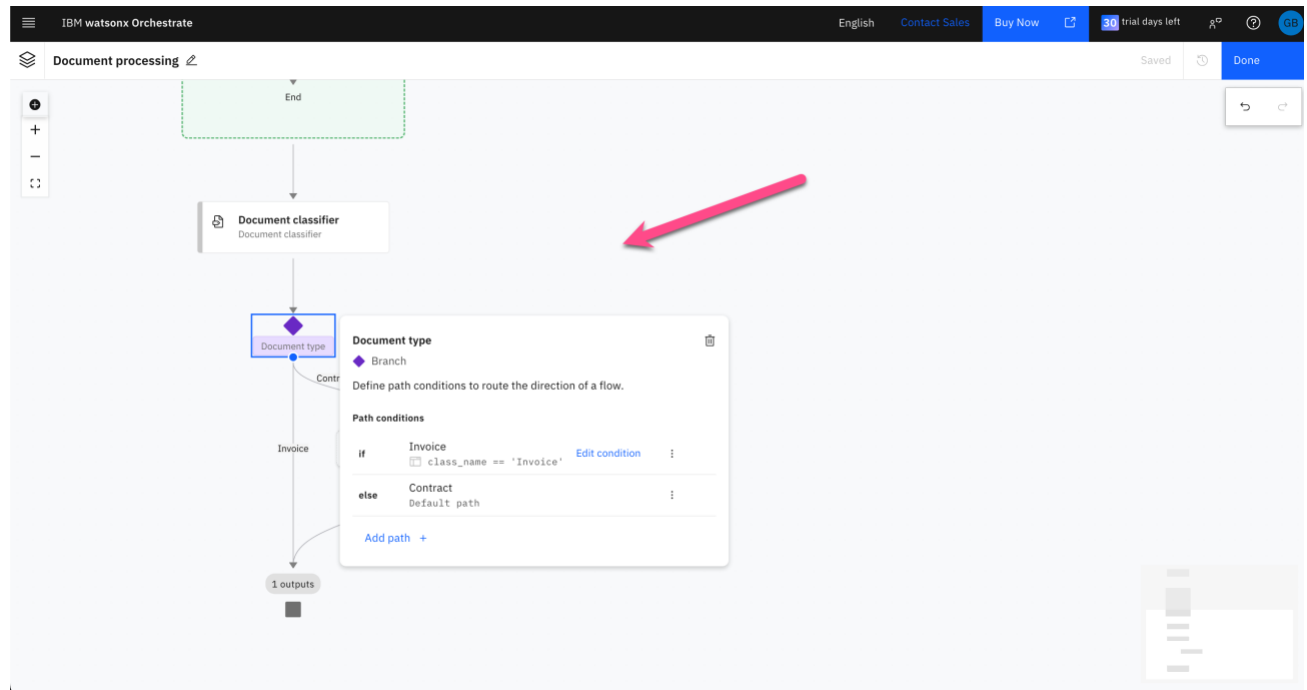
15. Click over **Path 2** (A).



16. Type 'Contract' (A) then hit Enter.



17. Click anywhere on the canvas to close the **Document type** branch panel.



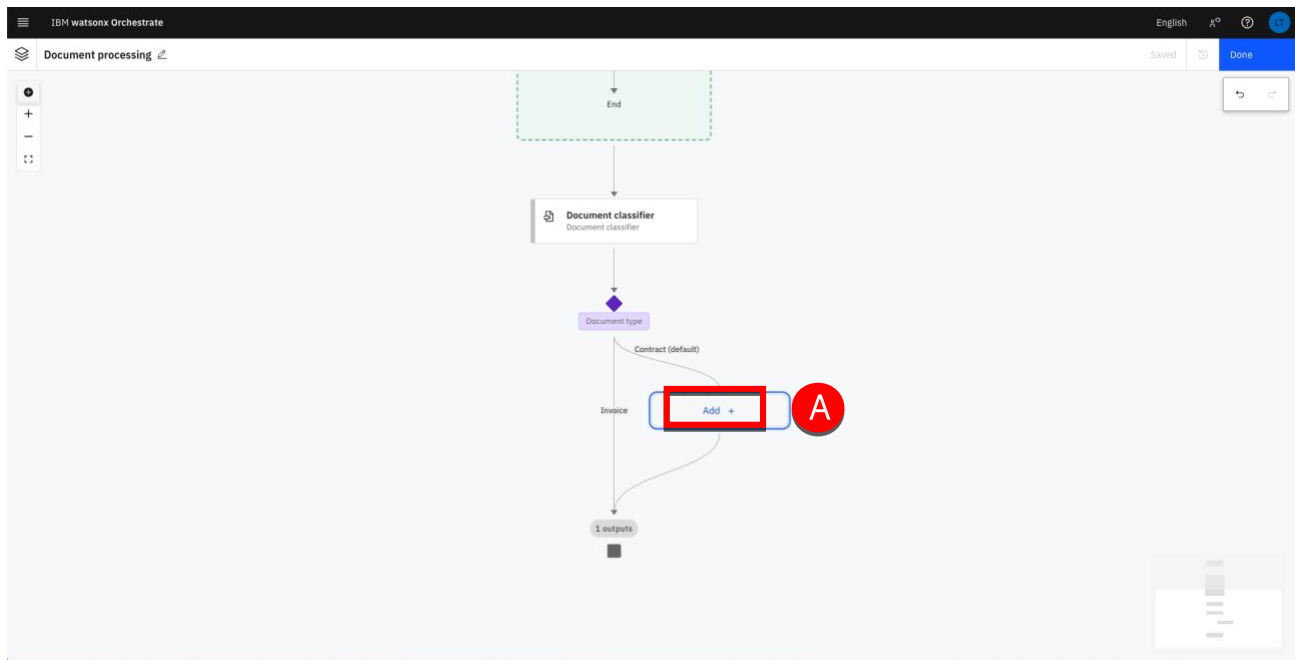
You are now ready to add the 2 document extractors, one for each document type.

Add a document extractor

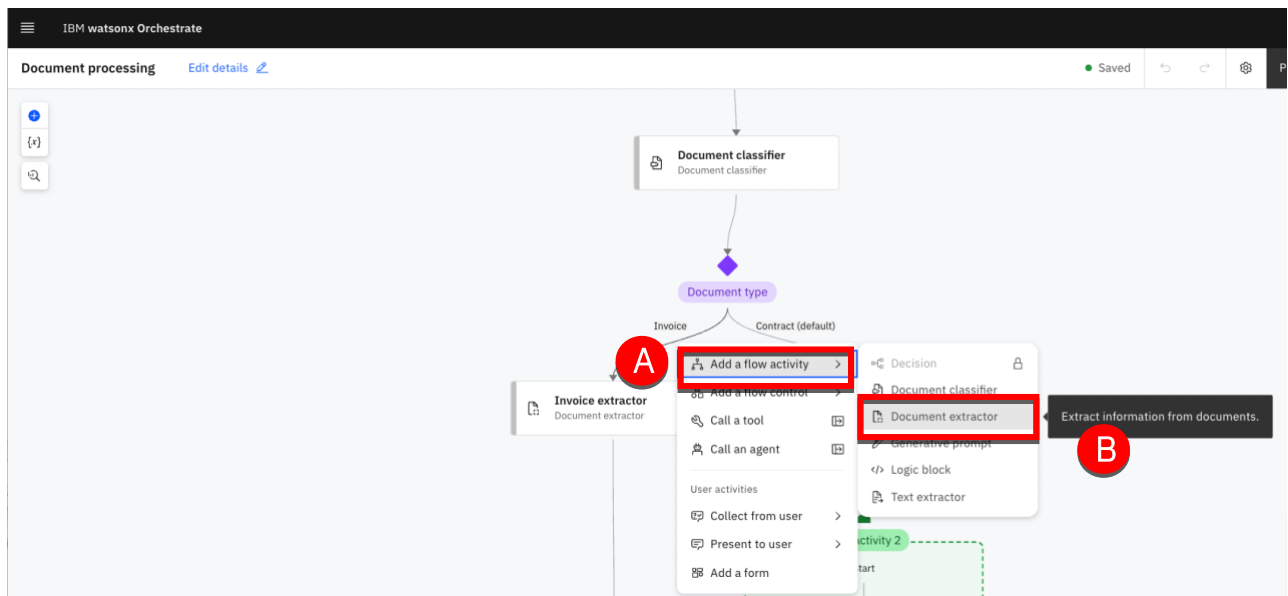
In this section, you will create two document extractors. Each document extractor will be responsible for extracting specific data from each document class. You will start with the Contract type of documents for which you will extract the following fields:

- Buyer
- Supplier
- Effective date

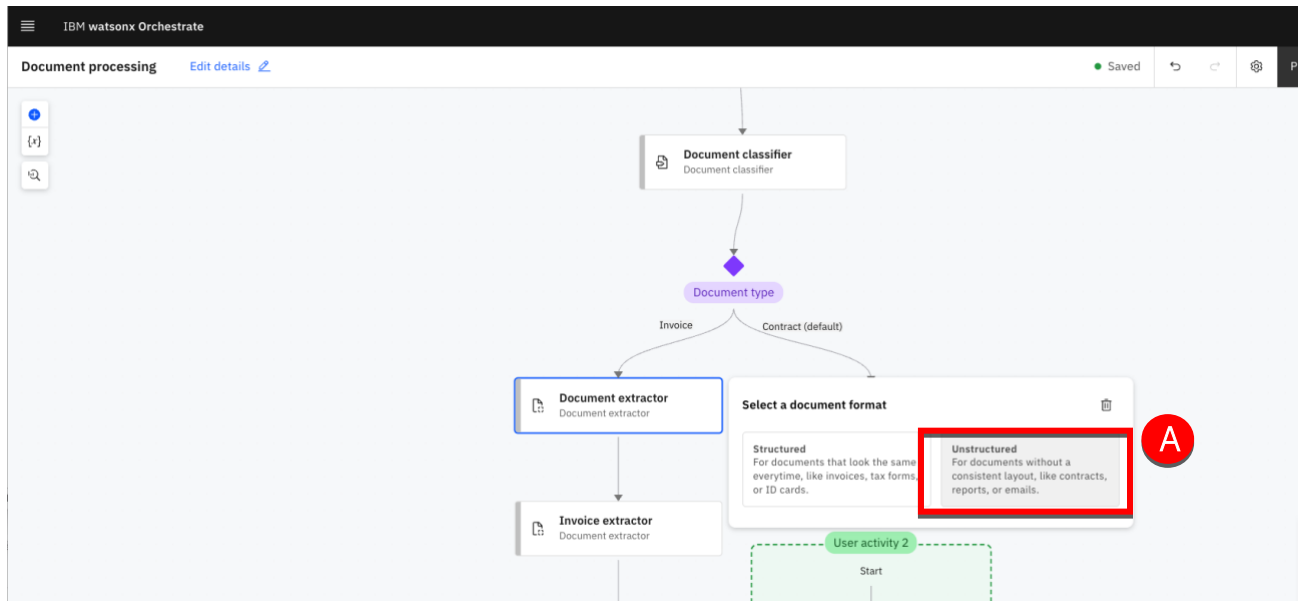
1. Click **Add +** (A).



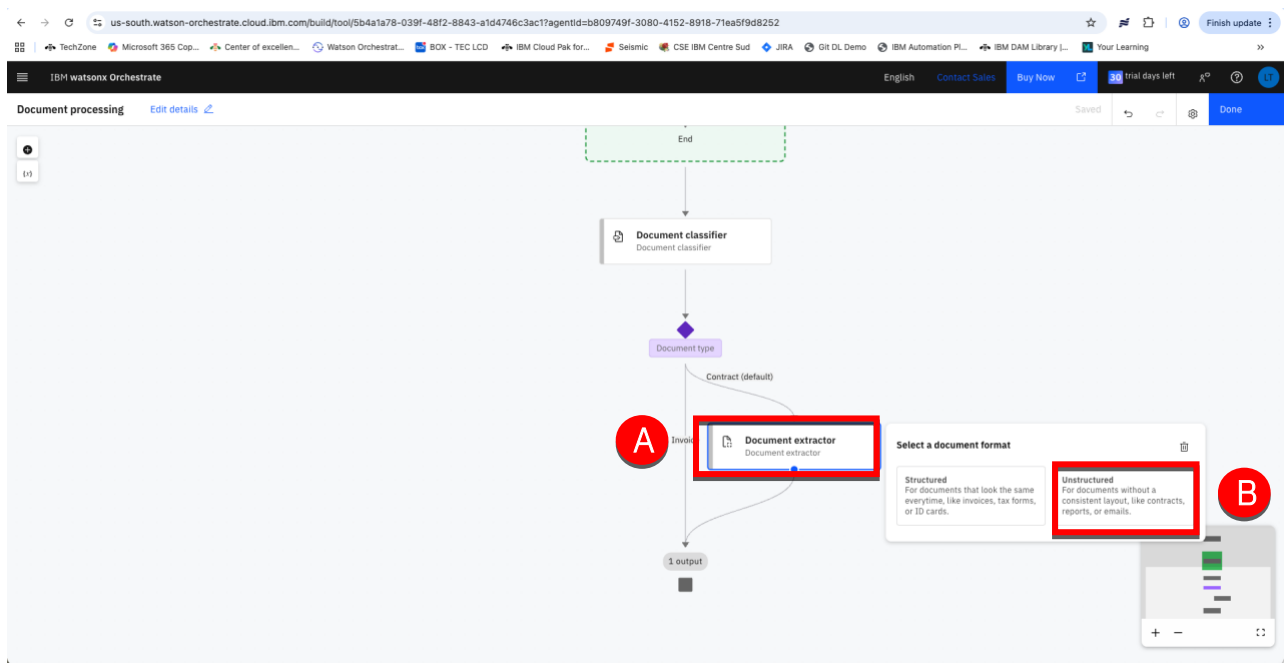
2. Select **Add a flow activity** (A) then click **Document extractor** (B).



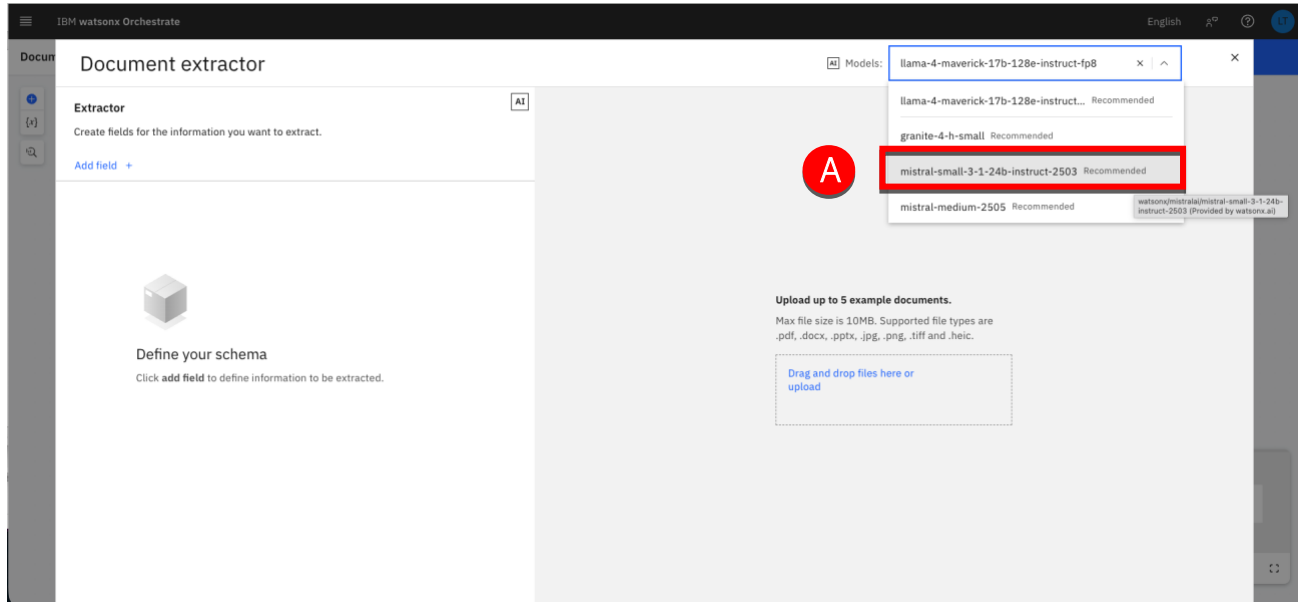
3. Click **Unstructured** (A)



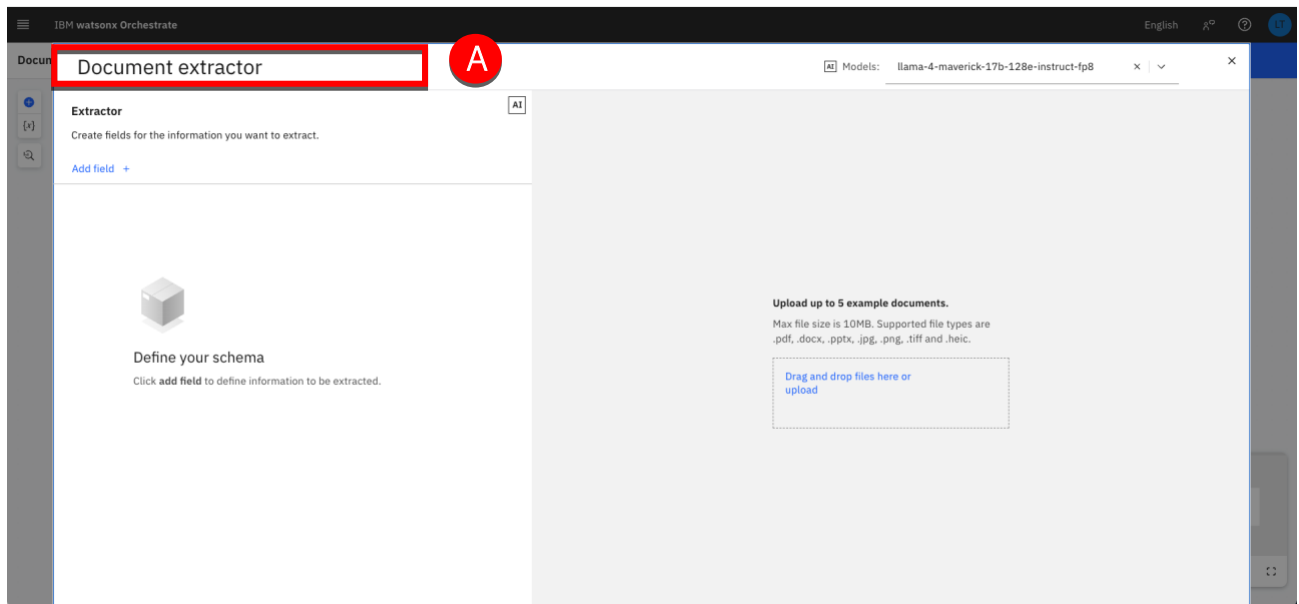
4. Click the **Document extractor** node (A) then click **Unstructured** (B).



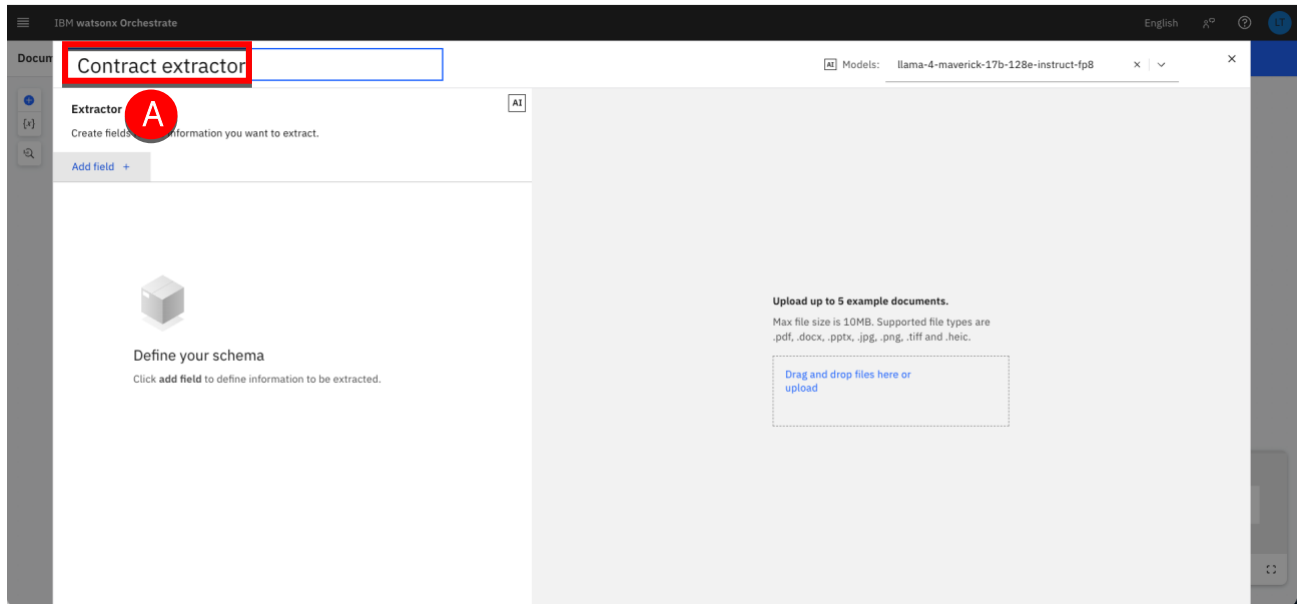
5. Select the llama-4-maverick-17b-128e-instruct-fp8 model (A).



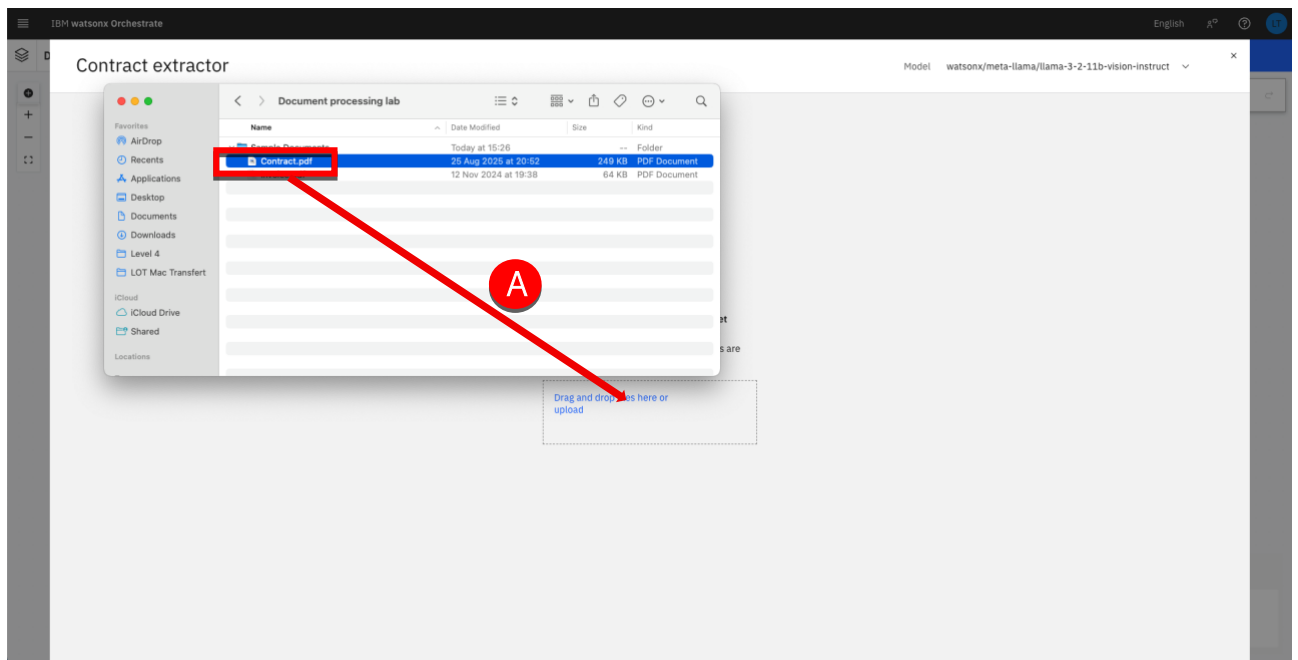
6. Click the Document extractor name (A) to edit it.



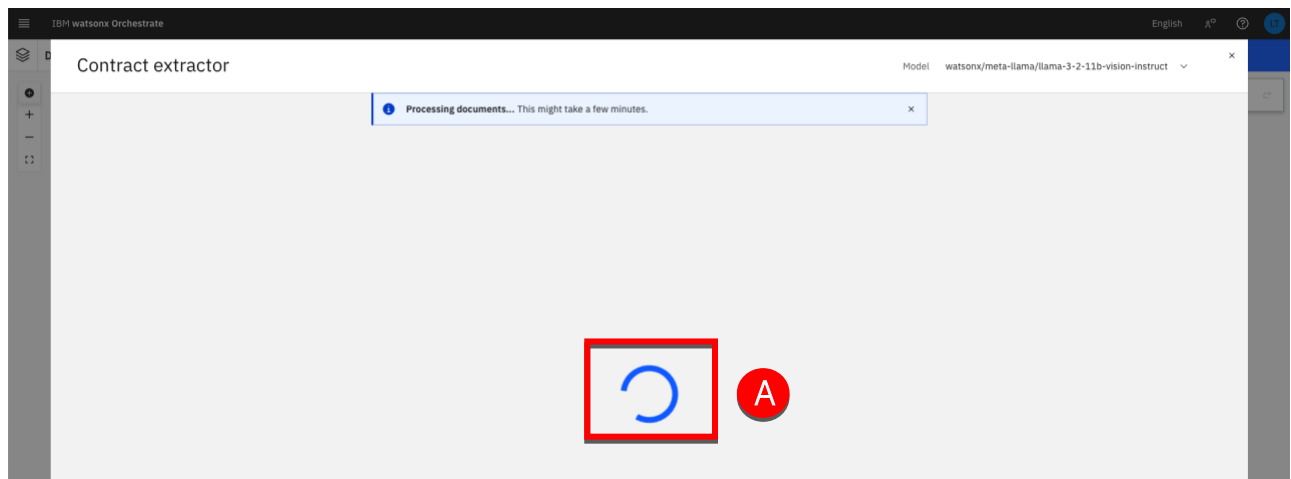
7. Type 'Contract extractor' (A) as a new name then click the save icon (B).



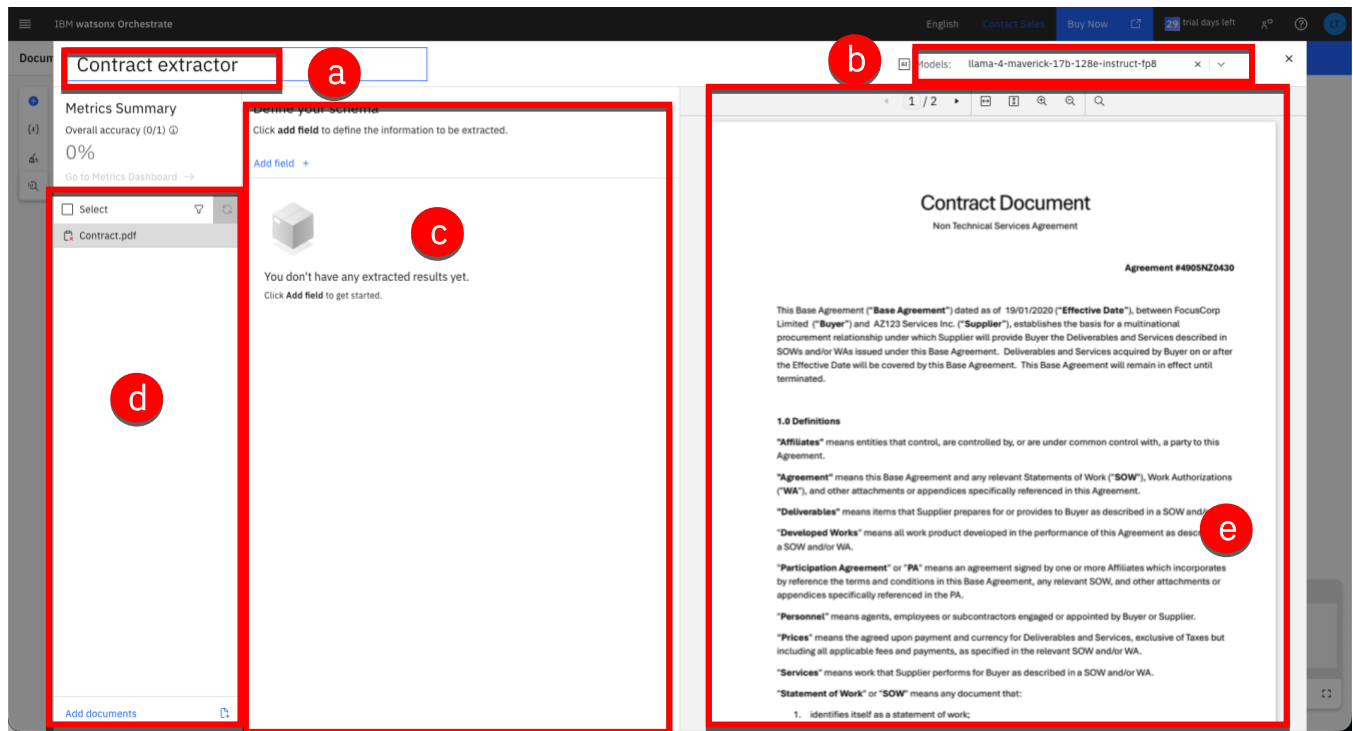
8. Drag and drop the **Contract.pdf** file (A) from your computer to the dropping area.



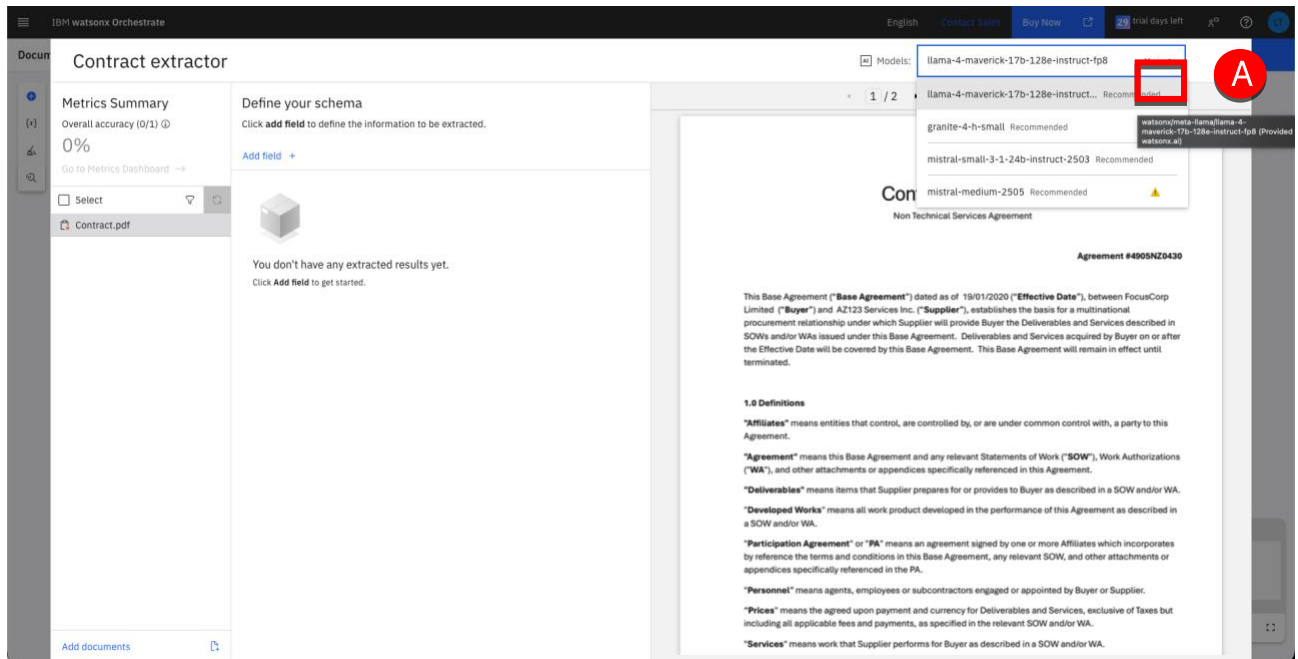
9. Wait for the document to be processed.



10. Once uploaded, the sample document is displayed, and the page is divided into the following key segments:
- a. Activity Name – Displays the name of the document extractor activity. Rename it to reflect the sample document you've uploaded.
 - b. LLM Model Selection – Choose your preferred LLM model from the drop-down menu. You can switch models at any time while prompting them for document extraction.
 - c. Field Definitions – Define the list of fields you want to extract from the sample document. The selected LLM will retrieve the corresponding values.
 - d. Document Navigation & Upload – Use the drop-down to navigate between uploaded documents. You can upload up to five additional documents for prompting and testing.
 - e. Document Viewer: Shows the uploaded document with the specified fields for extraction highlighted.



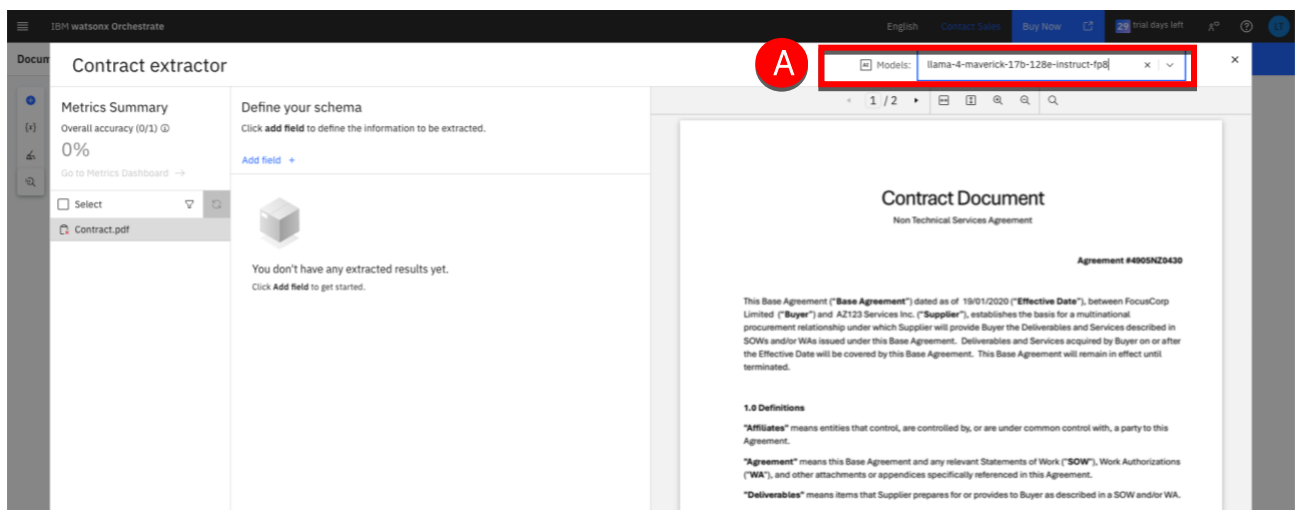
11. Click the model expand icon (A)



Note: You can change the LLM model by clicking on the 'Model' drop-down. Here you'll see all the available models that you can select. You can identify the current model that is being used, as you'll see a tick icon.

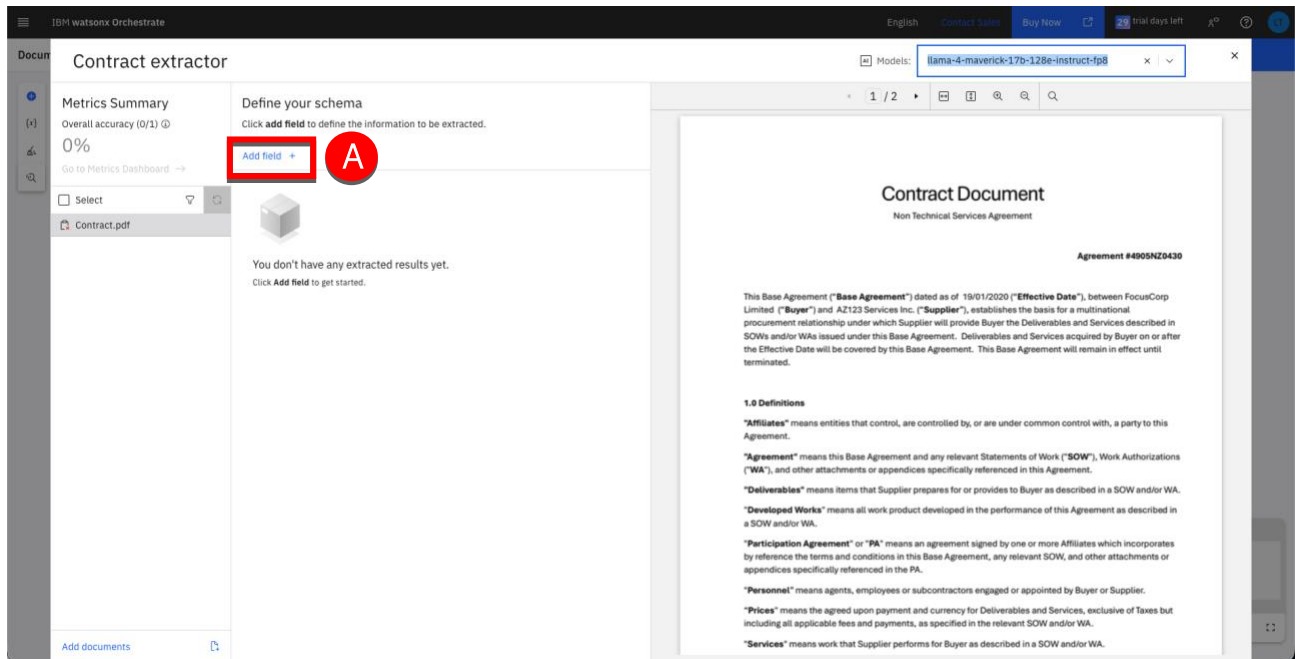
You can change the model at any time while you test to see which one is most accurate at extracting fields

12. Keep the llama-4-maverick-17b-128e-instruct-fp8 model (A).

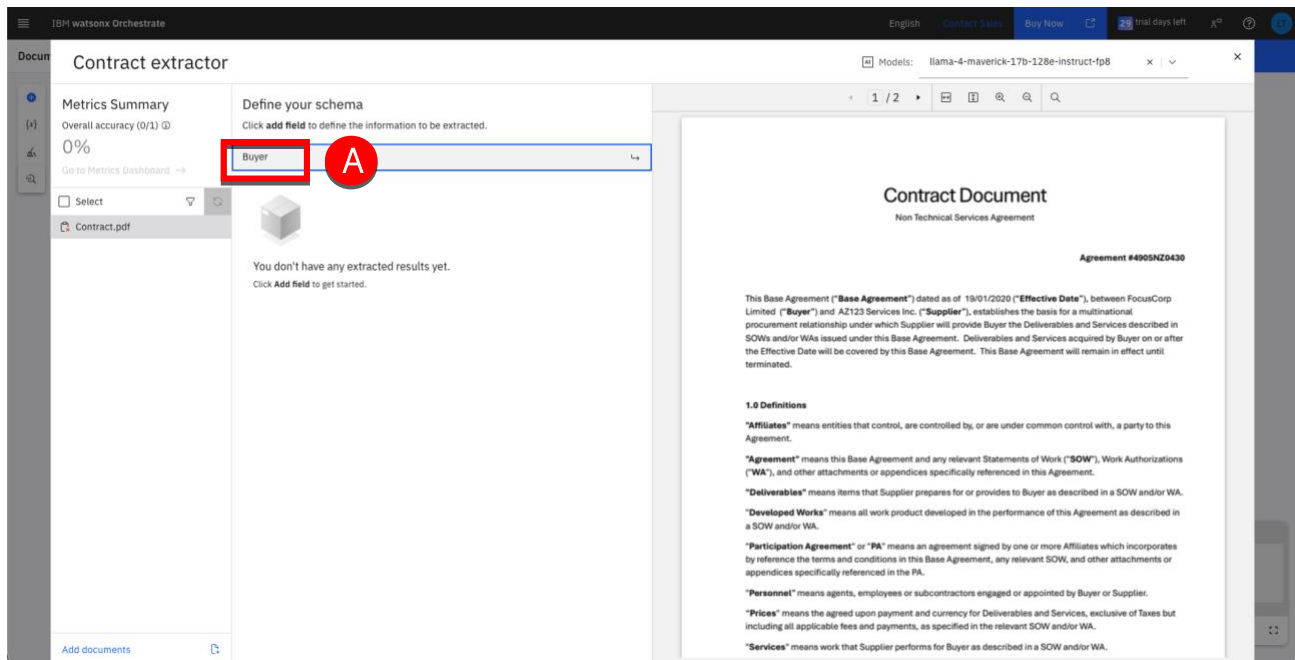


Let's now define the fields to extract.

13. Click Add field (A).



14. Type Buyer (A) and hit Enter.



Note: The LLM will retrieve a value associated with the key you define here (A) , which will auto-highlight in the document preview (B).

Contract extractor

Contract.pdf Unverified

Review each extracted field and enter their correct value if needed.

Add field +

Field Name	Extracted Value	Correct Value
Buyer	FocusCorp Limited	FocusCorp Limited

Contract Document

Non Technical Services Agreement

Agreement #4905NZ0430

This Base Agreement ("Base Agreement") dated as of 19/01/2020 ("Effective Date"), between FocusCorp Limited ("Buyer") and AZ123 Services Inc. ("Supplier"), establishes the basis for a multinational procurement relationship under which Supplier will provide Buyer the Deliverables and Services described in SOWs and/or WAs issued under this Base Agreement. Deliverables and Services acquired by Buyer on or after the Effective Date will be covered by this Base Agreement. This Base Agreement will remain in effect until terminated.

1.0 Definitions

"Affiliates" means entities that control, are controlled by, or are under common control with, a party to this Agreement.

"Agreement" means this Base Agreement and any relevant Statements of Work ("SOW"), Work Authorizations ("WA"), and other attachments or appendices specifically referenced in this Agreement.

"Deliverables" means items that Supplier prepares for or provides to Buyer as described in a SOW and/or WA.

"Developed Works" means all work product developed in the performance of this Agreement as described in a SOW and/or WA.

"Participation Agreement" or "PA" means an agreement signed by one or more Affiliates which incorporates by reference the terms and conditions in this Base Agreement, any relevant SOW, and other attachments or appendices specifically referenced in the PA.

"Personnel" means agents, employees or subcontractors engaged or appointed by Buyer or Supplier.

"Prices" means the agreed upon payment and currency for Deliverables and Services, exclusive of Taxes but including all applicable fees and payments, as specified in the relevant SOW and/or WA.

"Services" means work that Supplier performs for Buyer as described in a SOW and/or WA.

15. In the same way, add the 2 following fields:

- Supplier
- Effective date

Your screen should look like:

Contract extractor

Contract.pdf Unverified

Review each extracted field and enter their correct value if needed.

Add field +

Field Name	Extracted Value	Correct Value
Effective date	19/01/2020	19/01/2020
Supplier	AZ123 Services Inc.	AZ123 Services Inc.
Buyer	FocusCorp Limited	FocusCorp Limited

Contract Document

Non Technical Services Agreement

Agreement #4905NZ0430

This Base Agreement ("Base Agreement") dated as of 19/01/2020 ("Effective Date"), between FocusCorp Limited ("Buyer") and AZ123 Services Inc. ("Supplier"), establishes the basis for a multinational procurement relationship under which Supplier will provide Buyer the Deliverables and Services described in SOWs and/or WAs issued under this Base Agreement. Deliverables and Services acquired by Buyer on or after the Effective Date will be covered by this Base Agreement. This Base Agreement will remain in effect until terminated.

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"Affiliates" means entities that control, are controlled by, or are under common control with, a party to this Agreement.

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"Deliverables" means items that Supplier prepares for or provides to Buyer as described in a SOW and/or WA.

"Developed Works" means all work product developed in the performance of this Agreement as described in a SOW and/or WA.

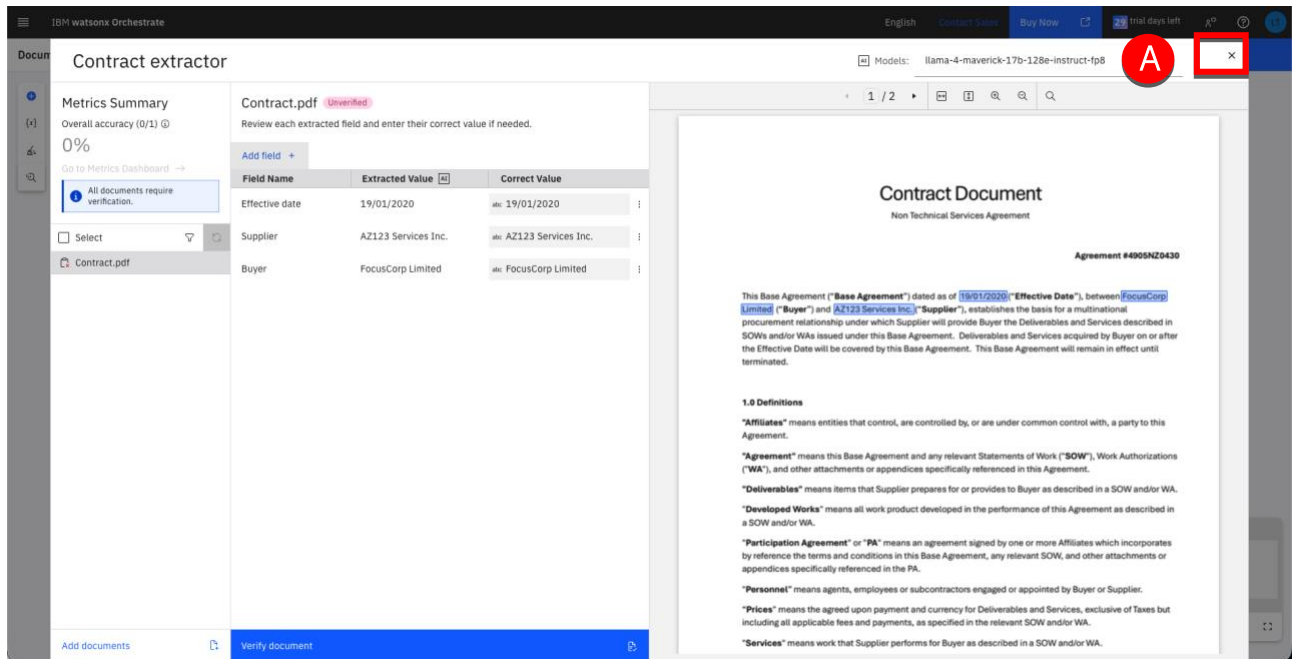
"Participation Agreement" or "PA" means an agreement signed by one or more Affiliates which incorporates by reference the terms and conditions in this Base Agreement, any relevant SOW, and other attachments or appendices specifically referenced in the PA.

"Personnel" means agents, employees or subcontractors engaged or appointed by Buyer or Supplier.

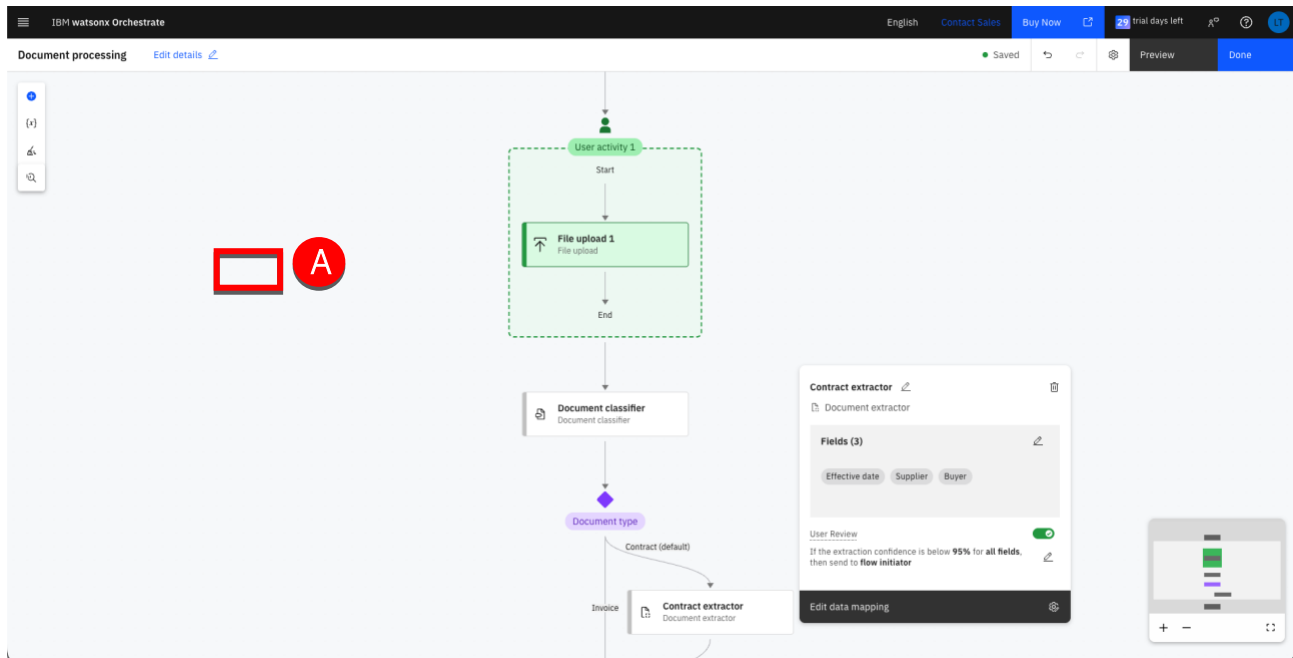
"Prices" means the agreed upon payment and currency for Deliverables and Services, exclusive of Taxes but including all applicable fees and payments, as specified in the relevant SOW and/or WA.

"Services" means work that Supplier performs for Buyer as described in a SOW and/or WA.

16. Click x (A) to close the extractor.



17. Click on the diagram background to close the **Contract extractor** property view (A).

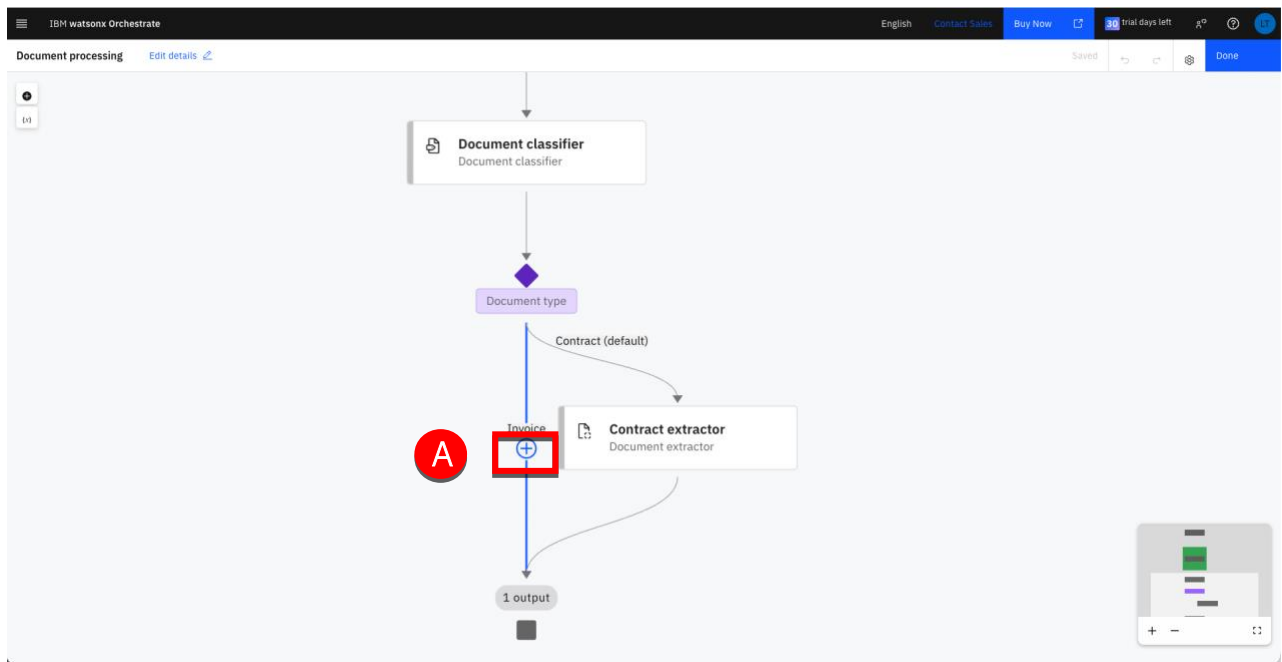


Next, you will create the Invoice extractor in the corresponding branch of the process. This extractor will be more complex and harder to implement. You will learn how to train the model to extract values from different kind of documents (PDF files or images).

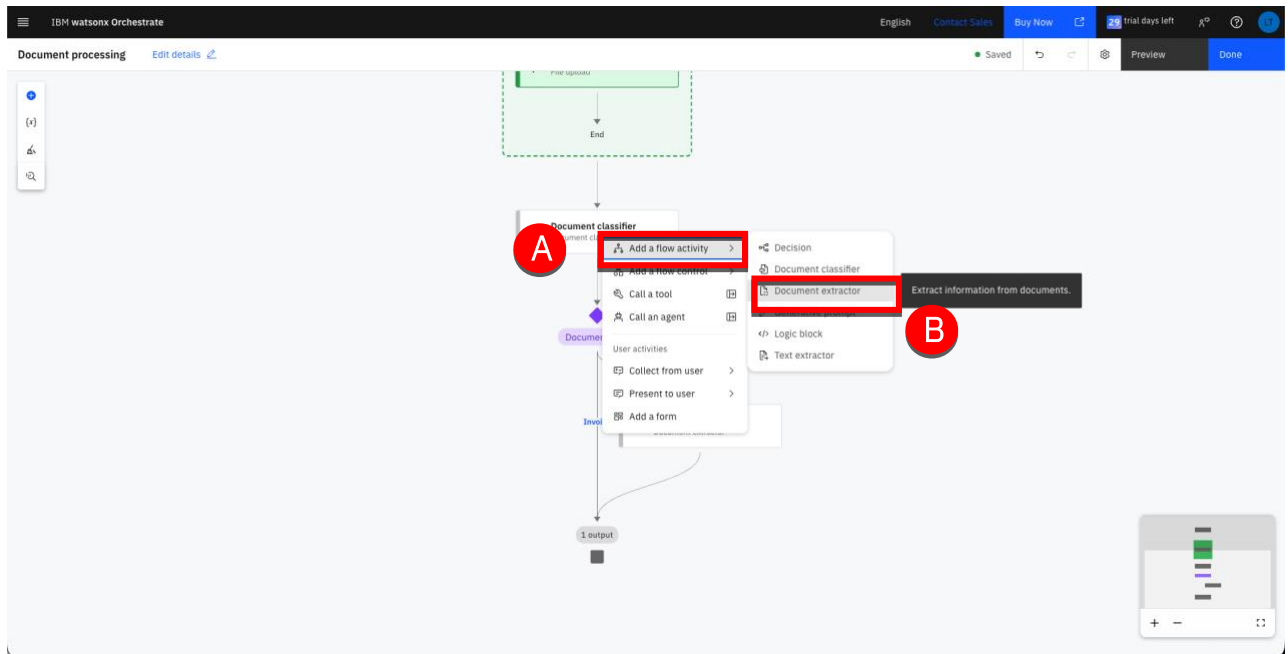
For the invoice documents, you will extract the following fields:

- Customer name
- Customer number
- Customer address
- Amount due
- Due date
- Monthly usage
- Year to year usage

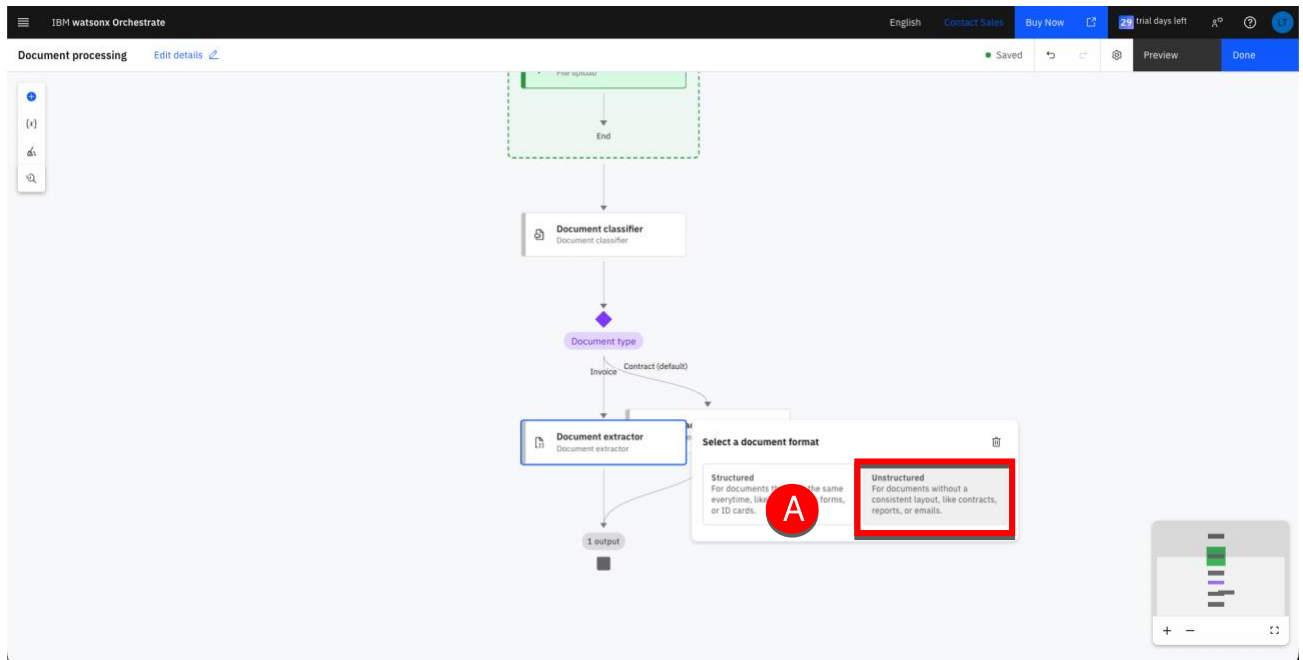
18. Move the mouse over the **Invoice** branch and click + (A)



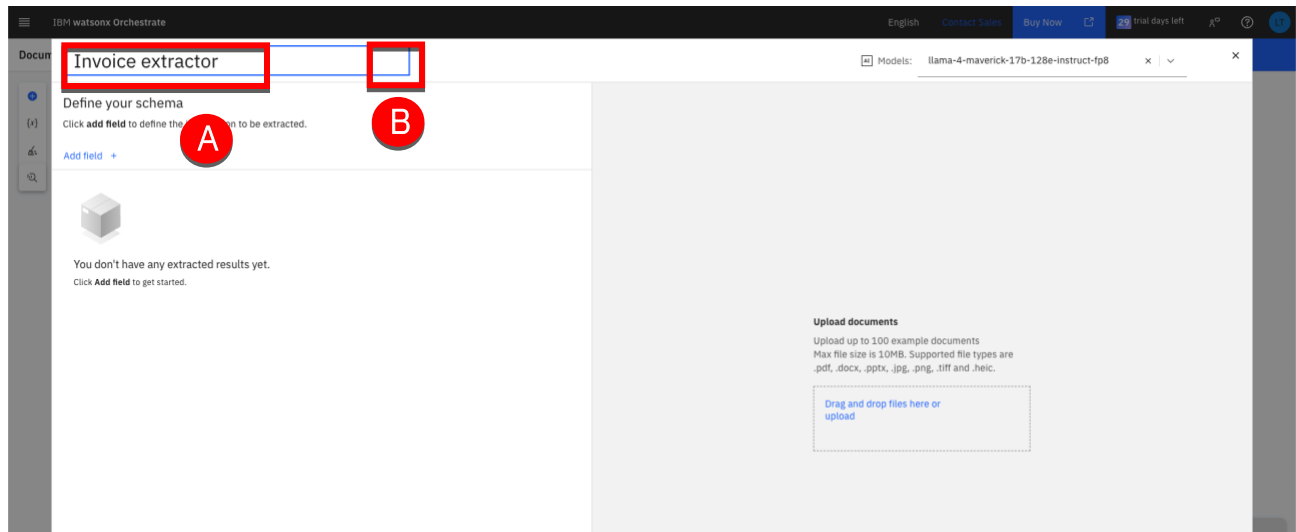
19. Select **Add a flow activity (A)** then **Document extractor (A)**.



20. Select **Unstructured (A)**.

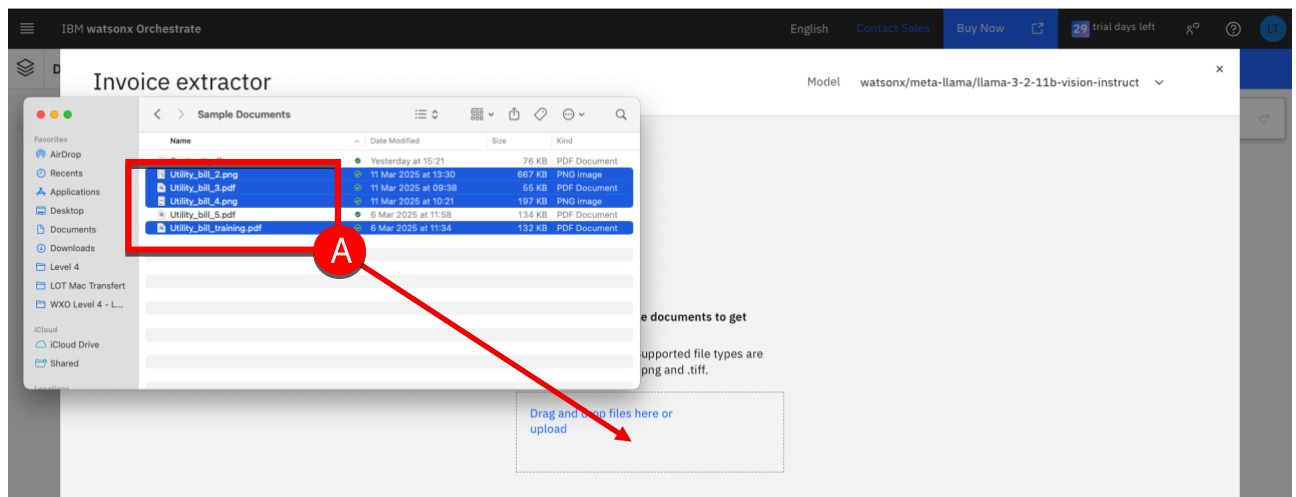


21. Type 'Invoice extractor' in the name field (A). Click the save icon (B).

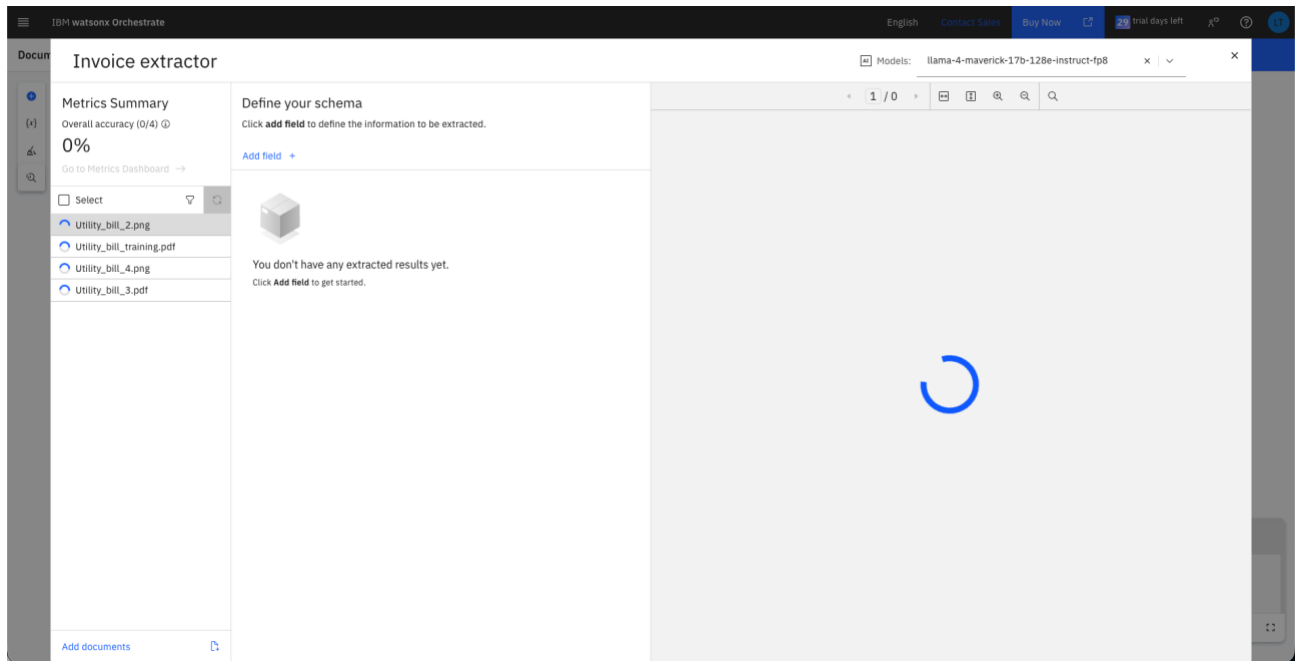


22. Drag and drop the four following files in the upload area (A)

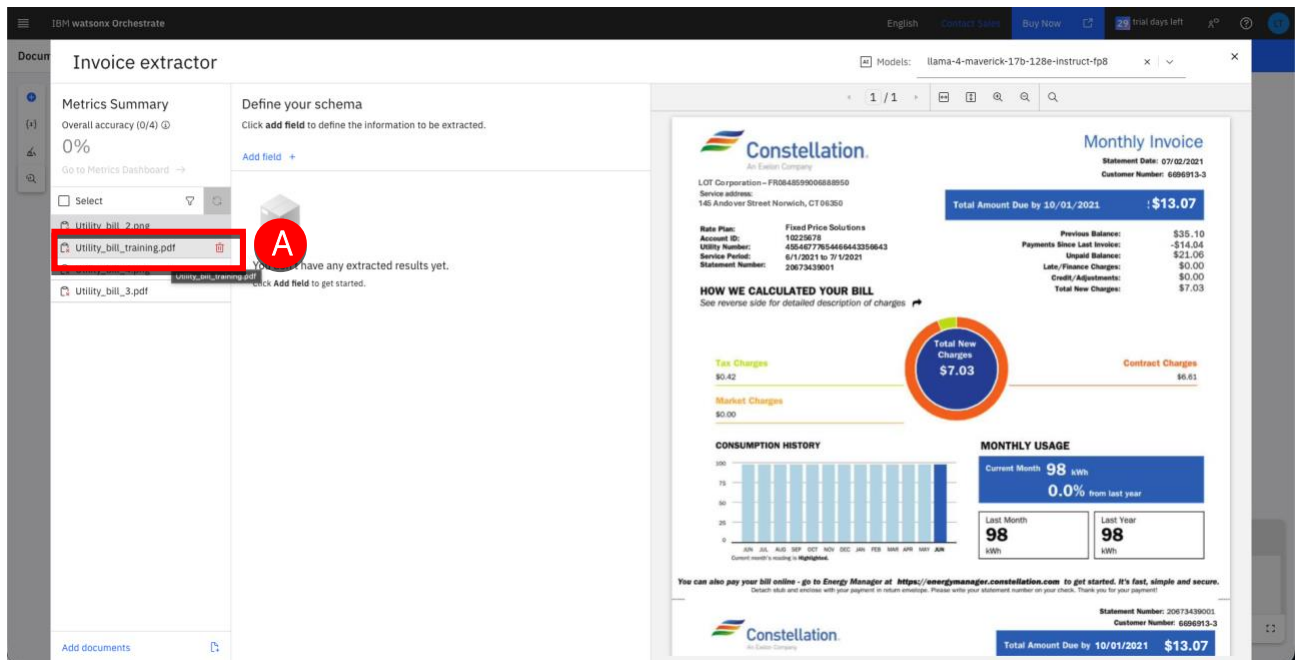
- Utility_bill_training.pdf
- Utility_bill_2.png
- Utility_bill_3.pdf
- Utility_bill_4.png



23. Wait for the 4 files to be processed.



24. Click the `Utility_bill_training.pdf` file (A).



Define the fields to extract

You are now ready to specify the fields to be extracted.

1. Click **Add field** (A).

The screenshot shows the IBM watsonx Orchestrate interface. On the left, the 'Invoice extractor' document is selected. The 'Define your schema' section has a red box around the 'Add field' button, labeled with a red circle 'A'. The document preview on the right shows a 'Monthly Invoice' from Constellation with various financial details and usage statistics.

2. Type 'Customer name' into the field name (A) then press Enter.

The screenshot shows the IBM watsonx Orchestrate interface. On the left, the 'Invoice extractor' document is selected. The 'Define your schema' section has a red box around the 'Customer name' text input, labeled with a red circle 'A'. The document preview on the right shows a 'Monthly Invoice' from Constellation with various financial details and usage statistics.

Note: After a few seconds, the Customer name should appear with 'ABC Corporation'.

- Click **Add field** (A).

The screenshot shows the IBM Watsonx Orchestrate interface for the 'Invoice extractor' agent. On the left, the 'Metrics Summary' shows 0% overall accuracy. Below it, a list of documents includes 'Utility_bill_training.pdf'. In the center, a table for extracted fields is visible, with the 'Add field' button highlighted by a red box and labeled 'A'. The right pane displays a sample 'Monthly Invoice' from Constellation, showing details like 'Total Amount Due by 10/01/2021: \$28.09' and 'Monthly Usage: 98 kWh'.

Note: The extractor may not recognize the field directly. We will see in the next step how to train the extractor.

- Enter 'Statement date' into the field name (A) then press Enter.

This screenshot shows the same interface as the previous one, but with the text 'Statement date' entered into the field name input, which is highlighted by a red box and labeled 'A'. The rest of the interface, including the document list and the sample invoice, remains the same.

5. Review the extractor results:

- The **Statement date** was recognized (A).
- The **Customer name** was recognized (B).

The screenshot shows the 'Invoice extractor' interface. On the left, a 'Metrics Summary' shows 'Overall accuracy (0/4)' at '0%'. Below it, a list of documents includes 'Utility_bill_training.pdf'. The main area displays the extraction results for 'Utility_bill_training.pdf' (Unverified). A table shows extracted fields:

Field Name	Extracted Value	Correct Value
Statement date	07/02/2021	07/02/2021
Customer	ABC Corporation	ABC Corporation

Red circles with letters 'A' and 'B' are placed over the 'Statement date' and 'Customer' rows respectively. Red arrows point from these circles to the corresponding fields in the original invoice image on the right.

The original invoice image on the right is a 'Monthly Invoice' from Constellation. It includes the following information:

- Service address: 145 Andover Street Norwich, CT 06360
- Customer Number: 6886913-3
- Total Amount Due by 10/01/2021: \$28.09
- Rate Plan: Fixed Price Solutions
- Account ID: 10229879
- Utility Number: 45545779546564305843
- Service Period: 6/1/2021 to 7/1/2021
- Statement Number: 20673438001
- How We Calculated Your Bill: See reverse side for detailed description of charges.
- Charges breakdown: Total New Charges \$7.03, Contract Charges \$6.61, Tax Charges \$0.42, Market Charges \$0.00.
- Consumption History: A bar chart showing usage from June 2021 to June 2022.
- Monthly Usage: Current Month 98 kWh, Last Month 98 kWh, Last Year 98 kWh.
- Payment summary: Previous Balance \$35.10, Payments Since Last Invoice -\$14.04, Unpaid Balance \$21.06, Late/Finesse Charges \$0.00, Credit Adjustments \$0.00, Total New Charges \$7.03.

Note: The recognition results you see may vary from these screenshots depending on the maturity of the LLM model and its ability to recognize utility bill fields as LLM models are evolving rapidly.

We will add additional fields described in the next section before training the model to extract the missing fields.

6. Repeat the steps above to add the following fields (A):

- Customer number
- Customer address
- Amount due
- Due date
- Monthly usage
- Year to year usage

The screenshot displays the IBM watsonx Orchestrate interface for an 'Invoice extractor' workflow. On the left, a 'Metrics Summary' sidebar shows 'Overall accuracy (0/4)' at 0%. Below this, a document list includes 'Utility_bill_training.pdf' (Unverified) and several other utility bills. The main area features a table of extracted fields, with a red box highlighting the following data:

Field Name	Extracted Value	Correct Value
Due date	10/01/2021	10/01/2021
Year to year usage	0.0%	0.0%
Monthly usage	98 kwh	98 kwh
Amount due	\$28.09	\$28.09
Customer address	P.O. Box 9005 Laurel, MD ...	P.O. Box 9005 Laurel, M...
Customer number	6696913-3	6696913-3
Statement date	07/02/2021	07/02/2021
Customer name	ABC Corporation	ABC Corporation

To the right, a sample 'Monthly Invoice' from Constellation is shown. A red circle with the letter 'A' highlights the 'Total New Charges' field, which is \$7.03. The invoice also displays 'Total Amount Due by 10/01/2021' as \$28.09 and 'Monthly Usage' as 98 kwh.

Note: In this example, the extractor was able to correctly recognize all the fields. This may not be true in your situation. We will see in the next chapter how to improve the model.

7. Once all the values are correctly extracted, click Verify document (A).

Invoice extractor

Metrics Summary

Overall accuracy (0/4) 0%

Go to Metrics Dashboard →

All documents require verification.

Select

Utility_bill_2.png

Utility_bill_training.pdf

Utility_bill_4.png

Utility_bill_3.pdf

Add documents

Utility_bill_training.pdf Unverified

Review each extracted field and enter their correct value if needed.

Add field +

Field Name	Extracted Value	Correct Value
Due date	No value	etc
Year to year usage	No value	etc 0.0%
Monthly usage	No value	etc 98 kwh
Amount due	No value	etc \$28.09
Customer address	No value	etc P.O. Box 9005 Laurel, M...
Customer number	No value	etc 6696913-3
Statement date	No value	etc 07/02/2021
Customer name	No value	etc ABC Corporation

Verify document

Constellation Monthly Invoice

Statement Date: 07/02/2021 Customer Number: 6696913-3

Total Amount Due by 10/01/2021: \$28.09

Rate Plan: Fixed Price Solutions

Account ID: 1022878

Utility Number: 45545770546664335843

Service Period: 6/1/2021 to 7/1/2021

Statement Number: 20673439001

Previous Balance: \$35.10

Payments Since Last Invoice: -\$18.54

Unpaid Balance: \$21.06

Let's Finance Charges: \$0.00

Credit Adjustment: \$0.00

Total New Charges: \$7.03

HOW WE CALCULATED YOUR BILL

See reverse side for detailed description of charges.

Total New Charges: \$7.03

Contract Charges: \$0.61

Market Charges: \$0.00

CONSUMPTION HISTORY

MONTHLY USAGE

Current Month: 98 kwh

0.0% from last year

Last Month: 98 kwh

Last Year: 98 kwh

you can also pay your bill online - go to Energy Manager at <https://www.energymanager.constellation.com> to get started. It's fast, simple and secure.

Statement Number: 20673439001 Customer Number: 6696913-3

Total Amount Due by 10/01/2021: \$28.09

Note: After verification:

- The document receives a **Verified** status.
- An accuracy percentage appears next to the document name.
- The document is included in evaluation metrics.
- The **Metrics Summary** is automatically updated.

8. Click Go to Metrics Dashboard (A).

Invoice extractor

Models: llama-4-maverick-17b-128e-instruct-fp8

Metrics Summary
Overall accuracy (1/4) 100%
[Go to Metrics Dashboard](#) (A)

Utility_bill_training.pdf Verified
Review each extracted field and enter their correct value if needed.

Field Name	Extracted Value	Correct Value
Due date	10/01/2021	10/01/2021
Year to year usage	0.0%	0.0%
Monthly usage	98 kwh	98 kwh
Amount due	\$28.09	\$28.09
Customer address	P.O. Box 9005 Laurel, MD ...	P.O. Box 9005 Laurel, M...
Customer number	6696913-3	6696913-3
Statement date	07/02/2021	07/02/2021
Customer name	ABC Corporation	ABC Corporation

Monthly Invoice
Constellation Energy Company
Statement Date: 07/02/2021
Customer Number: 6696913-3
Total Amount Due by 10/01/2021: \$28.09

HOW WE CALCULATED YOUR BILL
See reverse side for detailed description of charges.

CONSUMPTION HISTORY
Bar chart showing usage from Jan to Jun.

MONTHLY USAGE
Current Month: 98 kwh
Last Month: 98 kwh
Last Year: 98 kwh
0.0% from last year

9. Review the various metrics on the extraction quality.

Invoice extractor

Models: llama-4-maverick-17b-128e-instruct-fp8

Field Evaluation Metrics [Back to Builder](#)

Accuracy 100% F1 score 100% Precision 100% Recall 100%

Field or column	Documents with warning	Accuracy	F1 score	Precision	Recall
Due date	0 documents	100%	100%	100%	100%
Customer name	0 documents	100%	100%	100%	100%
Customer number	0 documents	100%	100%	100%	100%
Amount due	0 documents	100%	100%	100%	100%
Statement date	0 documents	100%	100%	100%	100%
Monthly usage	0 documents	100%	100%	100%	100%
Customer address	0 documents	100%	100%	100%	100%
Year to year usage	0 documents	100%	100%	100%	100%

10. Click **Back to Builder** (A).

The screenshot shows the 'Invoice extractor' interface with 'Field Evaluation Metrics'. The metrics for Accuracy, F1 score, Precision, and Recall are all 100%. A table below lists fields like 'Due date', 'Customer name', etc., all with 100% accuracy. A red circle 'A' highlights the 'Back to Builder' button in the top right corner.

Field or column	Documents with warning	Accuracy	F1 score	Precision	Recall
Due date	0 documents	100%	100%	100%	100%
Customer name	0 documents	100%	100%	100%	100%
Customer number	0 documents	100%	100%	100%	100%
Amount due	0 documents	100%	100%	100%	100%
Statement date	0 documents	100%	100%	100%	100%
Monthly usage	0 documents	100%	100%	100%	100%
Customer address	0 documents	100%	100%	100%	100%
Year to year usage	0 documents	100%	100%	100%	100%

Test on an image file

In the previous step, the document extractor has been trained using a pdf document. Scanned images can also be used in the same way. Let's use an image of the same document populated with different data.

11. Select **Utility bill 2.png** (A).

The screenshot shows the 'Invoice extractor' interface with a document list on the left and a preview of a 'Monthly Invoice' from Constellation on the right. A red circle 'A' highlights 'Utility_bill_2.png' in the document list. The preview shows a utility bill with a total amount due of \$28.09.

Field Name	Extracted Value	Correct Value
Due date	10/01/2021	10/01/2021
Yes	0.0%	0.0%
Monthly usage	98 kwh	98 kwh
Amount due	\$28.09	\$28.09
Customer address	P.O. Box 9005 Laurel, MD	P.O. Box 9005 Laurel, M...
Customer number	6696913-3	6696913-3
Statement date	07/02/2021	07/02/2021
Customer name	ABC Corporation	ABC Corporation

12. In some situations, some fields may not have been recognized with this image version of the document. In this example, all the fields have been correctly extracted from the image (A).

Invoice extractor

Metrics Summary
Overall accuracy (1/4) 100%

Utility_bill_2.png **Unverified**
Review each extracted field and enter their correct value if needed.

Field Name	Extracted Value	Correct Value
Due date	10/01/2021	10/01/2021
Year to year usage	0.0%	0.0%
Monthly usage	98 kWh	98 kWh
Amount due	\$13.07	\$13.07
Customer address	P.O. Box9005 Laurel, MD ...	P.O. Box9005 Laurel, MD...
Customer number	6696913-3	6696913-3
Statement date	07/02/2021	07/02/2021
Customer name	LOT Corporation	LOT Corporation

Constellation Monthly Invoice

Statement Date: 07/02/2021
Customer Number: 6696913-3
Total Amount Due by 10/01/2021: **\$13.07**

HOW WE CALCULATED YOUR BILL

Total New Charges: **\$7.03**
Contract Charges: \$6.61

MONTHLY USAGE

Current Month: 98 kWh
0.0% from last year
Last Month: 98 kWh
Last Year: 98 kWh

Note: It may take a few seconds before the screen is refreshed with the extracted values. If all the fields are showing “No result found”, refresh the view by switching to another document, then back again.

13. Click **Verify document** when all values are correctly extracted (A).

Invoice extractor

Metrics Summary
Overall accuracy (1/4) 100%

Utility_bill_2.png **Unverified**
Review each extracted field and enter their correct value if needed.

Field Name	Extracted Value	Correct Value
Due date	10/01/2021	10/01/2021
Year to year usage	0.0%	0.0%
Monthly usage	98 kWh	98 kWh
Amount due	\$13.07	\$13.07
Customer address	P.O. Box9005 Laurel, MD ...	P.O. Box9005 Laurel, MD...
Customer number	6696913-3	6696913-3
Statement date	07/02/2021	07/02/2021
Customer name	LOT Corporation	LOT Corporation

Constellation Monthly Invoice

Statement Date: 07/02/2021
Customer Number: 6696913-3
Total Amount Due by 10/01/2021: **\$13.07**

HOW WE CALCULATED YOUR BILL

Total New Charges: **\$7.03**
Contract Charges: \$6.61

MONTHLY USAGE

Current Month: 98 kWh
0.0% from last year
Last Month: 98 kWh
Last Year: 98 kWh

Not all invoices will have the same structure. In this section, you will test your document extractor on a utility invoice from a different energy company.

IBM watsonx Orchestrate

Docu

Metrics Summary

Overall accuracy (2/4) 100%

Go to Metrics Dashboard →

2 documents require verification.

Select

Utility_bill_2.png 100%

Utility_bill_4.png 100%

Utility_bill_2.png

Utility_bill_4.png

Utility_bill_2.png Verified

Review each extracted field and enter their correct value if needed.

Add field +

Field Name	Extracted Value	Correct Value
Due date	10/01/2021	alt: 10/01/2021
Year to year usage	0.0%	alt: 0.0%
Monthly usage	98 kWh	alt: 98 kWh
Due	\$13.07	alt: \$13.07
Customer address	P.O. Box9005 Laurel, MD ...	alt: P.O. Box9005 Laurel, MD...
Customer number	6696913-3	alt: 6696913-3
Statement date	07/02/2021	alt: 07/02/2021
Customer name	LOT Corporation	alt: LOT Corporation

Add documents Verify document

Models: llama-4-maverick-17b-128e-instruct-lp8

1 / 1

Utility_bill_4.png

Monthly Invoice

Constellation

lot corporation

LOT Corporation

POB4855906880550

Service address:

145 Andover Street Norwalk, CT 06850

Rate Plan:

Account ID:

Utility Number:

Service Period:

Statement Number:

Fixed Price Solutions

10223678

4584577954456643358643

6/1/2021 to 7/1/2021

20673436011

Monthly Invoice

Statement Date: 07/02/2021

Customer Number: 6696913-3

Total Amount Due by 10/01/2021: \$13.07

Previous Balance: \$35.10

Payments Since Last Invoice: -\$14.04

Unpaid Balance: \$21.06

Late/Fee/Finance Charge: \$0.00

Credit/Adjustment: \$0.00

Total New Charges: \$7.03

HOW WE CALCULATED YOUR BILL

See reverse side for detailed description of charges

Tax Charges \$0.42

Market Charges \$0.00

CONSUMPTION HISTORY

MONTHLY USAGE

Current Month 98 kWh

0.0% from last year

Last Month 98 kWh

Last Year 98 kWh

Total Amount Due by 10/01/2021 \$13.07

You can also pay your bill online - go to Energy Manager at <https://energymanager.constellation.com> to get started. It's fast, simple and secure.

Document label and content with your program in mind. Please note your document number on your invoice. Thank you for your payment.

Constellation

lot corporation

Statement Number: 20673436011

Customer Number: 6696913-3

Invoice extractor

Metrics Summary

Overall accuracy (2/4) ①

100%

Go to Metrics Dashboard →

2 documents require verification.

Select

Utility_bill_2.png100%

Utility_bill_training.pdf100%

Utility_bill_4.png

Utility_bill_3.pdf

Utility_bill_4.pngUnverified

Review each extracted field and enter their correct value if needed.

Add field +

Field Name	Extracted Value	Correct Value
Due date	May 15 th , 2025	aka May 15 th , 2025
Year-to-year usage	+24%	aka +24%
Monthly usage	124 kWh	aka 124 kWh
Amount due	\$45.99	aka \$45.99
Customer address	Gran plaza D 3rd Floor-Ap...	aka Gran plaza D 3rd Floor-A...
Customer number	8C-45678	aka 8C-45678
Statement date	May 5 th , 2025	aka May 5 th , 2025

Monthly Invoice

Date: May 5th, 2025

Customer nameMr. Smith | all

Customer addressGran plaza D
8th Floor -Appt 345
234 Main Street
CA 1234, Los Angeles

Contract
Customer Number: FC-45678
Account ID : 34567844
Rate plan: Adjustable rates

Period: April 2005

Amount Due\$45.99

Due byMay 15th, 2025

Monthly Usage

Period: April 1st, 2025 to April 30th, 2025

Consumption: 124 kWh (year to year +24%)

Last month: 136 kWh
Last year: 100 kWh

Payments

Method: Automatic transfer

Previous Balance: \$44.09
Last payment: \$40.00
Unpaid Balance: \$4.09
New charges: \$7.09

The document extractor may have been unable to recognize and extract all the fields with this new version of the document, and some errors may also be present. You will improve the results from the extractor in the next section by giving better descriptions of the fields and providing examples.

Note: Based on your LLM history and version, un-identified fields or errors may differ from the one presented in this lab. You may have to adapt the following script to correct errors encountered with your extractor.

Improve your extractor

In this section, you will apply techniques to improve your results, helping the model to better identify and extract relevant data from your unstructured documents.

Add field descriptions

In our example (that may differ from yours), the Statement date is extracted but requires a reformatting. We would like all the dates to be formatted the same way with the mm/dd/yyyy format. Let's add a precise description of the field that will be used by the LLM to better extract the information in the format we want.

1. Identify a field to improve (A) then click the **Options** icon (B).

Invoice extractor

Metrics Summary
Overall accuracy (2/4) 100%
Go to Metrics Dashboard →
2 documents require verification.

Utility_bill_4.png (Unverified)
Review each extracted field and enter their correct value if needed.

Field Name	Extracted Value	Correct Value	Options
Due date	May 15, 2025	May 15, 2025	
Year to year usage	+24%	+24%	
Monthly usage	124 kWh	124 kWh	
Amount due	\$45.99	\$45.99	
Customer address	Gran plaza D 3rd Floor-Ap...	Gran plaza D 3rd Floor-A...	
Customer number	FC-45678	FC-45678	
Statement date	May 5th, 2025	May 5th, 2025	
Customer name	Mr. Smith bill	Mr. Smith bill	

Monthly Invoice
Date: May 5th, 2025

Customer name: Mr. Smith bill
Customer address: Capitol plaza building, 1234 South Beach avenue, CA 34678, San Francisco
Contract: FC-45678
Customer Number: FC-45678
Account ID: 34567544
Rate plan: Adjustable rates

Amount Due: \$45.99
Due by: May 15th, 2025

Monthly Usage
Period: April 1st, 2025 to April 30th, 2025
Consumption: 124 kWh (year to year +24%)
Last month: 136 kWh
Last year: 100 kWh

Payments
Method: Automatic transfer
Previous Balance: \$44.09
Last payment: \$40.00
Unpaid Balance: \$4.09
New charges: \$7.09

2. Click **Edit** (A).

Invoice extractor

Models: llama-4-maverick-17b-128e-instruct-1p8

1 / 1

Utility_bill_4.png Unverified

Review each extracted field and enter their correct value if needed.

Add field +

Field Name	Extracted Value	Correct Value
Due date	May 15', 2025	etc. May 15', 2025
Year to year usage	+24%	etc. +24%
Monthly usage	124 kWh	etc. 124 kWh
Amount due	\$45.99	etc. \$45.99
Customer address	Gran plaza D 3rd Floor-Ap...	etc. Gran plaza D 3rd Floor-A...
Customer number	FC-45678	etc. FC-45678
Statement date	May 5'h, 2025	etc. May 5'h, 2025
Customer name	Mr. Smith bill	etc. Mr. Smith bill

Monthly Invoice
Date: May 5th, 2025

Customer name: Mr. Smith bill
Customer address: Gran plaza D, 3rd Floor-Appt 345, 234 Main Street, CA 1234, Los Angeles
Contract: Customer Number: FC-45678, Account ID: 34567544, Rate plan: Adjustable rates

Period: April 2005

Amount Due: \$45.99 Due by: May 15th, 2025

Monthly Usage
Period: April 1st, 2025 to April 30th, 2025
Consumption: 124 kWh (year to year +24%)
Last month: 136 kWh
Last year: 100 kWh

Payments
Method: Automatic transfer
Previous Balance: \$44.09
Last payment: \$40.00
Unpaid Balance: \$4.09
New charges: \$7.09

Add documents Verify document

3. Enter 'The statement date is the date the invoice was issued. Return the date using in the MM/DD/YYYYYY format. May 1st, 2025 will be displayed as 05/01/2025.' into the **Description** field (A). Click **Submit** (B).

Invoice extractor

Models: llama-4-maverick-17b-128e-instruct-1p8

1 / 1

Configure 'Statement date'

Enter the field details used to guide the model when extracting values from document.

Name: Statement date
The term that is used to identify this field.

Data type: String
How the information is presented in the document.

Description
The statement date is the date the invoice was issued. Return the date using in the MM/DD/YYYYYY format. May 1st, 2025 will be displayed as 05/01/2025.

Examples (Optional)
Add 3-5 examples to help the LLM understand what output you're expecting.
Enter example +

Delete field

Cancel Submit

Monthly Invoice
Date: May 5th, 2025

Customer name: Mr. Smith bill
Customer address: Gran plaza D, 3rd Floor-Appt 345, 234 Main Street, CA 1234, Los Angeles
Contract: Customer Number: FC-45678, Account ID: 34567544, Rate plan: Adjustable rates

Period: April 2005

Amount Due: \$45.99 Due by: May 15th, 2025

Monthly Usage
Period: April 1st, 2025 to April 30th, 2025
Consumption: 124 kWh (year to year +24%)
Last month: 136 kWh
Last year: 100 kWh

Payments
Method: Automatic transfer
Previous Balance: \$44.09
Last payment: \$40.00
Unpaid Balance: \$4.09
New charges: \$7.09

4. The field is now correctly identified and formatted the right way (A).

Invoice extractor

Metrics Summary
Overall accuracy (0/4) 0%

Go to Metrics Dashboard →

All documents require verification.

Select

Utility_bill_2.png
Utility_bill_training.pdf
Utility_bill_4.png
Utility_bill_3.pdf

Add documents

Verify document

Utility_bill_4.png Unverified

Review each extracted field and enter their correct value if needed.

Add field +

Field Name	Extracted Value	Correct Value
Due date	May 15', 2025	May 15', 2025
Year to year usage	+24%	+24%
Monthly usage	124 kWh	124 kWh
Amount due	\$45.99	\$45.99
Customer address	Gran plaza D 3rd Floor-Ap...	Gran plaza D 3rd Floor-A...
Customer number	FC-45678	FC-45678
Statement date	05/05/2025	05/05/2025
Customer name	Mr. Smith bill	Mr. Smith bill

Monthly Invoice

Date: May 5th, 2025

Customer name
Mr. Smith bill

Customer address
Gran plaza D
3rd Floor-Apt 345
1234 Main Street
CA 1234, Los Angeles

Contract
Customer Number: FC-45678
Account ID: 34567544
Rate plan: Adjustable rates

Period: April 2005

Amount Due \$45.99 Due by May 15th, 2025

Monthly Usage

Period: April 1st, 2025 to April 30th, 2025

Consumption: 124 kWh (year to year +24%)
Last month: 136 kWh
Last year: 100 kWh

Payments

Method: Automatic transfer

Previous Balance: \$44.09
Last payment: \$40.00
Unpaid Balance: \$4.09
New charges: \$7.09

Note: If your extractor does not show the correctly formatted date, click the <- icon, then return back to the field details view to refresh the values.

5. Repeat the instruction definition (adapt it to the due date) so that the Due date is also displayed in the MM/DD/YYYY format (A). Click **Submit** (B).

Invoice extractor

Metrics Summary
Overall accuracy (0/4) 0%

Go to Metrics Dashboard →

All documents require verification.

Select

Utility_bill_2.png
Utility_bill_training.pdf
Utility_bill_4.png
Utility_bill_3.pdf

Add documents

Configure 'Due date'

Enter the field details used to guide the model when extracting values from document.

Name

Due date

The term that is used to identify this field.

Data type

String

How the information is presented in the document.

Description

The due date is the date the invoice was issued. Return the date using the MM/DD/YYYY format. May 1st, 2025 will be displayed as 05/01/2025.

Describe the information you want to extract.

Examples (Optional)

Add 3-5 examples to help the LLM understand what output you're expecting.

Enter example +

Delete field

Cancel

Submit

Monthly Invoice

Date: May 5th, 2025

Customer name
Mr. Smith bill

Customer address
Gran plaza D
3rd Floor-Apt 345
1234 Main Street
CA 1234, Los Angeles

Contract
Customer Number: FC-45678
Account ID: 34567544
Rate plan: Adjustable rates

Period: April 2005

Amount Due \$45.99 Due by May 15th, 2025

Monthly Usage

Period: April 1st, 2025 to April 30th, 2025

Consumption: 124 kWh (year to year +24%)
Last month: 136 kWh
Last year: 100 kWh

Payments

Method: Automatic transfer

Previous Balance: \$44.09
Last payment: \$40.00
Unpaid Balance: \$4.09
New charges: \$7.09

Note: If for some reasons the model suddenly loses all the fields as shown below, select another model to force a refresh and come back to the model of your choice.

The screenshot shows the IBM watsonx Orchestrate interface for the 'Invoice extractor' agent. The 'Metrics Summary' on the left indicates an overall accuracy of 0% (0/4 documents). The 'Utility_bill_4.png' document is selected, and the 'Extracted Value' table is empty, suggesting the model has lost its state. The 'Verify document' button at the bottom is highlighted with a red box. The right pane displays a sample 'Monthly Invoice' from FCE Energies.

Field Name	Extracted Value	Correct Value
Due date	No value	etc
Year to year usage	No value	etc
Monthly usage	No value	etc
Amount due	No value	etc
Customer address	No value	etc
Customer number	No value	etc
Statement date	No value	etc
Customer name	No value	etc

6. Once all the values are correctly extracted, click **Verify document** (A).

The screenshot shows the IBM watsonx Orchestrate interface for the 'Invoice extractor' agent. The 'Metrics Summary' on the left indicates an overall accuracy of 0% (0/4 documents). The 'Utility_bill_4.png' document is selected, and the 'Extracted Value' table is populated with correct values. The 'Verify document' button at the bottom is highlighted with a red box and labeled 'B'. The right pane displays a sample 'Monthly Invoice' from FCE Energies.

Field Name	Extracted Value	Correct Value
Due date	05/15/2025	etc 05/15/2025
Year to year usage	100 kWh	etc 100 kWh
Monthly usage	124 kWh (year to year +24%)	etc 124 kWh (year to year +24%)
Amount due	\$45.99	etc \$45.99
Customer address	Gran plaza D 3rd Floor-Ap...	etc Gran plaza D 3rd Floor-A...
Customer number	FC-45678	etc FC-45678
Statement date	05/05/2025	etc 05/05/2025
Customer name	Mr. Smith bill	etc Mr. Smith bill

7. Changing the model (A) may lead to different extraction quality results (B) .

Invoice extractor

Models: granite-4-h-small

Utility_bill_4.png Verified

Review each extracted field and enter their correct value if needed.

Add field +

Field Name | Extracted Value | Correct Value

Due date	05/15/2025	05/15/2025
Year to year usage	100 kWh	100 kWh
Monthly usage	124 kWh (year to year +24%)	124 kWh (year to year +24%)
Amount due	\$45.99	\$45.99
Customer address	Gran plaza D 3rd Floor-Ap...	Gran plaza D 3rd Floor-Ap...
Customer number	FC-45678	FC-45678
Statement date	05/05/2025	05/05/2025
Customer name	Mr. Smith bill	Mr. Smith bill

Customer name: Mr. Smith bill

Customer address: Gran plaza D, 3rd Floor-Appl 345, 234 Main Street, CA 1234, Los Angeles

Contract: Customer Number: FC-45678, Account ID: 34567544, Rate plan: Adjustable rates

Period: April 2005

Amount Due: \$45.99 Due by: May 15th, 2025

Monthly Usage: Period: April 1st, 2025 to April 30th, 2025. Consumption: 124 kWh (year to year +24%), Last month: 136 kWh, Last year: 100 kWh

Payments: Method: Automatic transfer. Previous Balance: \$44.09, Last payment: \$40.00, Unpaid Balance: \$4.09, New charges: \$7.09

Note: Work with different models to find the most appropriate extractor for your situation.

8. Click x (A) to close the Extractor view.

Invoice extractor

Models: granite-4-h-small

Utility_bill_4.png Verified

Review each extracted field and enter their correct value if needed.

Add field +

Field Name | Extracted Value | Correct Value

Due date	05/15/2025	05/15/2025
Year to year usage	100 kWh	100 kWh
Monthly usage	124 kWh (year to year +24%)	124 kWh (year to year +24%)
Amount due	\$45.99	\$45.99
Customer address	Gran plaza D 3rd Floor-Ap...	Gran plaza D 3rd Floor-Ap...
Customer number	FC-45678	FC-45678
Statement date	05/05/2025	05/05/2025
Customer name	Mr. Smith bill	Mr. Smith bill

Customer name: Mr. Smith bill

Customer address: Gran plaza D, 3rd Floor-Appl 345, 234 Main Street, CA 1234, Los Angeles

Contract: Customer Number: FC-45678, Account ID: 34567544, Rate plan: Adjustable rates

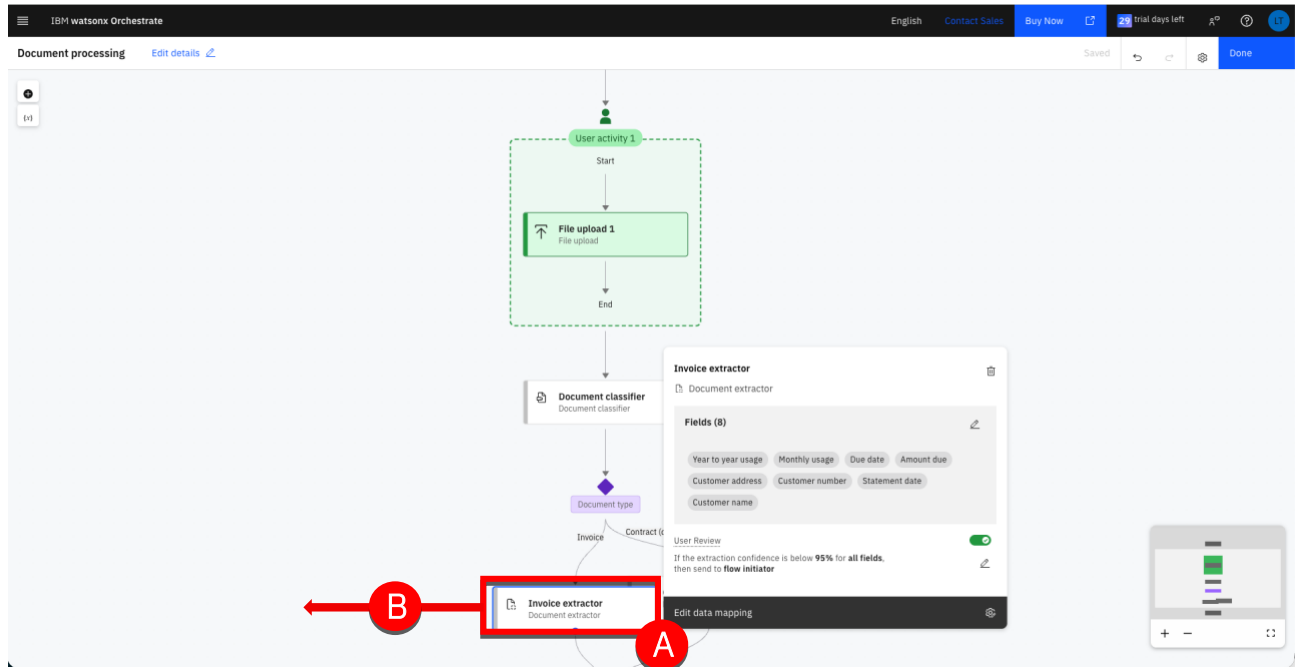
Period: April 2005

Amount Due: \$45.99 Due by: May 15th, 2025

Monthly Usage: Period: April 1st, 2025 to April 30th, 2025. Consumption: 124 kWh (year to year +24%), Last month: 136 kWh, Last year: 100 kWh

Payments: Method: Automatic transfer. Previous Balance: \$44.09, Last payment: \$40.00, Unpaid Balance: \$4.09, New charges: \$7.09

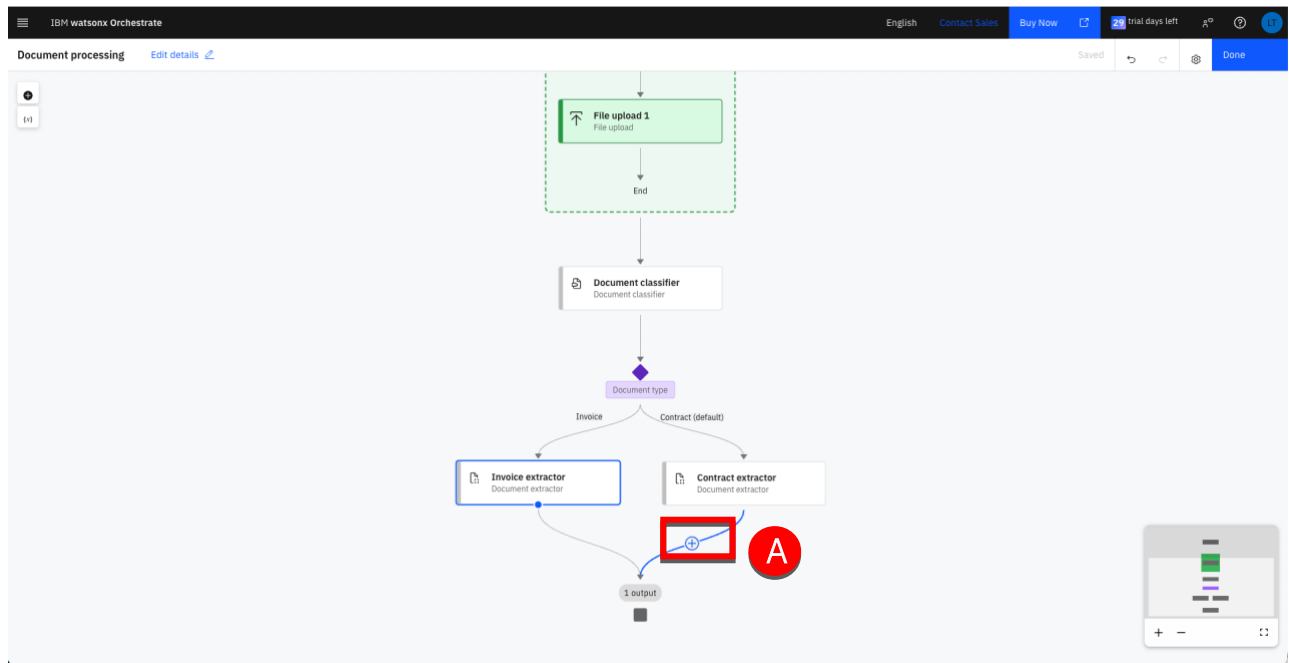
- Click and hold the Invoice extractor node (a) and move it on the left to improve the diagram display (B).



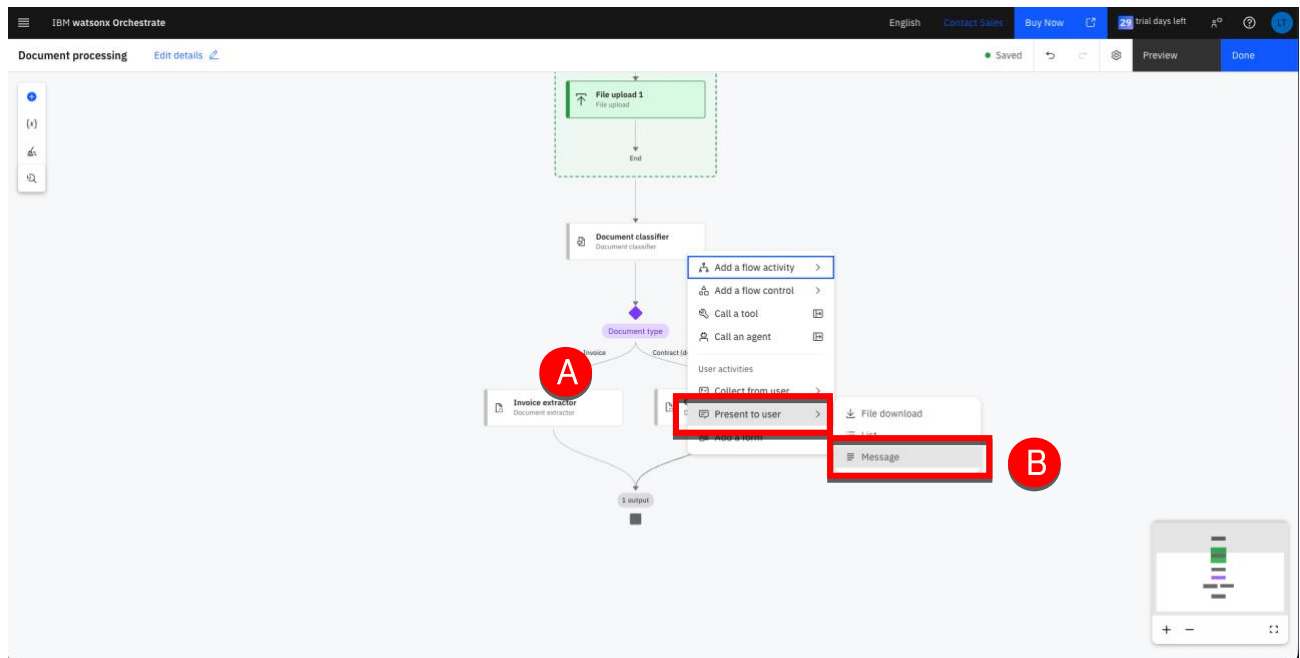
Add a Display to user activity

To complete the flow, we need to add a final activity that presents the extracted values to the user. This step allows us to format and display the results clearly, based on the core fields that we defined when we trained each extractor.

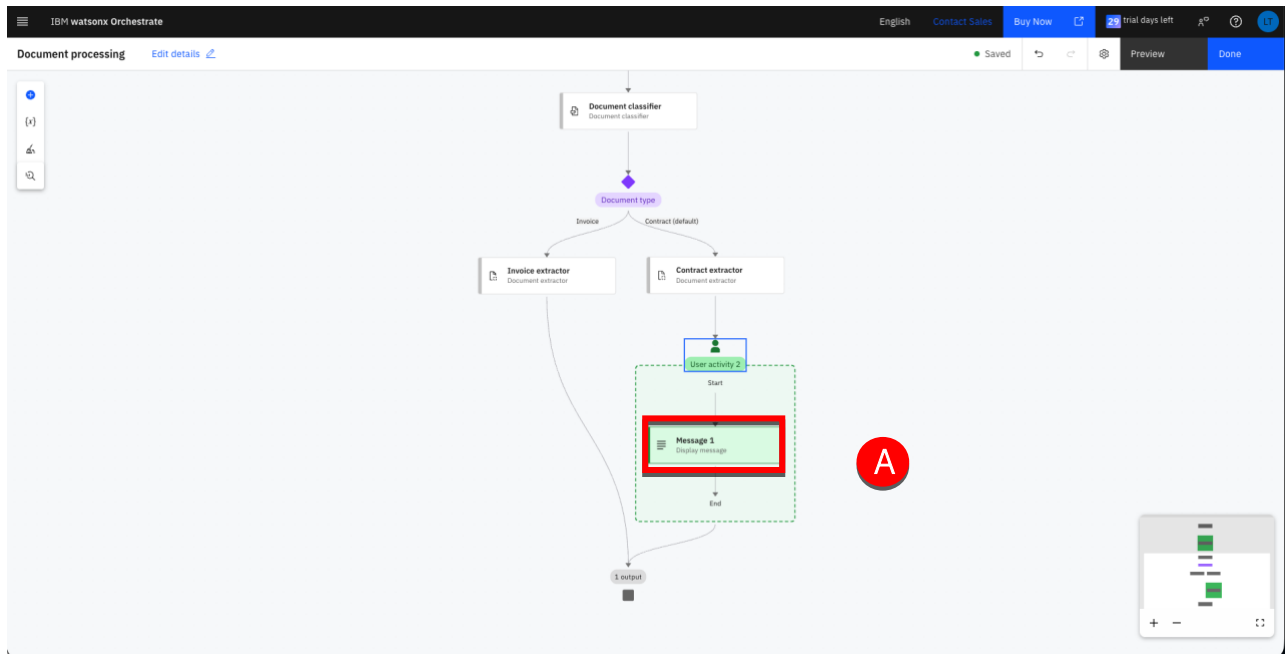
1. Hover the last link in the **Contract extractor** branch and click + (A).



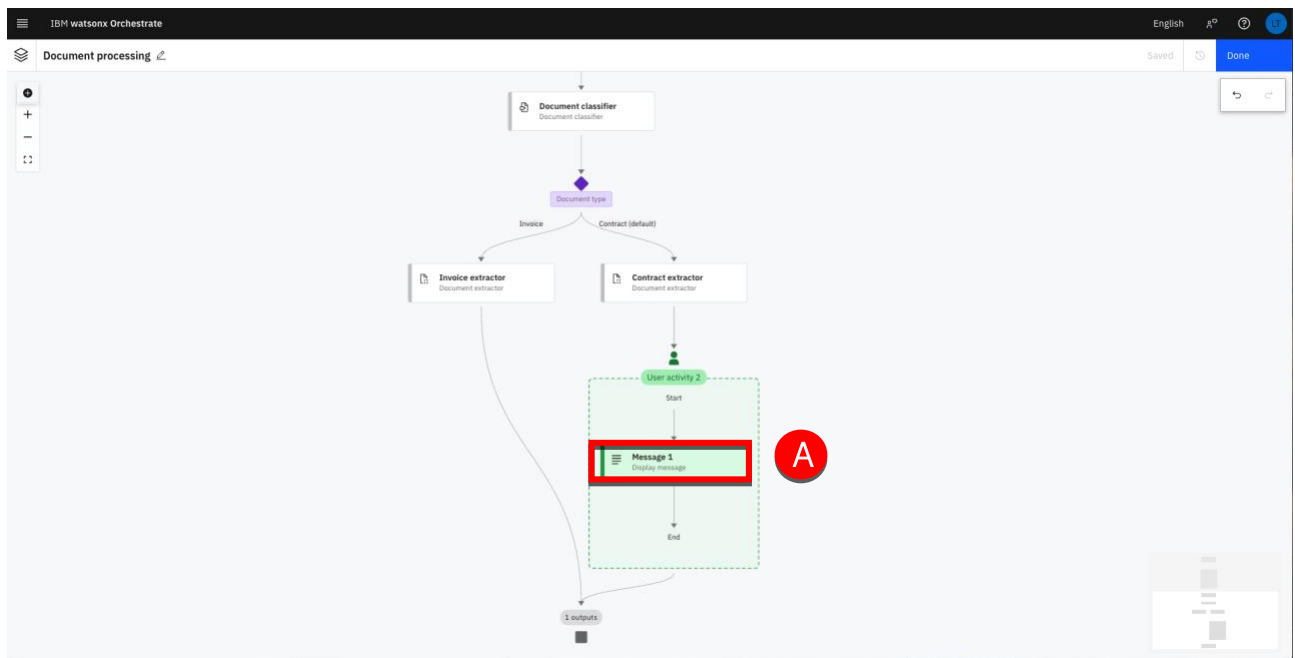
2. Select **Present to user** (A) then click **Message** (B).



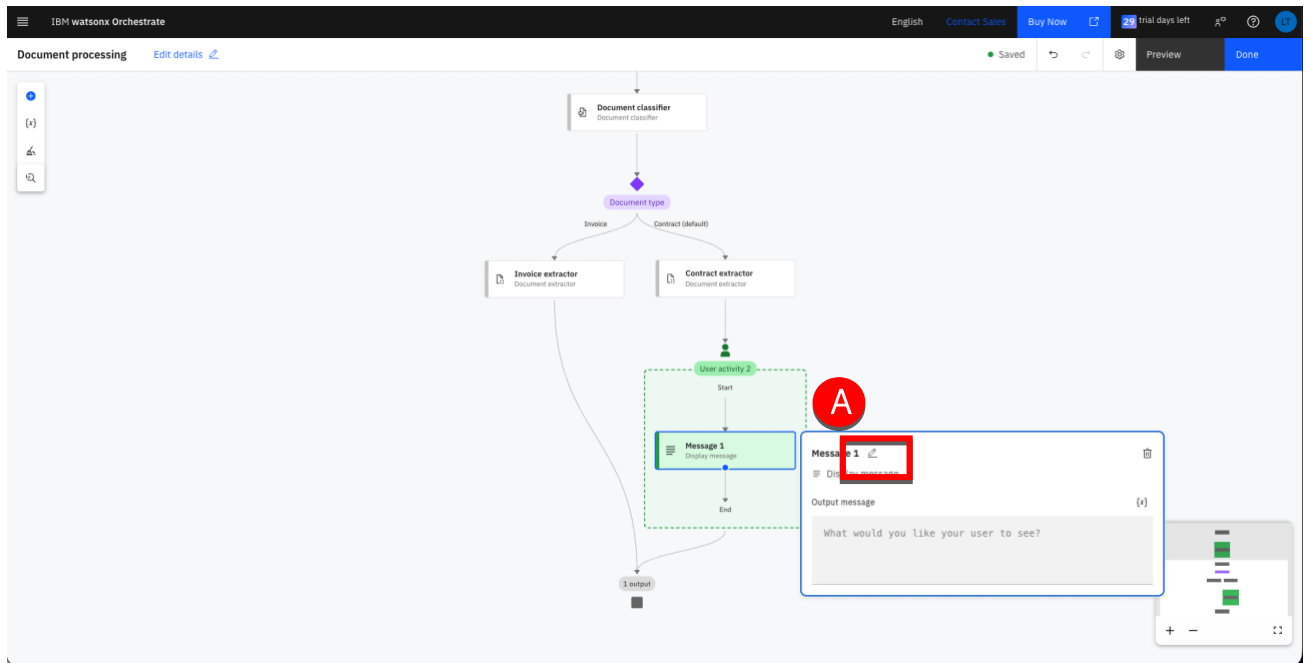
- Click **Message** (A) to add a new action in the user activity node.



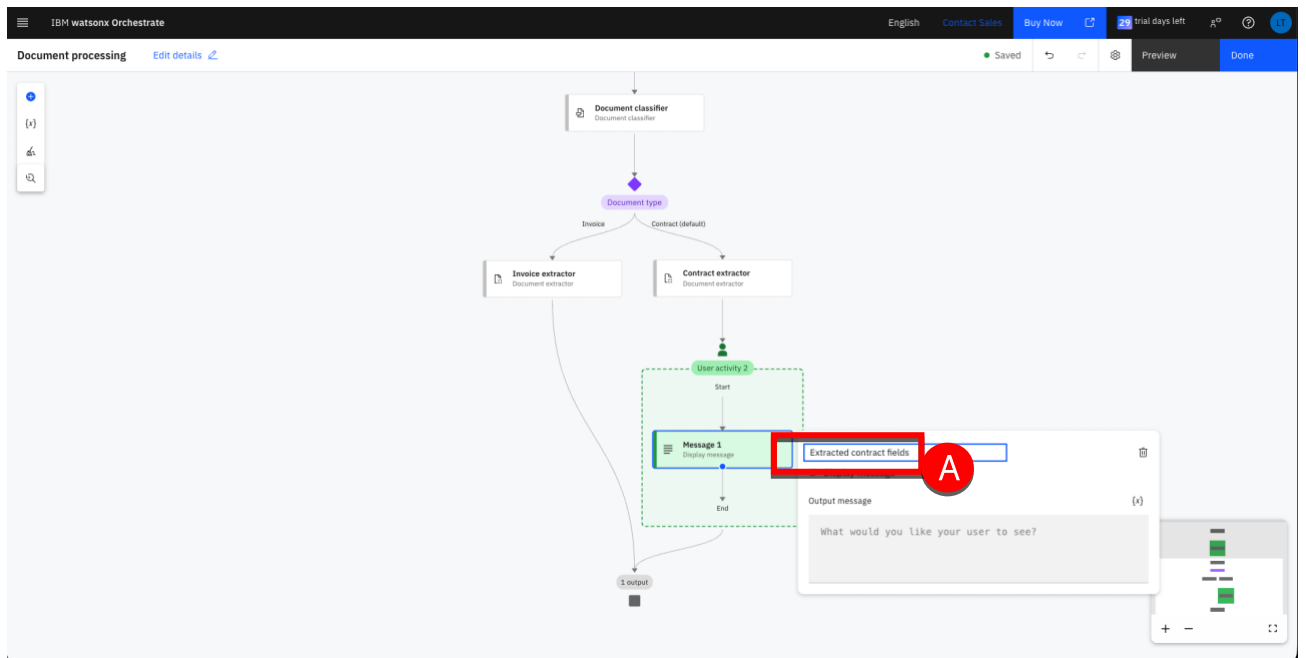
- Click the **Message** box to edit its content (A).



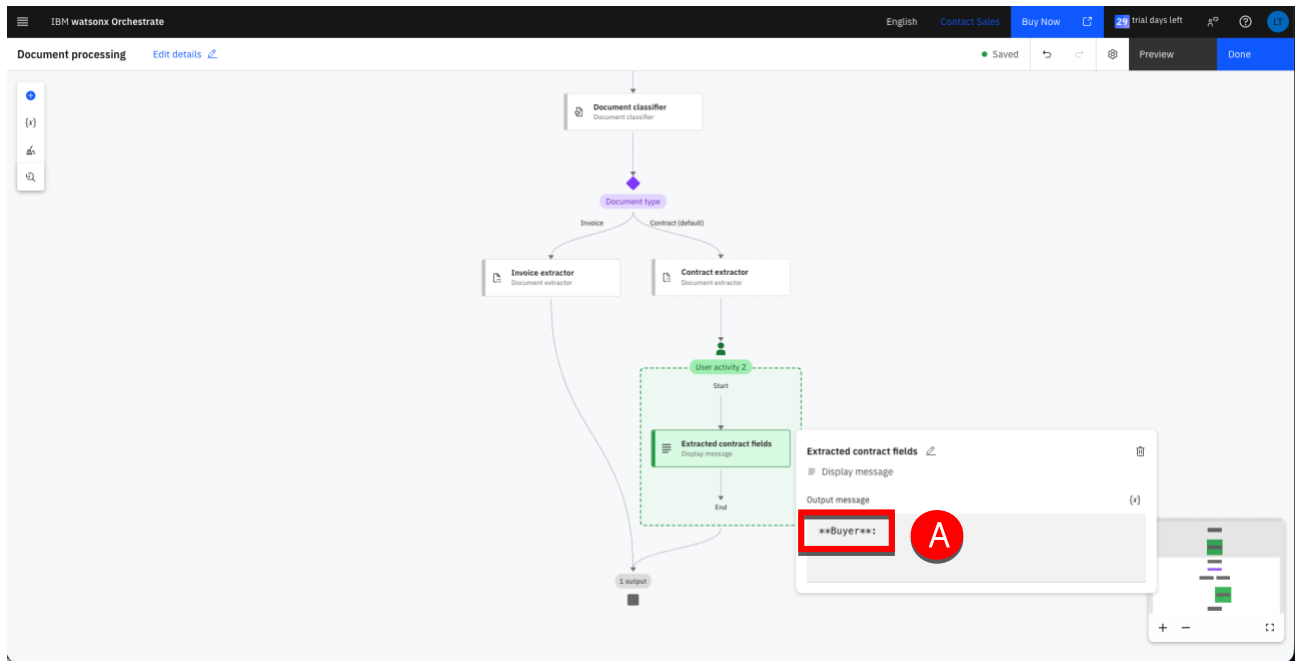
- Click the edit button (B) to rename it.



- Type 'Extracted contract fields' (A) and hit Enter.

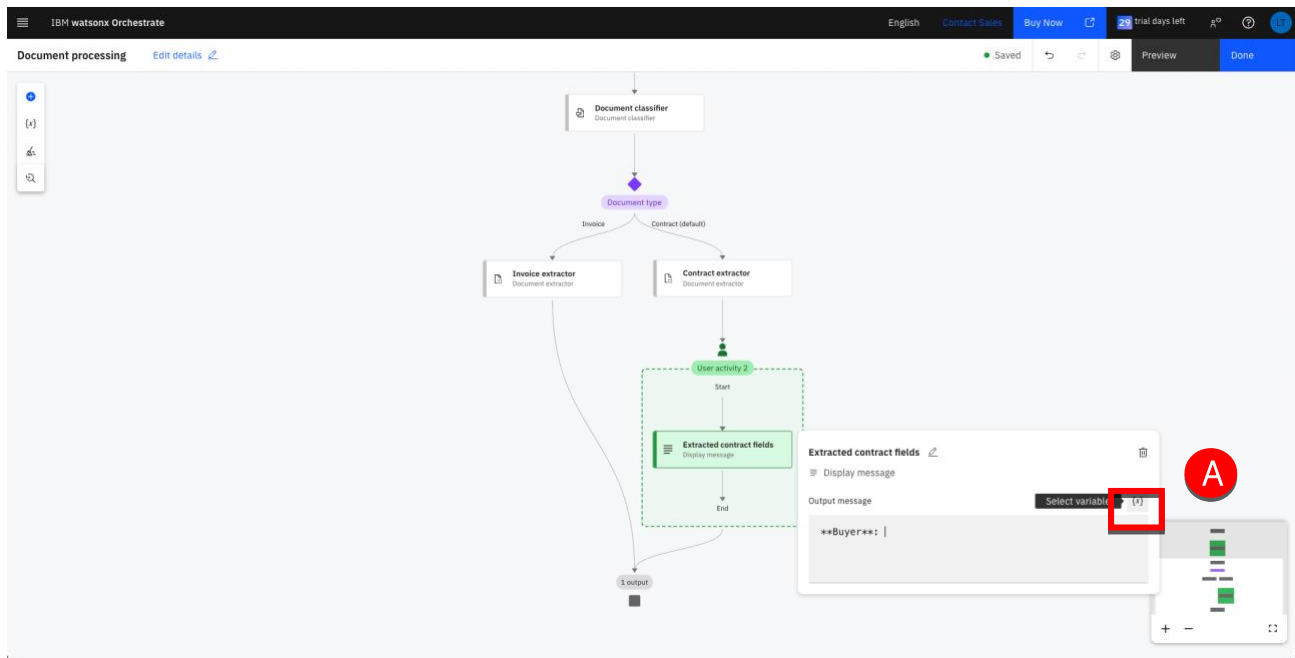


7. Type “**Buyer**:” in the Output message field (A)

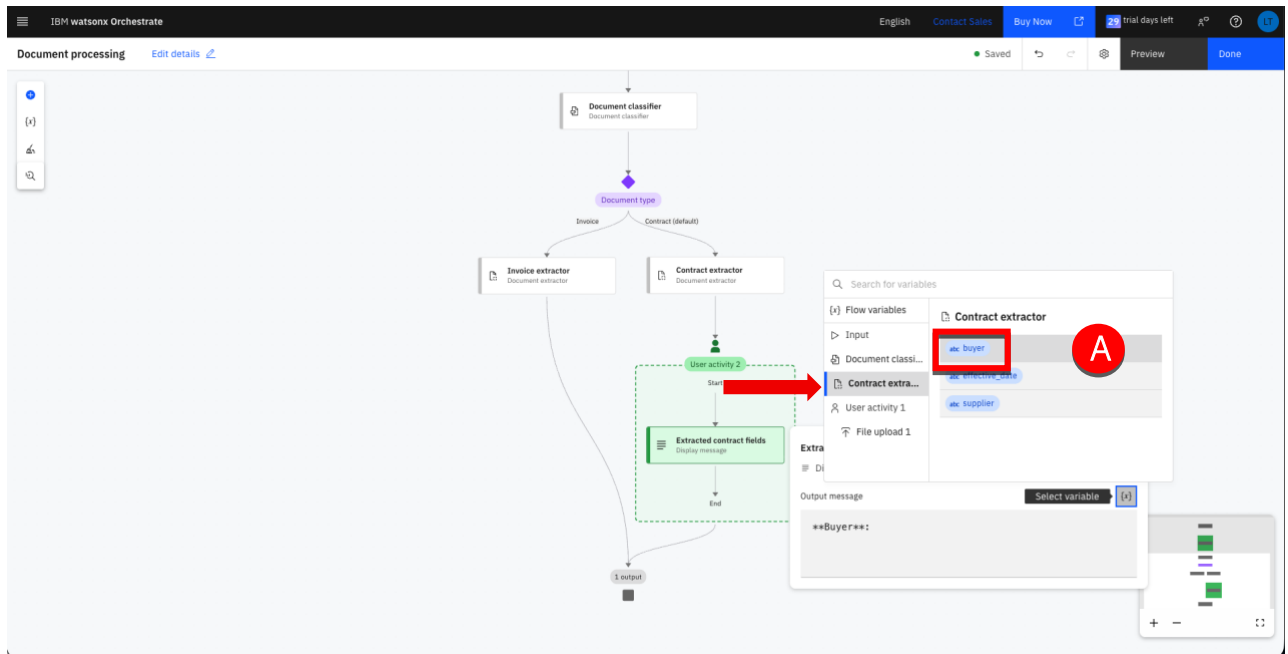


Note: the '**' are just here to specify that the label must be displayed in Bold.

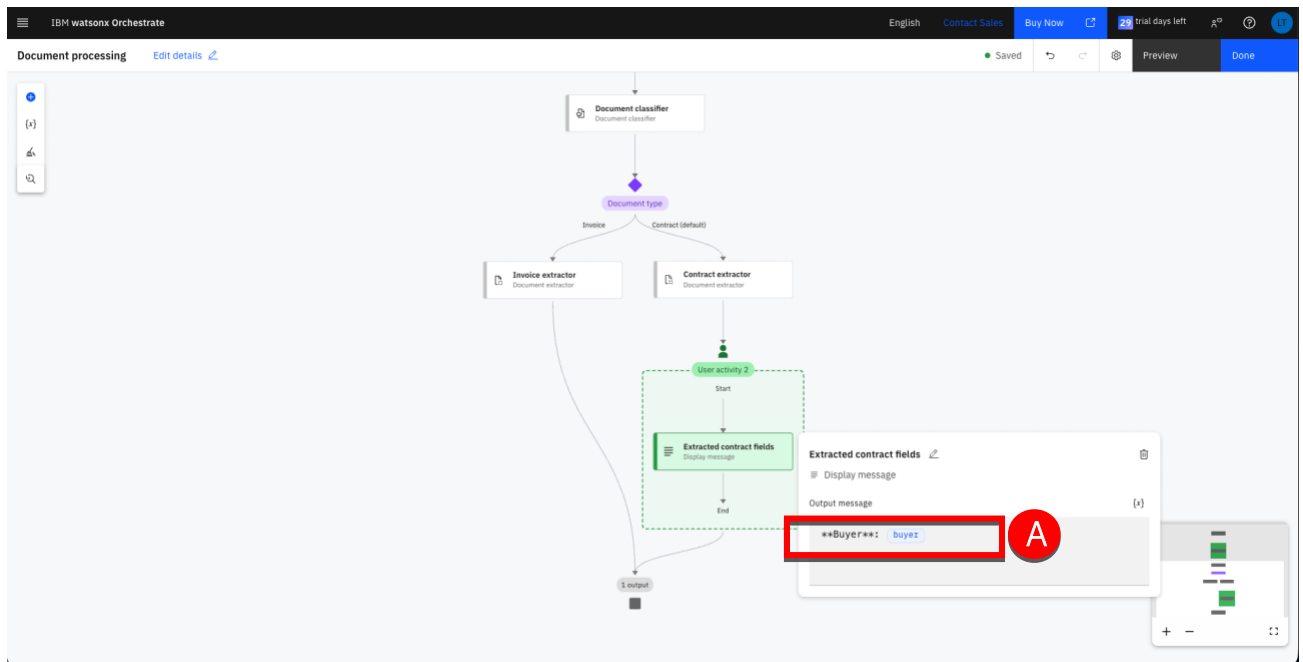
8. To display the results for the field, 'Buyer' you must assign a variable to it. Click on the Select variable '[x]' button.



9. Under the 'Contract extractor' menu click on the **buyer** variable (A).



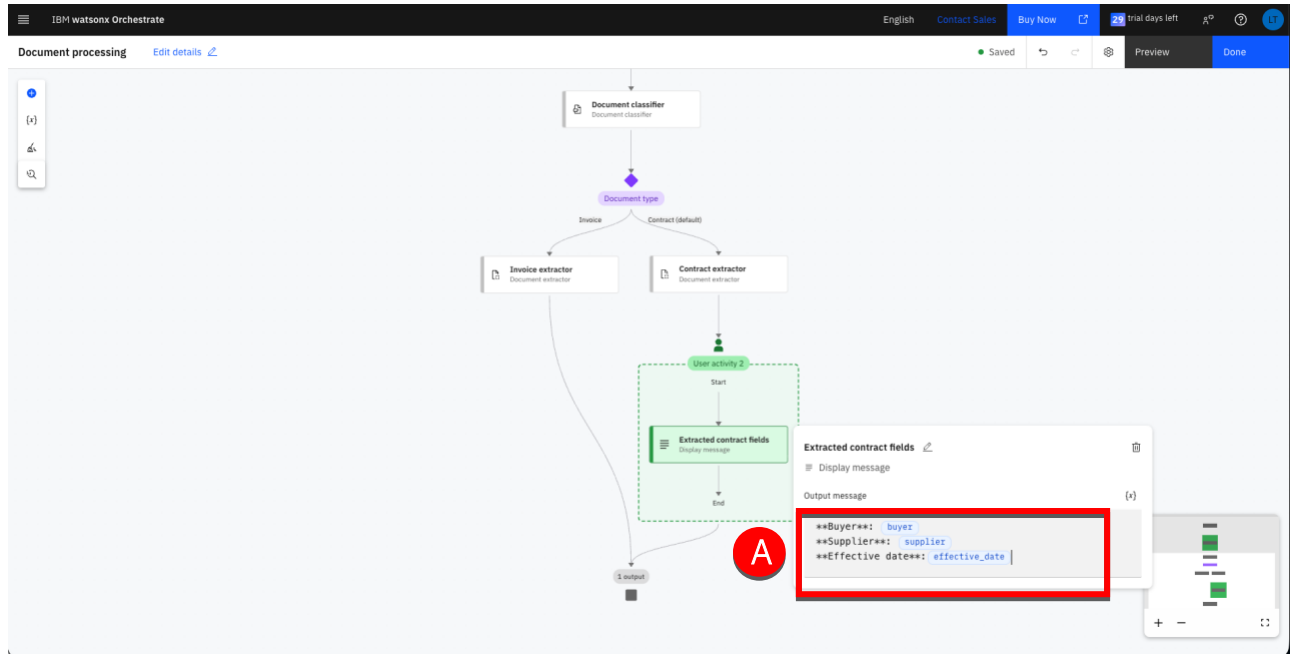
10. Your screen should look like (A):



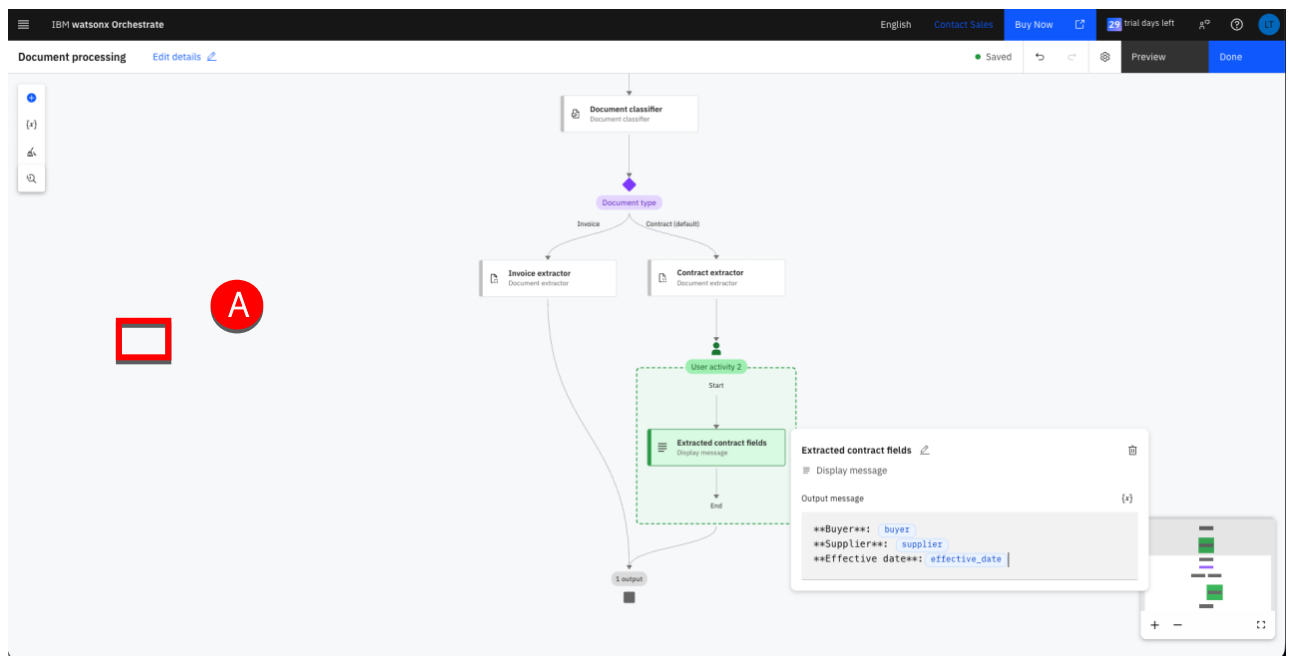
11. Repeat from step 9 to add the remaining fields defined in the Contract extractor:

- Supplier
- Effective date

Your screen should look like (A):

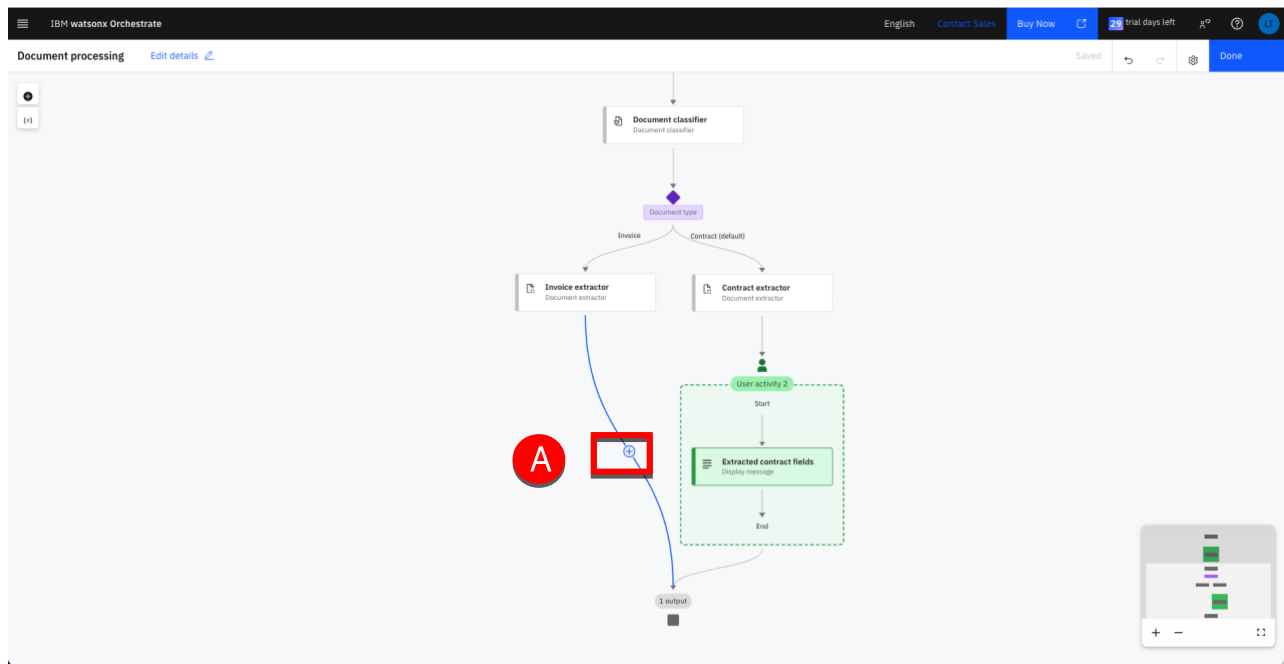


12. Click the background to close the **Extracted contract fields** panel (A).



13. Repeating from Step 1, create a user activity in the **Invoice** branch (A) to display the following fields:

- Customer name
- Customer number
- Customer address
- Amount due
- Due date
- Monthly usage
- Year to year usage



Note: The invoice variables will be under the Invoice extractor folder.

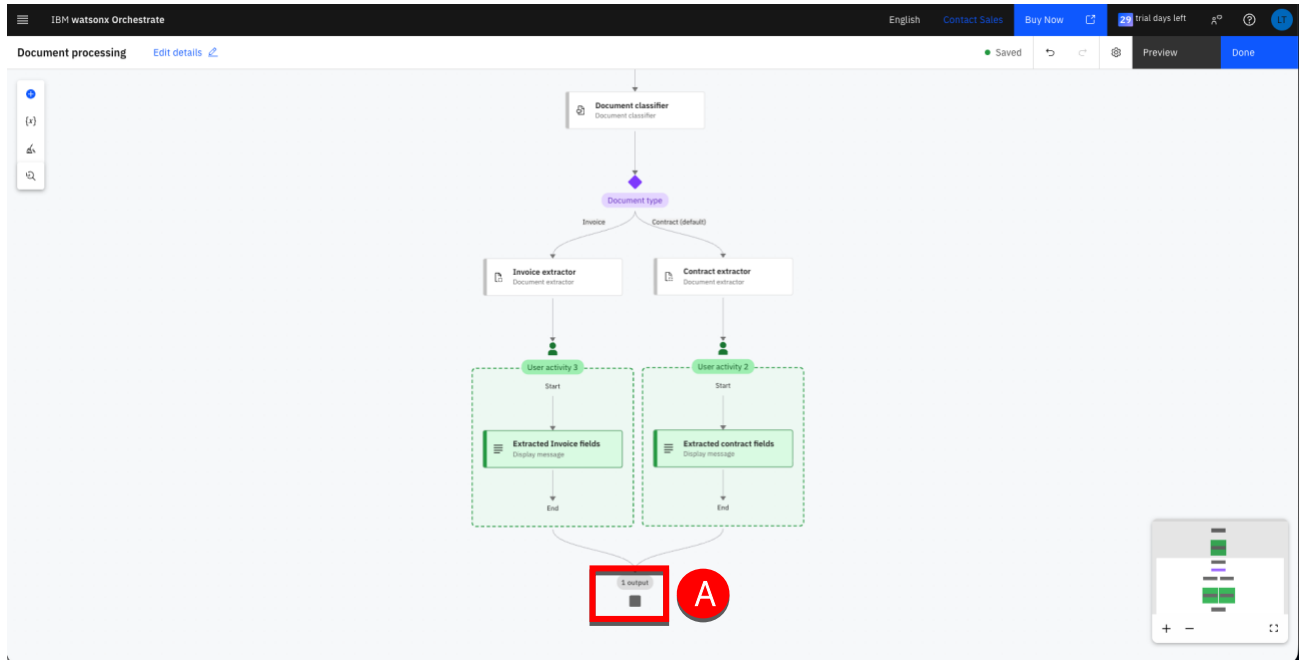
You should get the following result:



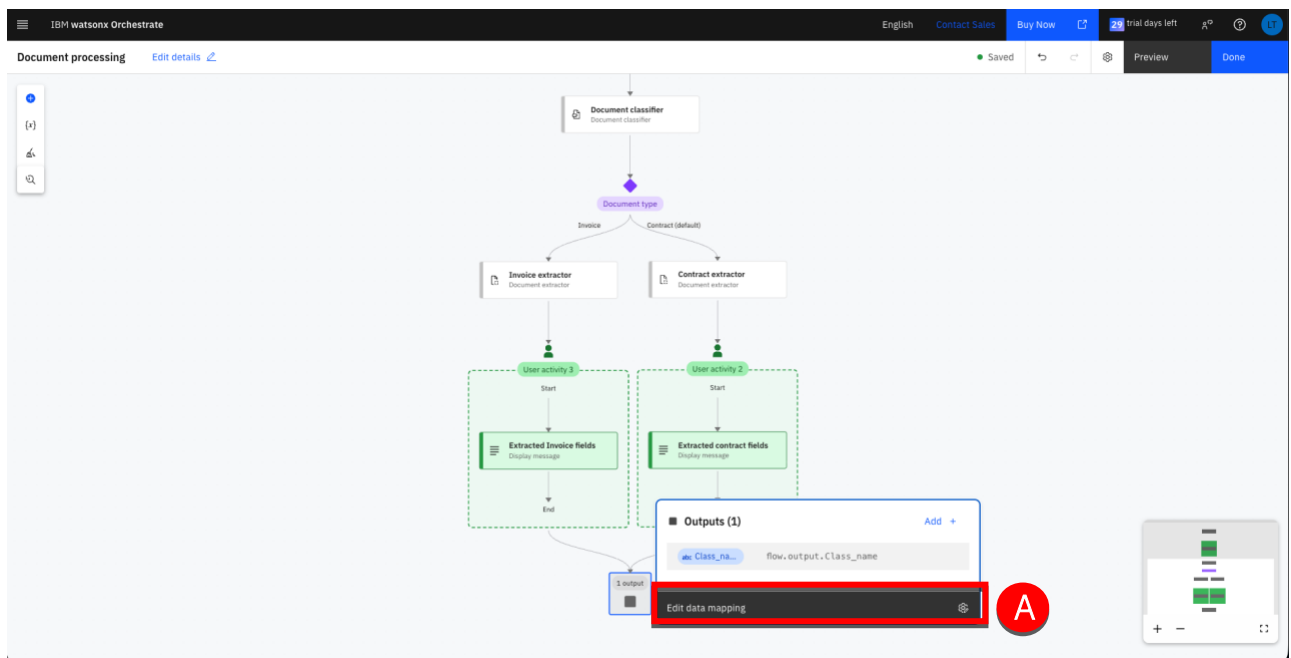
Define the end flow

Now that the output message has been defined for both file types, we can now conclude the flow by updating the end node. The end node will just return the Document type (i.e. Contract/Invoice).

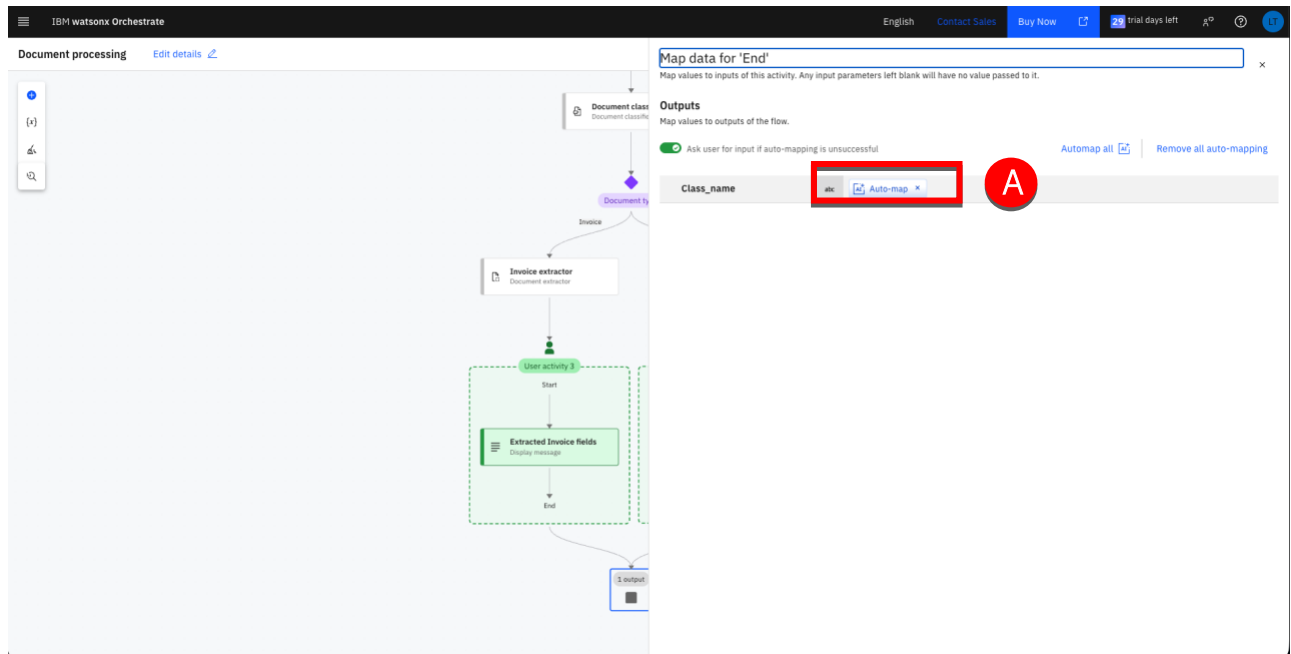
1. Click the **End** node (A).



2. Click **Edit data mapping** (A).

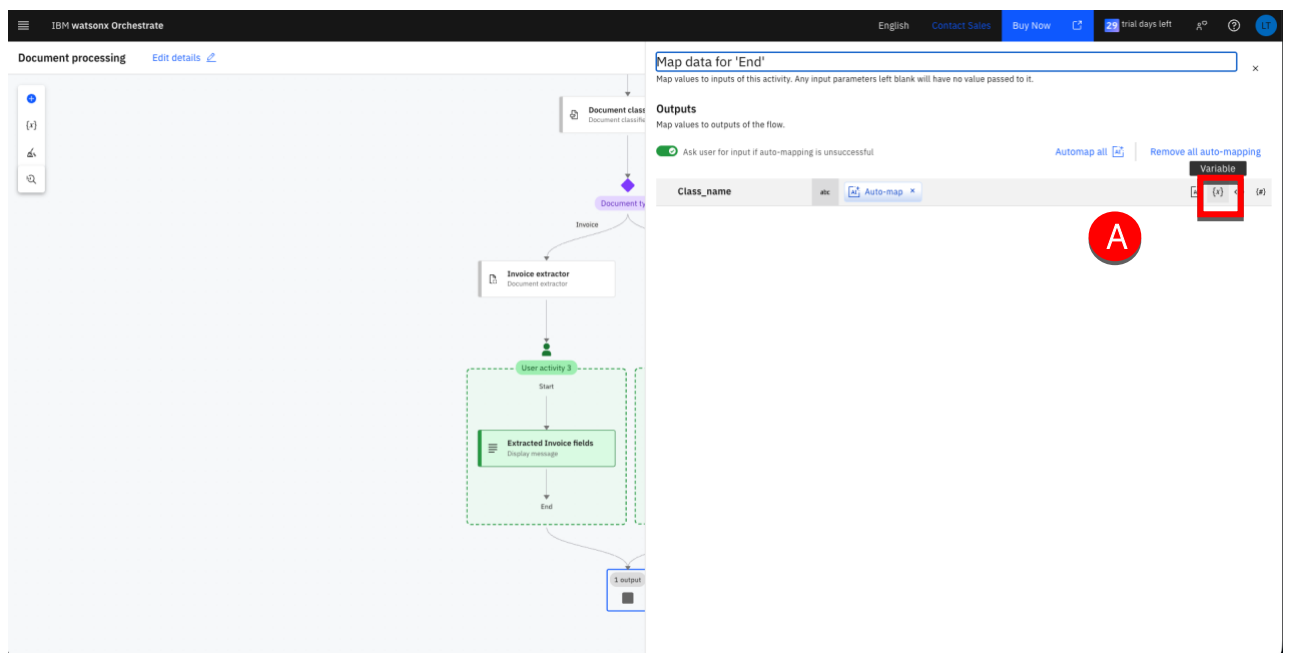


Note that the output, 'class_name' is auto-mapped by the LLM (A). This means that the LLM will attempt to automatically link the input/outputs.



Note: The auto-map feature works effectively for those flows that are less complex. But for the more complex flows, (for instance if you had two 'class_name' outputs) then explicit mapping performs best.

3. Hover the variable row and click on the **Variable** icon (A).



4. Click the Document classifier (A).

The screenshot shows the IBM watsonx Orchestrate interface. On the left, a flow diagram is visible with a 'Document type' node branching into 'Invoice' and 'Contract (default)'. The 'Invoice' branch leads to an 'Invoice extractor' node, which then leads to a 'User activity 3' node. A red circle labeled 'A' is placed over the 'Document classifier' node in the flow diagram. On the right, the 'Map data for End' panel is open. It shows a table with 'Class_name' and 'File upload 1' columns. A red circle labeled 'A' is placed over the 'Document classifier' node in the table.

5. Select class_name (A).

The screenshot shows the IBM watsonx Orchestrate interface. On the left, a flow diagram is visible with a 'Document type' node branching into 'Invoice' and 'Contract (default)'. The 'Invoice' branch leads to an 'Invoice extractor' node, which then leads to a 'User activity 3' node. A red circle labeled 'A' is placed over the 'Document classifier' node in the flow diagram. On the right, the 'Map data for End' panel is open. It shows a table with 'Class_name' and 'File upload 1' columns. A red circle labeled 'A' is placed over the 'class_name' column header in the table.

6. The output 'class_name' is now assigned to the 'class_name' variable (A). Click x (B) to close the window.

The screenshot shows the IBM watsonx Orchestrate interface. On the left, a flow diagram is visible with a 'Document classifier' activity. The main panel on the right is titled 'Map data for End' and contains a table for mapping outputs to variables. The table has two columns: 'Class_name' and 'Variable'. The 'Class_name' column contains the text 'class_name'. The 'Variable' column contains the text 'class_name'. A red circle labeled 'A' is positioned over the 'class_name' variable in the 'Variable' column. A red circle labeled 'B' is positioned over the 'x' button in the top right corner of the dialog. The dialog also includes a search bar and a list of variables to choose from.

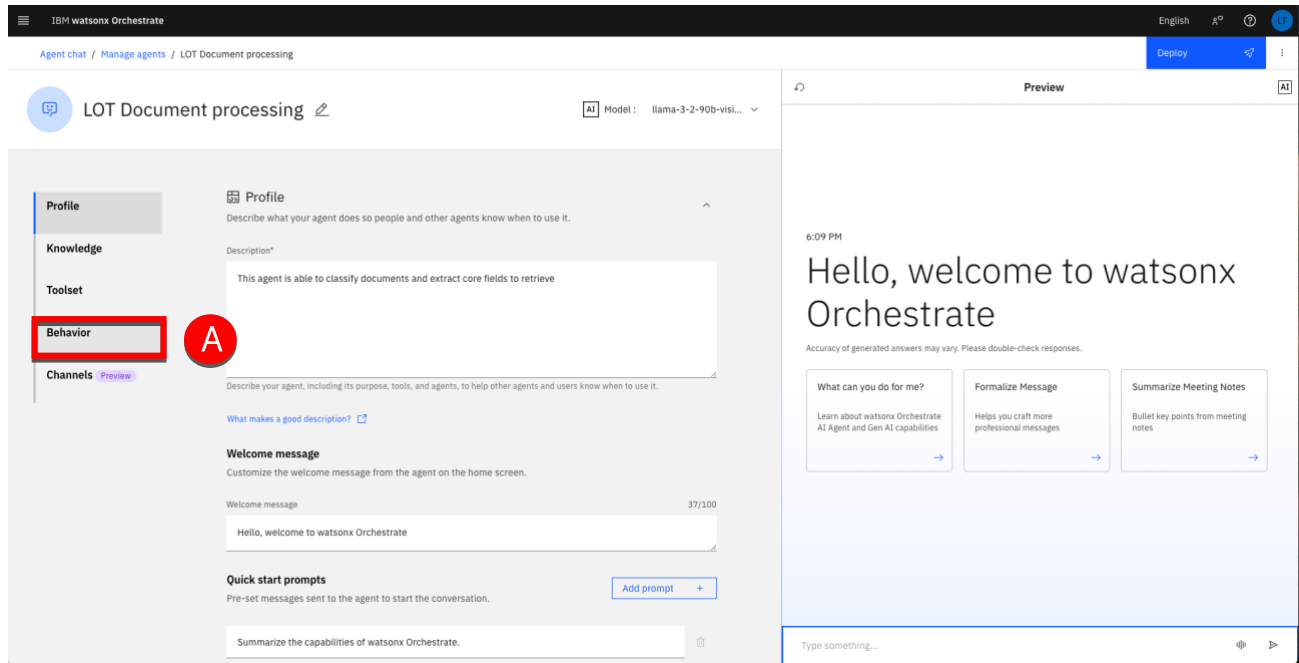
7. Click Done (A).

The screenshot shows the IBM watsonx Orchestrate interface with the flow diagram updated. The flow starts with a 'Document classifier' activity, which branches into two paths: 'Invoice' and 'Contract (default)'. The 'Invoice' path leads to an 'Invoice extractor' activity, followed by 'User activity 3' and 'Extracted Invoice Fields'. The 'Contract (default)' path leads to a 'Contract extractor' activity, followed by 'User activity 2' and 'Extracted contract Fields'. Both paths converge at a final 'Output' activity. A red circle labeled 'A' is positioned over the 'Done' button in the top right corner of the interface.

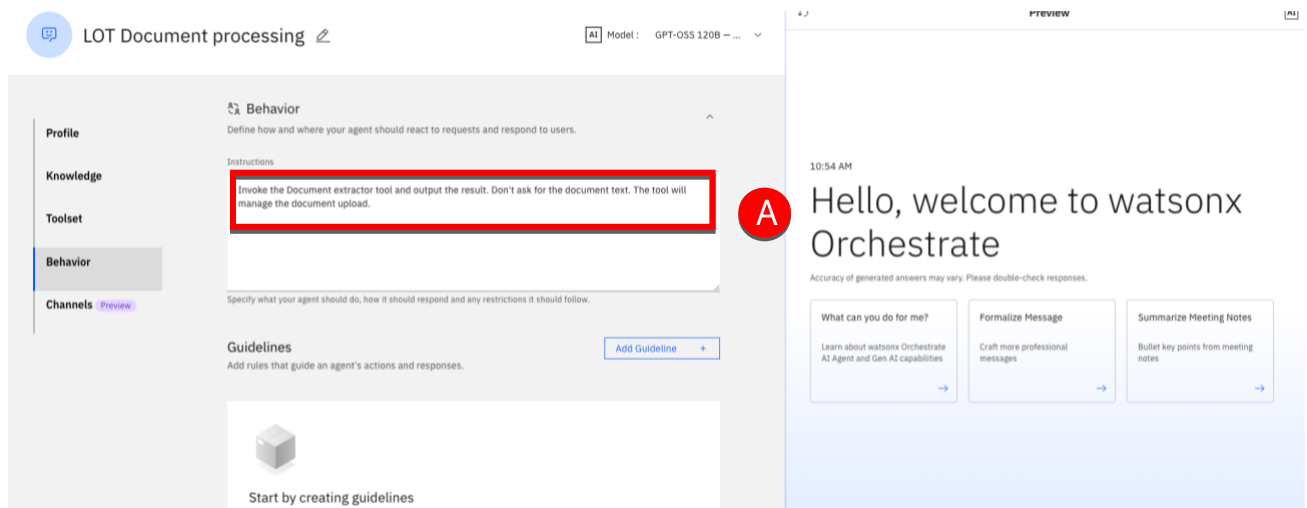
Add a behavior

To trigger the tool you just created, you must update the agent behavior.

1. Click **Behavior** (A).



2. Type 'Invoke the Document extractor tool and output the result. Don't ask for the document text. The tool will manage the document upload.' (A) in the Instructions.



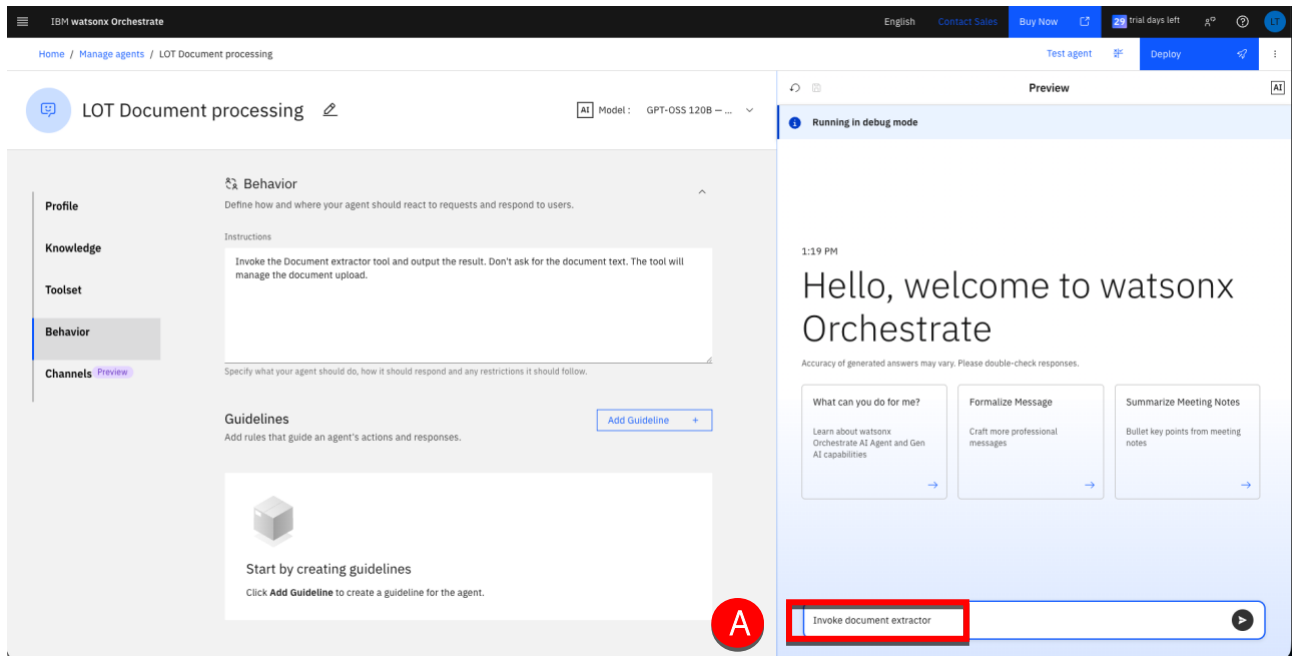
Your agent is ready to be tested.

Test your agent

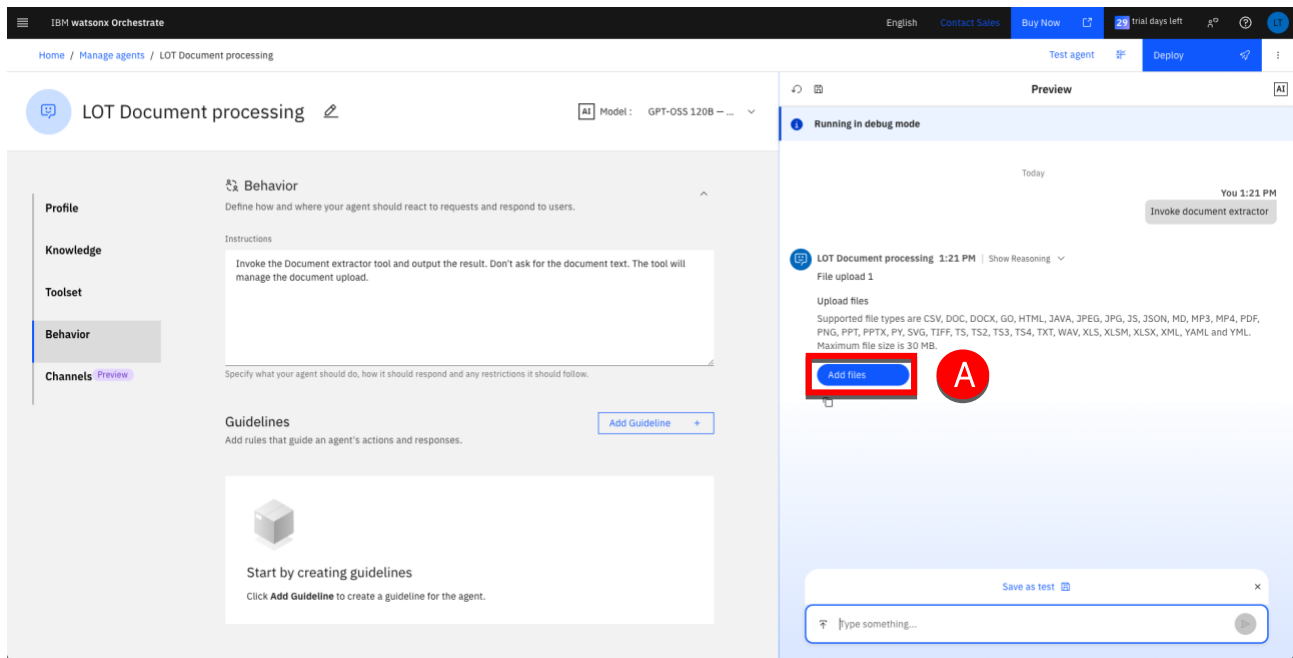
Let's now test the agent using documents that were not used in the different extractor trainings.

Test without manual validation

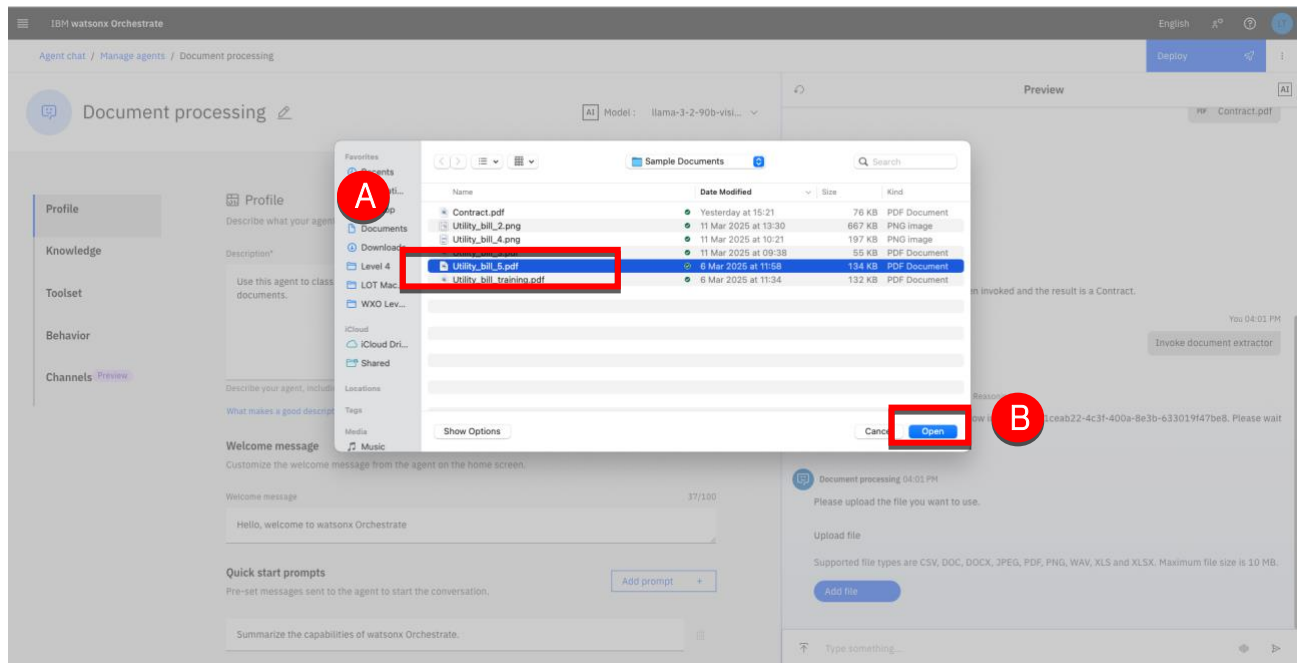
1. Type 'Invoke document extractor' (A) in the Preview instructions area.



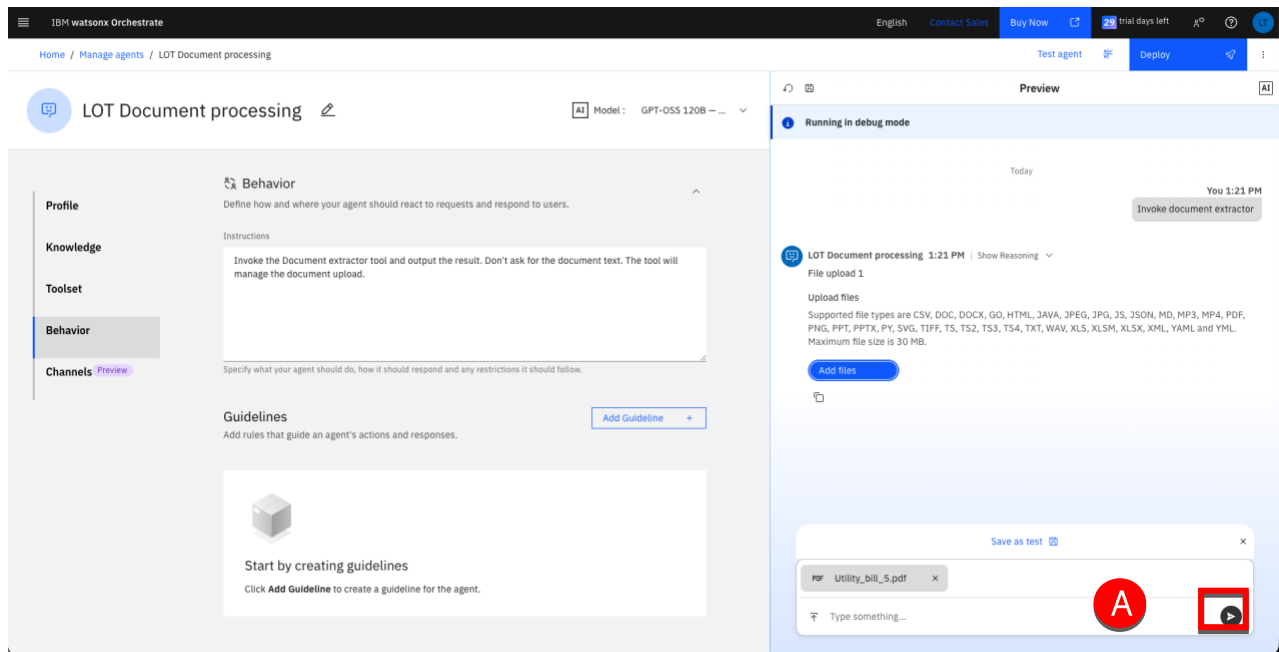
2. Click Add file (A).



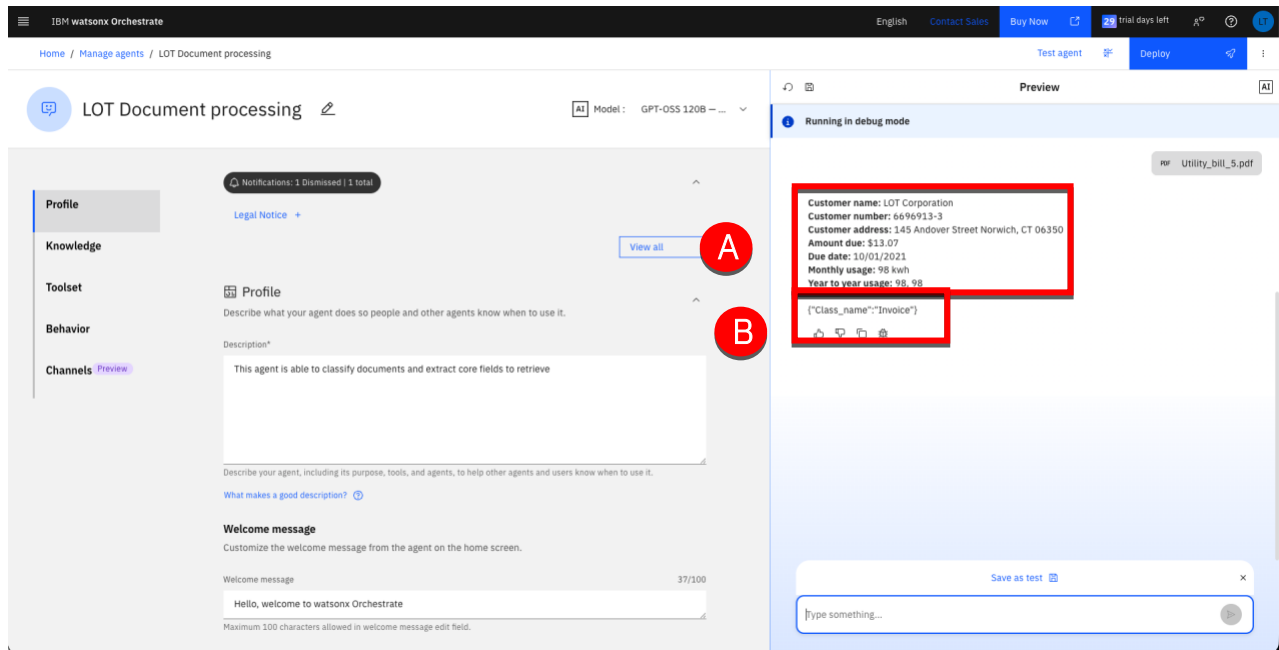
3. Navigate to your **Utility_bill_5.pdf** location and select it (A). Click **Open** (B).



4. Click the send message icon (A).



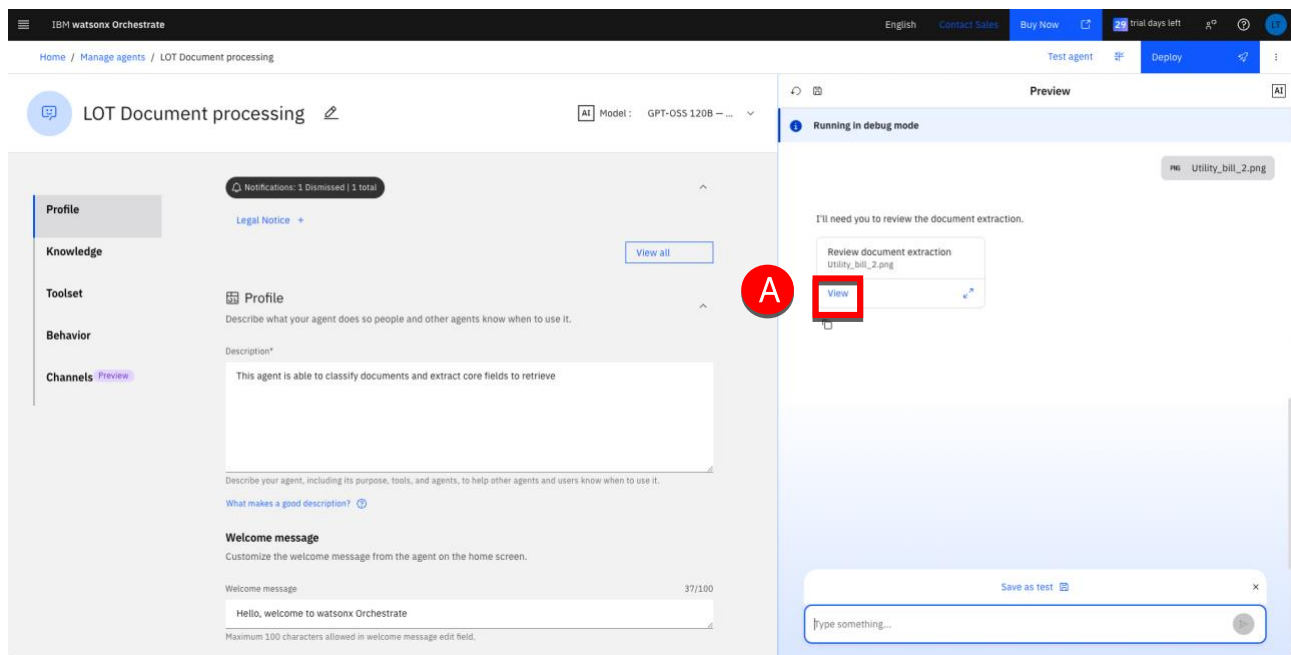
- The extracted fields are displayed (A) as specified in the user activity. The workflow also displays its output that is the document class (B).



Test requiring a manual validation

In some situations, the extraction confidence score may be too low requiring a manual review. This section will guide you through the review process steps.

- When the document the confidence score is too low for automatic extraction, the agent will ask you to validate some of the extracted fields. Click **View** (A).



2. In this example, 8 fields require a review (A):

The extraction looks good despite the low confidence score. Should you need to correct an extracted value, follow the following step:

a. Click the value you need to correct (A).

b. Draw a rectangle around the value in the document image (A) to capture the value.

The screenshot shows the IBM watsonx Orchestrate interface. On the left, there's a sidebar with 'Profile', 'Knowledge', 'Toolset', 'Behavior', and 'Channels'. The main area displays the 'LOT Document processing' agent profile. On the right, a 'Review document extraction' panel shows a document image of a 'Monthly Invoice' from Constellation. A red rectangle labeled 'A' highlights the value '13.07' in the document image. The right sidebar shows 'Needs review (8)' with fields for 'Due Date', 'Year To Year Usage', 'Monthly Usage', 'Amount Due', and 'Customer Address', each with a 'Missing value' status and an 'Enter missing value' button.

Note: You can also directly type in the value in the field.

c. Click **Submit** (A) when all the fields have been reviewed.

The screenshot shows the IBM watsonx Orchestrate interface. On the left, there's a sidebar with 'Profile', 'Knowledge', 'Toolset', 'Behavior', and 'Channels'. The main area displays the 'LOT Document processing' agent profile. On the right, a 'Review document extraction' panel shows a document image of a 'Monthly Invoice' from Constellation. A red rectangle labeled 'A' highlights the 'Submit' button in the bottom right corner of the review panel. The right sidebar shows 'Needs review (0)' with fields for 'Amount Due', 'Customer Address', 'Customer Number', 'Statement Date', and 'Customer Name', each with an 'Updated by user' status and a 'Submit' button.

d. Wait for the process to complete and display the final result (A).

The screenshot displays the IBM watsonx Orchestrate interface for the 'LOT Document processing' agent. The left sidebar contains navigation links: Profile, Knowledge, Toolset, Behavior, and Channels (highlighted in purple). The main area shows the agent's configuration, including a 'Legal Notice' link, a 'Profile' section with a description, and a 'Welcome message' field containing 'Hello, welcome to watsonx Orchestrate'. A red circle with the letter 'A' is placed over the 'Description' field.

The right sidebar shows the 'Preview' section, which is currently in 'Running in debug mode'. A red box highlights the output of the agent, which is a JSON object containing customer information:

```
{
  "Class_name": "Invoice",
  "Customer name": "LOT Corporation",
  "Customer number": "FROB48599006888950",
  "Customer address": "CONSTELLATION BEWENERGY, INC. POBOX 4640",
  "Amount due": "$13.07",
  "Due date": "10/01/2021",
  "Monthly usage": "98 kWh",
  "Year to year usage": "0.0%"
}
```

Below the preview, there is a 'Save as test' button and a text input field labeled 'Type something...'.

Summary

Congratulations on completing the Flow and Document Processing Lab guide.

You are now well equipped to create your own agents able to classify and extract data from documents.

In this lab, we have just used one document to train the Contract extractor. In real projects, it is recommended to train the model with multiple different documents with different formats and to test the extractor with documents that were not used for training as we did for the Invoice extractor.



